

Proof If we let $c = 1/|\mathbf{a}|$, then $c > 0$ and it follows from Definition (14.7) that the vector $\mathbf{u} = c\mathbf{a}$ has the same direction as $\mathbf{a}$. Moreover, by Theorem (14.8),

$$|\mathbf{u}| = |c\mathbf{a}| = |c| |\mathbf{a}| = \frac{1}{|\mathbf{a}|} |\mathbf{a}| = 1. \quad \square$$

Example 5 Find a unit vector $\mathbf{u}$ having the same direction as $3\mathbf{i} - 4\mathbf{j}$.

Solution Since $|\mathbf{a}| = \sqrt{9 + 16} = 5$ we have, from Theorem (14.12),

$$\mathbf{u} = \frac{1}{5}(3\mathbf{i} - 4\mathbf{j}) = \frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}. \quad \blacksquare$$

14.1 Exercises

- 1 (a) Prove (ii)–(iv) of Theorem (14.5).
(b) Give geometric arguments that justify the conclusions.
- 2 Prove (ii)–(v) of Theorem (14.9).

Prove the properties in Exercises 3–10, where $\mathbf{a}$, $\mathbf{b}$ are vectors and c is a scalar.

- 3 $(-1)\mathbf{a} = -\mathbf{a}$
- 4 $(-c)\mathbf{a} = -c\mathbf{a}$
- 5 $-(\mathbf{a} + \mathbf{b}) = -\mathbf{a} - \mathbf{b}$
- 6 $c(\mathbf{a} - \mathbf{b}) = c\mathbf{a} - c\mathbf{b}$
- 7 If $\mathbf{a} + \mathbf{b} = \mathbf{0}$, then $\mathbf{b} = -\mathbf{a}$.
- 8 If $\mathbf{a} + \mathbf{b} = \mathbf{a}$, then $\mathbf{b} = \mathbf{0}$.
- 9 If $c\mathbf{a} = \mathbf{0}$ and $c \neq 0$, then $\mathbf{a} = \mathbf{0}$.
- 10 If $c\mathbf{a} = \mathbf{0}$ and $\mathbf{a} \neq \mathbf{0}$, then $c = 0$.

In Exercises 11–20 find $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, $4\mathbf{a} + 5\mathbf{b}$, and $4\mathbf{a} - 5\mathbf{b}$.

- 11 $\mathbf{a} = \langle 2, -3 \rangle$, $\mathbf{b} = \langle 1, 4 \rangle$
- 12 $\mathbf{a} = \langle -2, 6 \rangle$, $\mathbf{b} = \langle 2, 3 \rangle$
- 13 $\mathbf{a} = -\langle 7, -2 \rangle$, $\mathbf{b} = 4\langle -2, 1 \rangle$
- 14 $\mathbf{a} = 2\langle 5, -4 \rangle$, $\mathbf{b} = -\langle 6, 0 \rangle$
- 15 $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$, $\mathbf{b} = 3\mathbf{i} - 5\mathbf{j}$
- 16 $\mathbf{a} = -3\mathbf{i} + \mathbf{j}$, $\mathbf{b} = -3\mathbf{i} + \mathbf{j}$
- 17 $\mathbf{a} = -(4\mathbf{i} - \mathbf{j})$, $\mathbf{b} = 2(\mathbf{i} - 3\mathbf{j})$
- 18 $\mathbf{a} = 8\mathbf{j}$, $\mathbf{b} = (-3)(-2\mathbf{i} + \mathbf{j})$

$$19 \quad \mathbf{a} = 2\mathbf{j}, \mathbf{b} = -3\mathbf{i}$$

$$20 \quad \mathbf{a} = \mathbf{0}, \mathbf{b} = \mathbf{i} + \mathbf{j}$$

In Exercises 21–26 find a vector $\mathbf{a}$ that has geometric representation $\overrightarrow{PQ}$. Sketch $\overrightarrow{PQ}$ and a position vector corresponding to $\mathbf{a}$.

- 21 $P(1, -4)$, $Q(5, 3)$
- 22 $P(7, -3)$, $Q(-2, 4)$
- 23 $P(2, 5)$, $Q(-4, 5)$
- 24 $P(-4, 6)$, $Q(-4, -2)$
- 25 $P(-3, -1)$, $Q(6, -4)$
- 26 $P(2, 3)$, $Q(-6, 0)$

In Exercises 27–30 sketch vectors corresponding to $\mathbf{a}$, $\mathbf{b}$, $\mathbf{a} + \mathbf{b}$, $\mathbf{a} - \mathbf{b}$, $2\mathbf{a}$, and $-3\mathbf{b}$.

- 27 $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j}$, $\mathbf{b} = -\mathbf{i} + 5\mathbf{j}$
- 28 $\mathbf{a} = -5\mathbf{i} + 2\mathbf{j}$, $\mathbf{b} = \mathbf{i} - 3\mathbf{j}$
- 29 $\mathbf{a} = \langle -4, 6 \rangle$, $\mathbf{b} = \langle -2, 3 \rangle$
- 30 $\mathbf{a} = \langle 2, 0 \rangle$, $\mathbf{b} = \langle -2, 0 \rangle$

In Exercises 31–38 find the magnitude of $\mathbf{a}$.

- 31 $\mathbf{a} = \langle 3, -3 \rangle$
- 32 $\mathbf{a} = \langle -2, -2\sqrt{3} \rangle$
- 33 $\mathbf{a} = \langle -5, 0 \rangle$
- 34 $\mathbf{a} = \langle 0, 10 \rangle$
- 35 $\mathbf{a} = -4\mathbf{i} + 5\mathbf{j}$
- 36 $\mathbf{a} = 10\mathbf{i} - 10\mathbf{j}$
- 37 $\mathbf{a} = -18\mathbf{j}$
- 38 $\mathbf{a} = 0\mathbf{i} + 0\mathbf{j}$

In Exercises 39–42 find a unit vector having (a) the same direction as $\mathbf{a}$; (b) the direction opposite to $\mathbf{a}$.

- 39 $\mathbf{a} = -8\mathbf{i} + 15\mathbf{j}$
- 40 $\mathbf{a} = 5\mathbf{i} - 3\mathbf{j}$
- 41 $\mathbf{a} = \langle 2, -5 \rangle$
- 42 $\mathbf{a} = \langle 0, 6 \rangle$

- 43 Find a vector that has the same direction as $\langle -6, 3 \rangle$ and (a) twice the magnitude; (b) one-half the magnitude.
- 44 Find a vector that has the opposite direction of $8\mathbf{i} - 5\mathbf{j}$ and (a) three times the magnitude; (b) one-third the magnitude.
- 45 Find a vector of magnitude 6 that has the same direction as $\mathbf{a} = 4\mathbf{i} - 7\mathbf{j}$.
- 46 Find a vector of magnitude 4 whose direction is opposite that of $\mathbf{a} = \langle 2, -5 \rangle$.
- 47 Demonstrate graphically that $|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$. Under what conditions is $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$?
- 48 (a) Let $\mathbf{a} = \langle a_1, a_2 \rangle$ be any nonzero vector and let $\overrightarrow{OA}$ be the position vector corresponding to $\mathbf{a}$. If θ is the smallest nonnegative angle from the positive x -axis to $\overrightarrow{OA}$, show that $\mathbf{a} = |\mathbf{a}|(\cos \theta \mathbf{i} + \sin \theta \mathbf{j})$.
(b) Show that every unit vector in V_2 can be expressed in the form $\cos \theta \mathbf{i} + \sin \theta \mathbf{j}$ for some θ .
- 49 If $\mathbf{a} = \langle 1, -4 \rangle$, $\mathbf{b} = \langle -2, 6 \rangle$, and $\mathbf{c} = \langle -2, 3 \rangle$, find scalars p and q such that $\mathbf{c} = p\mathbf{a} + q\mathbf{b}$.
- 50 Generalize Exercise 49 as follows. If $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$ are any nonzero, nonparallel vectors, and if $\mathbf{c}$ is any other vector, prove that there exist scalars p and q such that $\mathbf{c} = p\mathbf{a} + q\mathbf{b}$. Interpret this fact geometrically.
- 51 If $\mathbf{a} = \langle 4, -2 \rangle$, $\mathbf{b} = \langle -6, 3 \rangle$, and $\mathbf{c} = \langle 2, 5 \rangle$, prove that $\mathbf{c}$ cannot be expressed in the form $\mathbf{c} = p\mathbf{a} + q\mathbf{b}$, where p and q are scalars.
- 52 If $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j}$, find all real numbers c such that
(a) $|c\mathbf{a}| = 3$; (b) $|c\mathbf{a}| = -3$; (c) $|c\mathbf{a}| = 0$.
- 53 Let $\mathbf{r}_0 = \langle x_0, y_0 \rangle$, $\mathbf{r} = \langle x, y \rangle$, and $c > 0$. Describe the set of all points $P(x, y)$ such that $|\mathbf{r} - \mathbf{r}_0| = c$.
- 54 Let $\mathbf{r}_0 = \langle x_0, y_0 \rangle$, $\mathbf{r} = \langle x, y \rangle$, and $\mathbf{a} = \langle a_1, a_2 \rangle \neq \mathbf{0}$. Describe the set of all points $P(x, y)$ such that $\mathbf{r} - \mathbf{r}_0$ is a scalar multiple of $\mathbf{a}$.
- 55 If $P_1, P_2, \dots, P_n$ are arbitrary points in a coordinate plane, show that $\sum_{i=1}^{n-1} \overrightarrow{P_i P_{i+1}} + \overrightarrow{P_n P_1}$ is the zero vector. Illustrate this fact graphically if $n = 5$.
- 56 If $P_1, P_2, \dots, P_n$ are vertices of a regular polygon and if O is the center of the polygon, prove that $\sum_{i=1}^n \overrightarrow{OP_i}$ is the zero vector.

14.2 Rectangular Coordinate Systems in Three Dimensions

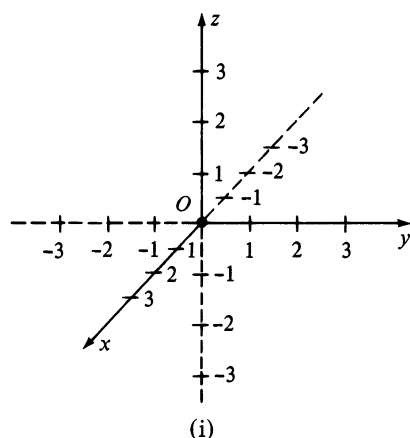


FIGURE 14.13 Right-handed coordinate system

If we wish to discuss geometric aspects of vectors in space, it is necessary to introduce a three-dimensional coordinate system. This is accomplished by means of ordered triples of real numbers. An **ordered triple** (a, b, c) is a set $\{a, b, c\}$ of three numbers in which a is considered as the first number of the set, b as the second number, and c as the third number. The totality of all such ordered triples is denoted by $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$. Two ordered triples (a_1, a_2, a_3) and (b_1, b_2, b_3) are said to be **equal**, and we write $(a_1, a_2, a_3) = (b_1, b_2, b_3)$, if and only if $a_1 = b_1$, $a_2 = b_2$, and $a_3 = b_3$.

To specify points in space we choose a fixed point O (called the **origin**) and consider three mutually perpendicular coordinate lines (called the x -, y -, and z -axes) with common origin O , as illustrated in (i) of Figure 14.13. To visualize this configuration, we may imagine that the y - and z -axes lie in the plane of the paper and that the x -axis projects out from the paper. A coordinate system of this type is called **right-handed**. If the x - and y -axes are interchanged, the resulting coordinate system is said to be **left-handed**. The coordinate plane determined by the x - and y -axes is called the **xy -plane**. Similarly, the coordinate plane determined by the y - and z -axes is called the **yz -plane** and that determined by the x - and z -axes is called the **xz -plane**. The terminology *right-handed* may be justified as follows. If, as in (ii) of Figure 14.13, the fingers of the right hand are curled in the direction of a 90° rotation of axes in the xy -plane (so that the positive x -axis is transformed into the positive y -axis), then the extended thumb points in the direction of the positive z -axis.

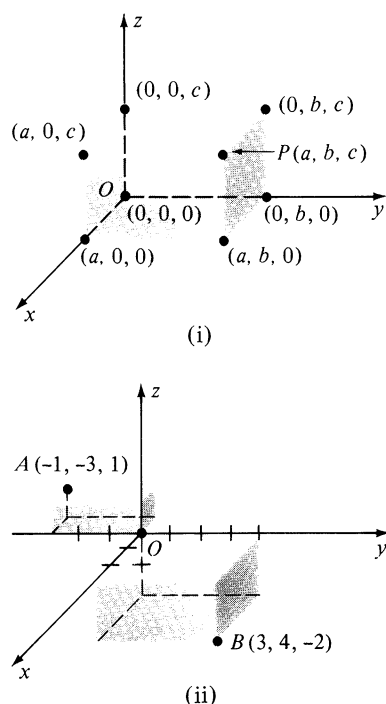


FIGURE 14.14

If P is a point, then the projection of P on the x -axis has some coordinate a , called the **x -coordinate** of P . The x -coordinate of P may also be thought of as a directed distance from the yz -plane to P . Similarly, the coordinates b and c of the projections of P on the y - and z -axes, respectively, are called the **y -coordinate** and **z -coordinate** of P . We shall use the ordered triple (a, b, c) to denote the coordinates of P , as well as P itself. The notation $P(a, b, c)$ will also be used for the point P with coordinates a, b , and c . If P is not on a coordinate plane, then the three planes through P which are parallel to the coordinate planes, together with the coordinate planes, determine a rectangular parallelepiped. This is illustrated in (i) of Figure 14.14, where the eight vertices are labeled in the manner just described. Conversely, to each ordered triple (a, b, c) of real numbers, there corresponds a point P having coordinates a, b , and c . The concept of plotting points is similar to that used in two dimensions. As an aid to plotting, it is sometimes convenient to construct a parallelepiped as shown in (i) of Figure 14.14. The points $A(-1, -3, 1)$ and $B(3, 4, -2)$ are plotted in (ii) of the figure.

The one-to-one correspondence between the points in space and ordered triples of real numbers we have described is called a **rectangular coordinate system in three dimensions**. The three coordinate planes partition space into eight parts, called **octants**. The part consisting of all points $P(a, b, c)$ whose three coordinates a, b , and c are positive is called the **first octant**.

It is not difficult to find a formula for the distance $d(P_1, P_2)$ between two points P_1 and P_2 . If P_1 and P_2 are on a line parallel to the z -axis, as illustrated in (i) of Figure 14.15, and if their projections on the z -axis are $A_1(0, 0, z_1)$ and $A_2(0, 0, z_2)$, respectively, then $d(P_1, P_2) = d(A_1, A_2) = |z_2 - z_1|$. Similar formulas hold if the line through P_1 and P_2 is parallel to the x - or y -axis.

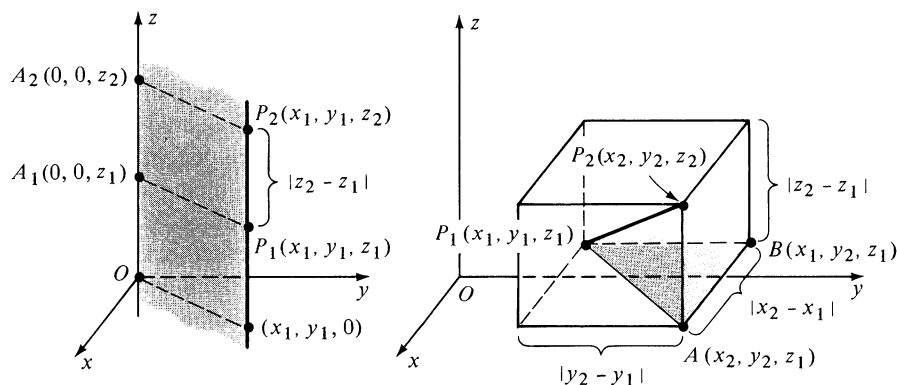


FIGURE 14.15

If we have a situation of the type illustrated in (ii) of Figure 14.15, then triangle P_1AP_2 is a right triangle, and hence by the Pythagorean Theorem,

$$d(P_1, P_2) = \sqrt{[d(P_1, A)]^2 + [d(A, P_2)]^2}.$$

Since P_1 and A are in a plane parallel to the xy -plane, it follows from the Distance Formula (1.10) that $[d(P_1, A)]^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$, whereas from our previous remarks $[d(A, P_2)]^2 = (z_2 - z_1)^2$. Substituting in the preceding formula for $d(P_1, P_2)$ gives us the following.

Distance Formula (14.13)

The distance between any two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Note that if P_1 and P_2 are on the xy -plane, so that $z_1 = z_2 = 0$, then (14.13) reduces to the two-dimensional Distance Formula.

Example 1 Find the distance between $A(-1, -3, 1)$ and $B(3, 4, -2)$.

Solution Points A and B are plotted in Figure 14.14. Using the Distance Formula,

$$\begin{aligned} d(A, B) &= \sqrt{(3 + 1)^2 + (4 + 3)^2 + (-2 - 1)^2} \\ &= \sqrt{16 + 49 + 9} = \sqrt{74}. \end{aligned}$$

■

By referring to Figure 14.14 and using similar triangles, the following result can be proved.

Midpoint Formula (14.14)

The midpoint of the line segment from $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right).$$

As an illustration of (14.14), if A and B are the points in Example 1, then the midpoint of segment AB has coordinates $(1, \frac{1}{2}, -\frac{1}{2})$.

The **graph of an equation** in three variables x , y , and z is defined as the set of all points $P(a, b, c)$ in a rectangular coordinate system such that the ordered triple (a, b, c) is a solution of the equation; that is, equality is obtained when a , b , and c are substituted for x , y , and z , respectively. The graph of such an equation is a **surface**. It is easy to derive an equation which has, as its graph, a sphere of radius r with center at the point $C(x_0, y_0, z_0)$. As illustrated in

Figure 14.16 a point $P(x, y, z)$ is on the sphere if and only if $[d(C, P)]^2 = r^2$. Applying the Distance Formula gives us the next theorem.

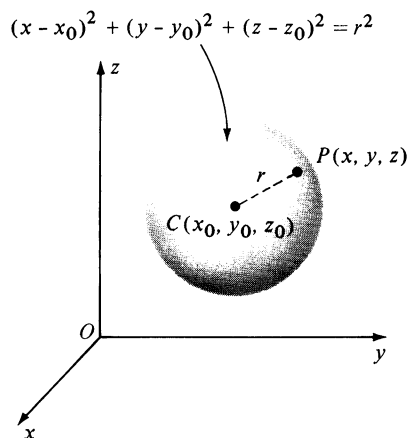


FIGURE 14.16

Theorem (14.15)

An equation of a sphere of radius r and center $C(x_0, y_0, z_0)$ is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2.$$

If we square the indicated expressions and simplify, then the equation in (14.15) may be written in the form

$$x^2 + y^2 + z^2 + ax + by + cz + d = 0$$

where the coefficients are real numbers. Conversely, if we begin with an equation of this form and if the graph exists, then by completing squares we can obtain the form (14.15), and hence the graph is a sphere or a point.

Example 2 Discuss the graph of the equation

$$x^2 + y^2 + z^2 - 6x + 8y + 4z + 4 = 0.$$

Solution We complete squares as follows:

$$(x^2 - 6x) + (y^2 + 8y) + (z^2 + 4z) = -4$$

$$(x^2 - 6x + 9) + (y^2 + 8y + 16) + (z^2 + 4z + 4) = -4 + 9 + 16 + 4$$

$$(x - 3)^2 + (y + 4)^2 + (z + 2)^2 = 25.$$

Comparing the last equation with (14.15) it follows that the graph is a sphere of radius 5 with center $C(3, -4, -2)$. ■

14.2 Exercises

In Exercises 1–6 plot the points A and B and find (a) $d(A, B)$; (b) the midpoint of AB .

- 1 $A(2, 4, -5), B(4, -2, 3)$ 2 $A(1, -2, 7), B(2, 4, -1)$
 3 $A(-4, 0, 1), B(3, -2, 1)$ 4 $A(0, 5, -4), B(1, 1, 0)$
 5 $A(1, 0, 0), B(0, 1, 1)$ 6 $A(0, 0, 0), B(-8, -1, 4)$

In Exercises 7 and 8 suppose A and B are opposite vertices of a parallelepiped having its faces parallel to the coordinate planes. Find the coordinates of the other vertices.

- 7 $A(2, 5, -3), B(-4, 2, 1)$ 8 $A(-3, 2, 6), B(1, 5, -1)$

In Exercises 9 and 10 prove that A, B , and C are vertices of a right triangle and find its area.

- 9 $A(2, 0, 1), B(3, 1, 2), C(1, 2, 0)$
 10 $A(4, -3, 2), B(6, -2, 1), C(7, -6, 5)$

In Exercises 11–14 find an equation of the sphere with center C and radius r .

- 11 $C(3, -1, 2), r = 3$ 12 $C(4, -5, 1), r = 5$
 13 $C(-5, 0, 1), r = 1/2$ 14 $C(0, -3, -6), r = \sqrt{3}$
 15 Find an equation of the sphere with center $(-2, 4, -6)$ that is tangent to (a) the yz -plane; (b) the xz -plane; (c) the xy -plane.
 16 Find an equation of the sphere having endpoints of a diameter at $A(1, 4, -2)$ and $B(-7, 1, 2)$.

In Exercises 17–22 find the center and radius of the sphere having the given equation.

- 17 $x^2 + y^2 + z^2 + 4x - 2y + 2z + 2 = 0$

18 $x^2 + y^2 + z^2 - 6x - 10y + 6z + 34 = 0$

19 $x^2 + y^2 + z^2 - 8x + 8z + 16 = 0$

20 $4x^2 + 4y^2 + 4z^2 - 4x + 8y - 3 = 0$

21 $x^2 + y^2 + z^2 + 4y = 0$

22 $x^2 + y^2 + z^2 - z = 0$

23 Find an equation for the set of all points equidistant from $A(2, -1, 3)$ and $B(-1, 5, 1)$. Describe the graph of the equation.

24 Work Exercise 23 using $A(5, 0, -4)$ and $B(2, -1, 7)$.

25 Describe the graphs of the equations (a) $z = 5$; (b) $y = 2$; and (c) $x = 0$.

26 Find an equation for the set of all points $P(x, y, z)$ that are three times as far from $A(2, -1, 3)$ as they are from $B(-1, 5, 1)$. Describe the graph of the equation.

In Exercises 27–34 describe the given region R in a three-dimensional coordinate system.

27 $R = \{(x, y, z): x^2 + y^2 + z^2 \leq 1\}$

28 $R = \{(x, y, z): x^2 + y^2 + z^2 > 1\}$

29 $R = \{(x, y, z): |x| \leq 1, |y| \leq 2, |z| \leq 3\}$ •

30 $R = \{(x, y, z): x^2 + y^2 = 1\}$

31 $R = \{(x, y, z): 0 < x^2 + y^2 \leq 1\}$

32 $R = \{(x, y, z): 4 < x^2 + y^2 + z^2 < 9\}$

33 $R = \{(x, y, z): x^2 + y^2 + z^2 - 4x - 2y + 1 < 0\}$

34 $R = \{(x, y, z): y = x\}$

14.3 Vectors in Three Dimensions

The next definition extends the concept of vectors in V_2 to three dimensions.

Definition (14.16)

The **three-dimensional vector space** V_3 is the set of all ordered triples $\langle x, y, z \rangle$ of real numbers, called **vectors**, such that if $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, and c is a scalar, then

(i) $\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$

(ii) $c\mathbf{a} = \langle ca_1, ca_2, ca_3 \rangle$.

The numbers a_1 , a_2 , and a_3 are called the **components** of the vector $\langle a_1, a_2, a_3 \rangle$. Following the same procedure used in V_2 , we define the **zero vector** $\mathbf{0}$, the **negative** $-\mathbf{a}$ of $\mathbf{a}$, the **magnitude** $|\mathbf{a}|$ of $\mathbf{a}$, and the **difference** $\mathbf{a} - \mathbf{b}$ as follows.

Definition (14.17)

- (i) $\mathbf{0} = \langle 0, 0, 0 \rangle$
- (ii) $-\mathbf{a} = -\langle a_1, a_2, a_3 \rangle = \langle -a_1, -a_2, -a_3 \rangle$
- (iii) $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$
- (iv) $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}) = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$

Properties of vectors in two dimensions may be extended without difficulty to V_3 by simply taking the third component into account. In particular, the properties listed in Theorems (14.5) and (14.9) are readily proved.

A vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ in V_3 may be represented in a rectangular coordinate system by a directed line segment $\overline{PQ}$ with arbitrary initial point $P(x, y, z)$ and terminal point $Q(x + a_1, y + a_2, z + a_3)$, as illustrated in Figure 14.17.

As before, directed line segments are also referred to as *vectors*. The vector $\overline{OP}$ from the origin to a point P is called the **position vector of P** . If P has coordinates (a_1, a_2, a_3) , then $\overline{OP}$ (see Figure 14.17) is called the **position vector corresponding to $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$** . The geometric interpretation of vector addition in three dimensions is exactly the same as that in two dimensions.

Note that the magnitude of $\mathbf{a}$ is the length of any of its geometric representations. Observe also that $|\mathbf{a}| \geq 0$, where $|\mathbf{a}| = 0$ if and only if $\mathbf{a} = \mathbf{0}$. As in two dimensions, it can be proved that $|c\mathbf{a}| = |c||\mathbf{a}|$.

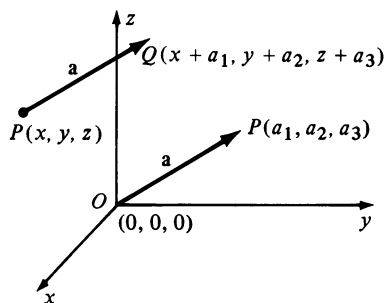


FIGURE 14.17

Example 1 If $\mathbf{a} = \langle 2, 5, -3 \rangle$ and $\mathbf{b} = \langle -4, 1, 7 \rangle$, find the vectors $\mathbf{a} + \mathbf{b}$, $2\mathbf{a} - 3\mathbf{b}$, and $|\mathbf{a}|$.

Solution

$$\mathbf{a} + \mathbf{b} = \langle 2, 5, -3 \rangle + \langle -4, 1, 7 \rangle = \langle -2, 6, 4 \rangle$$

$$\begin{aligned} 2\mathbf{a} - 3\mathbf{b} &= 2\langle 2, 5, -3 \rangle - 3\langle -4, 1, 7 \rangle = \langle 4, 10, -6 \rangle - \langle -12, 3, 21 \rangle \\ &= \langle 16, 7, -27 \rangle \end{aligned}$$

$$|\mathbf{a}| = \sqrt{(2)^2 + (5)^2 + (-3)^2} = \sqrt{38}$$

■

Exactly as for vectors in two dimensions, we say that two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ of V_3 have the *same direction* if $\mathbf{b} = c\mathbf{a}$ for some scalar $c > 0$, or *opposite directions* if $\mathbf{b} = c\mathbf{a}$ for some $c < 0$. The vectors $\mathbf{a}$ and $\mathbf{b}$ are *parallel* if $\mathbf{b} = c\mathbf{a}$ for some scalar c . To illustrate, given the vectors

$$\mathbf{a} = \langle 15, -6, 24 \rangle, \quad \mathbf{b} = \langle 5, -2, 8 \rangle, \quad \mathbf{c} = \langle -\frac{15}{2}, 3, -12 \rangle$$

the vectors $\mathbf{a}$ and $\mathbf{b}$ have the same direction since $\mathbf{a} = 3\mathbf{b}$ or, equivalently, $\mathbf{b} = \frac{1}{3}\mathbf{a}$; whereas $\mathbf{c}$ and $\mathbf{a}$ have opposite directions since $\mathbf{c} = -\frac{1}{2}\mathbf{a}$, or $\mathbf{a} = -2\mathbf{c}$. Each of these vectors is parallel to the other two.

Theorem (14.18)

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are points, then the vector $\mathbf{a}$ in V_3 that has geometric representation $\overrightarrow{P_1P_2}$ is

$$\mathbf{a} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle.$$

The proof is similar to that of Theorem (14.4).

Two vectors $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ are said to have the same (or the opposite) direction if their corresponding vectors in V_3 have the same (or the opposite) direction. If the magnitude $|\overrightarrow{PQ}|$ is defined as the distance between P and Q , then the notation $\overrightarrow{PQ} = \overrightarrow{RS}$ means that the indicated vectors have the same magnitude and direction. If $\mathbf{a}$ is the vector in V_3 corresponding to $\overrightarrow{PQ}$, and c is a scalar, then a geometric representation of $c\mathbf{a}$ is denoted by $c\overrightarrow{PQ}$. Such scalar multiples have the same geometric meaning as their counterparts in two dimensions. Moreover, the vectors $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ have the same direction if $\overrightarrow{PQ} = c\overrightarrow{RS}$ for some $c > 0$, and opposite directions if $\overrightarrow{PQ} = c\overrightarrow{RS}$ for some $c < 0$. In either case, we say that $\overrightarrow{PQ}$ and $\overrightarrow{RS}$ are *parallel*.

Example 2 Given the points $P_1(5, 6, -2)$ and $P_2(-3, 8, 7)$,

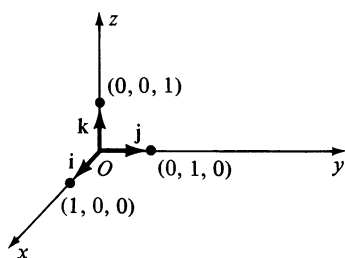
(a) find the position vector corresponding to $\overrightarrow{P_1P_2}$.

(b) find $|\overrightarrow{P_1P_2}|$.

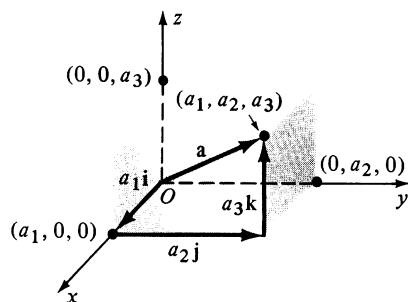
Solution

(a) By (14.18), the vector $\mathbf{a} = \langle -3 - 5, 8 - 6, 7 + 2 \rangle = \langle -8, 2, 9 \rangle$ has geometric representation $\overrightarrow{P_1P_2}$. Thus $A(-8, 2, 9)$ is the terminal point of the position vector $\overrightarrow{OA}$ corresponding to $\overrightarrow{P_1P_2}$.

(b) $|\overrightarrow{P_1P_2}| = |\mathbf{a}| = \sqrt{64 + 4 + 81} = \sqrt{149}$ ■



(i)



(ii)

FIGURE 14.18

A vector $\mathbf{a}$ is a **unit vector** if $|\mathbf{a}| = 1$. The special unit vectors

$$\mathbf{i} = \langle 1, 0, 0 \rangle, \quad \mathbf{j} = \langle 0, 1, 0 \rangle, \quad \mathbf{k} = \langle 0, 0, 1 \rangle$$

are important since any vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ can be expressed as a linear combination of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. Specifically,

$$\mathbf{a} = \langle a_1, a_2, a_3 \rangle = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}.$$

Geometric representations of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are sketched in (i) of Figure 14.18. Part (ii) of the figure illustrates how the position vector for $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ may be regarded as the sum of three vectors corresponding to $a_1\mathbf{i}$, $a_2\mathbf{j}$, and $a_3\mathbf{k}$.

As in Section 14.1, rules for addition, subtraction, and multiplication by scalars may be easily translated into the $\mathbf{i}, \mathbf{j}, \mathbf{k}$ notation. It is often convenient to regard V_2 as a subset of V_3 by identifying the vector $\langle a_1, a_2 \rangle$ with $\langle a_1, a_2, 0 \rangle$. If this is done, then there is no essential difference between the vectors $\mathbf{i} = \langle 1, 0 \rangle$, $\mathbf{j} = \langle 0, 1 \rangle$, and the vectors $\mathbf{i}, \mathbf{j}$ defined in V_3 .

The next concept to be introduced has many mathematical and physical applications. In the manner of our previous work we shall begin with an algebraic definition and give geometric and physical interpretations later in the section.

Definition (14.19)

The dot product $\mathbf{a} \cdot \mathbf{b}$ of $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ is

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

The symbol $\mathbf{a} \cdot \mathbf{b}$ is read “ $\mathbf{a}$ dot $\mathbf{b}$.” The dot product is also referred to as the **scalar product**, or **inner product**. It is important to note that $\mathbf{a} \cdot \mathbf{b}$ is a scalar, not a vector. For example,

$$\langle 2, 4, -3 \rangle \cdot \langle -1, 5, 2 \rangle = (2)(-1) + (4)(5) + (-3)(2) = 12$$

$$(3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot (4\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}) = (3)(4) + (-2)(5) + (1)(-2) = 0.$$

Some properties of the dot product are listed in the following theorem, where $\mathbf{a}, \mathbf{b}$, and $\mathbf{c}$ are any vectors, and c is a scalar.

Theorem (14.20)

- (i) $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$
- (ii) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- (iii) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$
- (iv) $(c\mathbf{a}) \cdot \mathbf{b} = c(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (c\mathbf{b})$
- (v) $\mathbf{0} \cdot \mathbf{a} = 0$

Proof If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$, then

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= \langle a_1, a_2, a_3 \rangle \cdot \langle b_1 + c_1, b_2 + c_2, b_3 + c_3 \rangle \\ &= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3) \\ &= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3) \\ &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}. \end{aligned}$$

This proves property (iii). Proofs of the remaining properties are left as exercises. $\square$

There is a close connection between dot products and the angle between two vectors, as defined below.

Definition (14.21)

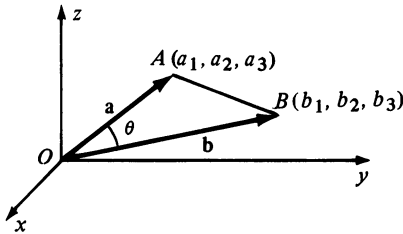


FIGURE 14.19

Let $\mathbf{a}$ and $\mathbf{b}$ be nonzero vectors. If $\mathbf{b}$ is not a scalar multiple of $\mathbf{a}$, and if $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are the position vectors corresponding to $\mathbf{a}$ and $\mathbf{b}$, respectively, then the angle θ between $\mathbf{a}$ and $\mathbf{b}$ (or between $\overrightarrow{OA}$ and $\overrightarrow{OB}$) is angle AOB of the triangle determined by points A , O , and B (see Figure 14.19). If $\mathbf{b} = c\mathbf{a}$ for some scalar c , then $\theta = 0$ or $\theta = \pi$ according as $c > 0$ or $c < 0$.

Note that if $\mathbf{a}$ and $\mathbf{b}$ are parallel then, by (14.21), $\theta = 0$ or $\theta = \pi$. The vectors $\mathbf{a}$ and $\mathbf{b}$ are said to be **orthogonal** if $\theta = \pi/2$. For convenience we shall assume that the zero vector $\mathbf{0}$ is parallel and orthogonal to every vector $\mathbf{a}$.

Theorem (14.22)

If θ is the angle between nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Proof If $\mathbf{b} \neq c\mathbf{a}$ we have a situation similar to that illustrated in Figure 14.19. Applying the Law of Cosines to triangle AOB gives us

$$|\overrightarrow{AB}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Consequently,

$$\begin{aligned} (b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2 \\ = a_1^2 + a_2^2 + a_3^2 + b_1^2 + b_2^2 + b_3^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta \end{aligned}$$

which reduces to

$$-2a_1b_1 - 2a_2b_2 - 2a_3b_3 = -2|\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Dividing both sides of the last equation by -2 gives us the desired conclusion.

If $\mathbf{b} = c\mathbf{a}$, then by properties (iv) and (i) of Theorem (14.20),

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot (c\mathbf{a}) = c(\mathbf{a} \cdot \mathbf{a}) = c|\mathbf{a}|^2.$$

Also, $|\mathbf{a}| |\mathbf{b}| \cos \theta = |\mathbf{a}| |c\mathbf{a}| \cos \theta = |c| |\mathbf{a}|^2 \cos \theta.$

If $c > 0$, then $|c| = c$, $\theta = 0$, and $|c| |\mathbf{a}|^2 \cos \theta$ reduces to $c|\mathbf{a}|^2$. Thus $|\mathbf{a}| |\mathbf{b}| \cos \theta = \mathbf{a} \cdot \mathbf{b}$. If $c < 0$, then $|c| = -c$, $\theta = \pi$, and again $|c| |\mathbf{a}|^2 \cos \theta$ reduces to $c|\mathbf{a}|^2$. This completes the proof of the theorem. $\square$

Corollary (14.23)If θ is the angle between nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}.$$

Example 3 Find the angle between $\mathbf{a} = \langle 4, -3, 1 \rangle$ and $\mathbf{b} = \langle -1, -2, 2 \rangle$.**Solution** Applying Corollary (14.23),

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{(4)(-1) + (-3)(-2) + (1)(2)}{\sqrt{16 + 9 + 1} \sqrt{1 + 4 + 4}} = \frac{4}{3\sqrt{26}} = \frac{4\sqrt{26}}{78} = \frac{2\sqrt{26}}{39},$$

or

$$\theta = \arccos(2\sqrt{26}/39).$$

Using a table or calculator gives us the approximations

$$\theta \approx 74.84^\circ \approx 1.31 \text{ radians.} \quad \blacksquare$$

The following theorem is an immediate consequence of Theorem (14.22).

Theorem (14.24)Two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if and only if $\mathbf{a} \cdot \mathbf{b} = 0$.**Example 4** Prove that the following pairs of vectors are orthogonal.(a) $\mathbf{i}, \mathbf{j}$ (b) $3\mathbf{i} - 7\mathbf{j} + 2\mathbf{k}, 10\mathbf{i} + 4\mathbf{j} - \mathbf{k}$ **Solution** The proof follows from Theorem (14.24). Thus

$$(a) \quad \mathbf{i} \cdot \mathbf{j} = \langle 1, 0, 0 \rangle \cdot \langle 0, 1, 0 \rangle = (1)(0) + (0)(1) + (0)(0) = 0.$$

$$(b) \quad (3\mathbf{i} - 7\mathbf{j} + 2\mathbf{k}) \cdot (10\mathbf{i} + 4\mathbf{j} - \mathbf{k}) = 30 - 28 - 2 = 0. \quad \blacksquare$$

Each of the next two results is true for all vectors $\mathbf{a}$ and $\mathbf{b}$.**Cauchy-Schwarz Inequality (14.25)**

$$|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|$$

Proof The result is trivial if either $\mathbf{a}$ or $\mathbf{b}$ is $\mathbf{0}$. If $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors, then by Theorem (14.22), $|\mathbf{a} \cdot \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| |\cos \theta|$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. Since $|\cos \theta| \leq 1$, it follows that $|\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|$. $\square$ **Triangle Inequality (14.26)**

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|$$

Proof Using properties of the dot product we may write

$$\begin{aligned} |\mathbf{a} + \mathbf{b}|^2 &= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) = \mathbf{a} \cdot \mathbf{a} + 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b} \\ &= |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2. \end{aligned}$$

Since $\mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a} \cdot \mathbf{b}| \leq |\mathbf{a}| |\mathbf{b}|$, we have

$$|\mathbf{a} + \mathbf{b}|^2 \leq |\mathbf{a}|^2 + 2|\mathbf{a}| |\mathbf{b}| + |\mathbf{b}|^2 = (|\mathbf{a}| + |\mathbf{b}|)^2.$$

Taking square roots gives us

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|.$$

□

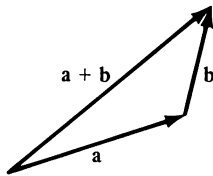


FIGURE 14.20

The reason for the name “Triangle Inequality” is apparent if we represent $\mathbf{a}$ and $\mathbf{b}$ geometrically as in Figure 14.20. In this case (14.26) states that the length of a side of a triangle is not greater than the sum of the lengths of the other two sides.

The dot product of two vectors $\overrightarrow{PQ}$ and $\overrightarrow{TR}$ may be obtained by first defining the angle between these vectors as the angle θ between their corresponding vectors $\mathbf{a}$ and $\mathbf{b}$ in V_3 . If $\theta = \pi/2$, then $\overrightarrow{PQ}$ and $\overrightarrow{TR}$ are **orthogonal**, or **perpendicular**. A typical angle is illustrated in Figure 14.21, where $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are position vectors corresponding to $\mathbf{a}$ and $\mathbf{b}$, respectively. The **dot product** of $\overrightarrow{PQ}$ and $\overrightarrow{TR}$ is defined by

$$\overrightarrow{PQ} \cdot \overrightarrow{TR} = \mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Since $|\overrightarrow{PQ}| = |\mathbf{a}|$ and $|\overrightarrow{TR}| = |\mathbf{b}|$, this may be written

$$\overrightarrow{PQ} \cdot \overrightarrow{TR} = |\overrightarrow{PQ}| |\overrightarrow{TR}| \cos \theta.$$

If $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ have the same initial point, and if S is the projection of Q on the line through P and R (see Figure 14.22), then the scalar $|\overrightarrow{PQ}| \cos \theta$ is called the **component of $\overrightarrow{PQ}$ along $\overrightarrow{PR}$** abbreviated **comp $_{\overrightarrow{PR}} \overrightarrow{PQ}$** .

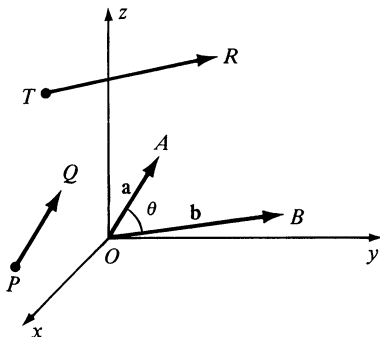
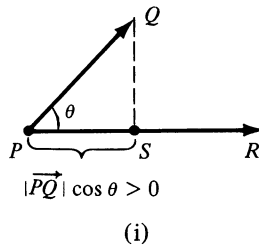
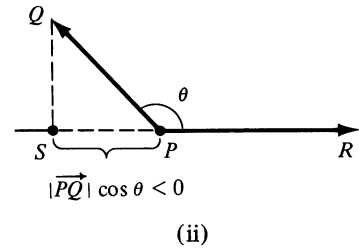


FIGURE 14.21



(i)



(ii)

 FIGURE 14.22 comp $_{\overrightarrow{PR}} \overrightarrow{PQ}$

Note that $|\overrightarrow{PQ}| \cos \theta$ is positive or negative if $0 \leq \theta < \pi/2$ or $\pi/2 < \theta \leq \pi$, respectively. If $\theta = \pi/2$, the component is 0. It follows that

$$\text{comp}_{\overrightarrow{PR}} \overrightarrow{PQ} = |\overrightarrow{PQ}| \cos \theta = \overrightarrow{PQ} \cdot \frac{1}{|\overrightarrow{PR}|} \overrightarrow{PR}.$$

One way to remember this fact is to use the following statement.

Theorem (14.27)

The component of $\overrightarrow{PQ}$ along $\overrightarrow{PR}$ equals the dot product of $\overrightarrow{PQ}$ with a unit vector having the same direction as $\overrightarrow{PR}$.

The preceding concept may be applied to vectors $\mathbf{a}$ and $\mathbf{b}$ of V_3 by representing them geometrically as $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, respectively. This gives us the following definition.

Definition (14.28)

Let $\mathbf{a}$ and $\mathbf{b}$ be vectors in V_3 . The **component of $\mathbf{a}$ along $\mathbf{b}$** , denoted by $\text{comp}_{\mathbf{b}} \mathbf{a}$, is

$$\text{comp}_{\mathbf{b}} \mathbf{a} = \mathbf{a} \cdot \frac{1}{|\mathbf{b}|} \mathbf{b}.$$

If $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, then by Definition (14.28)

$$\text{comp}_{\mathbf{i}} \mathbf{a} = \mathbf{a} \cdot \mathbf{i} = a_1, \quad \text{comp}_{\mathbf{j}} \mathbf{a} = \mathbf{a} \cdot \mathbf{j} = a_2, \quad \text{and} \quad \text{comp}_{\mathbf{k}} \mathbf{a} = \mathbf{a} \cdot \mathbf{k} = a_3.$$

Thus, the components of $\mathbf{a}$ along $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the same as the components a_1 , a_2 , and a_3 of $\mathbf{a}$.

Example 5 If $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, find

(a) $\text{comp}_{\mathbf{b}} \mathbf{a}$. (b) $\text{comp}_{\mathbf{a}} \mathbf{b}$.

Solution (a) Using Definition (14.28),

$$\begin{aligned} \text{comp}_{\mathbf{b}} \mathbf{a} &= \mathbf{a} \cdot \frac{1}{|\mathbf{b}|} \mathbf{b} = (4\mathbf{i} - \mathbf{j} + 5\mathbf{k}) \cdot \frac{1}{7} (6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}) \\ &= \frac{24 - 3 - 10}{7} = \frac{11}{7}. \end{aligned}$$

(b) Interchanging the roles of $\mathbf{a}$ and $\mathbf{b}$ in (14.28),

$$\begin{aligned} \text{comp}_{\mathbf{a}} \mathbf{b} &= \mathbf{b} \cdot \frac{1}{|\mathbf{a}|} \mathbf{a} = (6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}) \cdot \frac{1}{\sqrt{42}} (4\mathbf{i} - \mathbf{j} + 5\mathbf{k}) \\ &= \frac{24 - 3 - 10}{\sqrt{42}} = \frac{11}{\sqrt{42}}. \end{aligned}$$

■

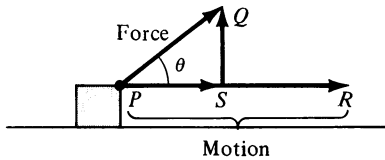


FIGURE 14.23

We shall conclude this section with an important physical interpretation for the dot product. Recall from Definition (6.9) that the work done if a constant force F is exerted through a distance d is given by $W = Fd$. This formula is very restrictive since it can only be used if the force is applied along the line of motion. More generally, suppose a vector $P\vec{Q}$ represents a force, and that its point of application moves along a vector $P\vec{R}$. This is illustrated in Figure 14.23, where a force $P\vec{Q}$ is used to pull an object along a level path from P to R . The vector $P\vec{Q}$ is the sum of the vectors $P\vec{S}$ and $S\vec{Q}$. Since $S\vec{Q}$ does not contribute to the horizontal movement, we may assume that the motion from P to R is caused by $P\vec{S}$ alone. Applying (6.9), the work W is found by multiplying the component of $P\vec{Q}$ in the direction of $P\vec{R}$ by the distance $|P\vec{R}|$, that is,

$$W = (|P\vec{Q}|\cos \theta)|P\vec{R}| = P\vec{Q} \cdot P\vec{R}.$$

This leads to the following definition.

Definition (14.29)

The work done by a constant force $P\vec{Q}$ as its point of application moves along the vector $P\vec{R}$ is $P\vec{Q} \cdot P\vec{R}$.

Example 6 The magnitude and direction of a constant force are given by $\mathbf{a} = 5\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$. Find the work done if the point of application of the force moves from $P(1, -1, 2)$ to $R(4, 3, -1)$.

Solution By Theorem (14.18) the vector in V_3 that corresponds to $P\vec{R}$ is $\mathbf{b} = \langle 3, 4, -3 \rangle$. If $P\vec{Q}$ is a geometric representation for $\mathbf{a}$, then by Definition (14.29) the work done is given by

$$P\vec{Q} \cdot P\vec{R} = \mathbf{a} \cdot \mathbf{b} = 15 + 8 - 18 = 5.$$

If, for example, the unit of length is feet and the magnitude of the force is measured in pounds, then the work done is 5 ft-lb. If the length is in meters and the force is in Newtons, then the work done is 5 joules. ■

14.3 Exercises

1 Complete the proof of Theorem (14.20).

2 Prove Theorems (14.5) and (14.9) for V_3 .

In Exercises 3–8 find $\mathbf{a} + \mathbf{b}$, $5\mathbf{a} - 4\mathbf{b}$, $\mathbf{a} \cdot \mathbf{b}$, $|\mathbf{a}|$, $|3\mathbf{a}|$, $|-3\mathbf{a}|$, $\mathbf{a} - \mathbf{b}$, and $|\mathbf{a} - \mathbf{b}|$.

3 $\mathbf{a} = \langle -2, 6, 1 \rangle$, $\mathbf{b} = \langle 3, -3, -1 \rangle$

4 $\mathbf{a} = \langle 1, 2, -3 \rangle$, $\mathbf{b} = \langle -4, 0, 1 \rangle$

5 $\mathbf{a} = 3\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = \mathbf{i} + 2\mathbf{j} - 5\mathbf{k}$

6 $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$, $\mathbf{b} = \mathbf{i} - \mathbf{k}$

7 $\mathbf{a} = \mathbf{i} + \mathbf{j}$, $\mathbf{b} = -\mathbf{j} + \mathbf{k}$

8 $\mathbf{a} = 2\mathbf{i}$, $\mathbf{b} = 3\mathbf{k}$

In Exercises 9 and 10 sketch geometric representations for $\mathbf{a}$, $\mathbf{b}$, $2\mathbf{a}$, $-3\mathbf{b}$, $\mathbf{a} + \mathbf{b}$, and $\mathbf{a} - \mathbf{b}$.

9 $\mathbf{a} = \langle 2, 3, 4 \rangle$, $\mathbf{b} = \langle 1, -2, 2 \rangle$

10 $\mathbf{a} = -\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, $\mathbf{b} = -2\mathbf{j} + \mathbf{k}$

57 The **direction angles** of a nonzero vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ are defined as the angles α , β , and γ between the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, respectively, and the vector $\mathbf{a}$. The **direction cosines** of $\mathbf{a}$ are $\cos \alpha$, $\cos \beta$, and $\cos \gamma$. Prove that

$$(a) \cos \alpha = \frac{a_1}{|\mathbf{a}|}, \cos \beta = \frac{a_2}{|\mathbf{a}|}, \cos \gamma = \frac{a_3}{|\mathbf{a}|}.$$

$$(b) \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1.$$

58 Refer to Exercise 57.

- Find the direction cosines of $\mathbf{a} = \langle -2, 1, 5 \rangle$.
- Find the direction angles and direction cosines of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.
- Find two unit vectors that have the same direction cosines.

59 Three nonzero numbers l , m , and n are called **direction numbers** of a nonzero vector $\mathbf{a}$ if they are proportional to the direction cosines, that is, if there exists a positive number k such that

$$l = k \cos \alpha, \quad m = k \cos \beta, \quad n = k \cos \gamma.$$

If l, m, n are direction numbers of $\mathbf{a}$ and $d = (l^2 + m^2 + n^2)^{1/2}$, prove that $\cos \alpha = l/d$, $\cos \beta = m/d$, $\cos \gamma = n/d$.

60 Refer to Exercise 59. If l_1, m_1, n_1 and l_2, m_2, n_2 are direction numbers of $\mathbf{a}$ and $\mathbf{b}$, respectively, prove the following.

(a) $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if and only if

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0.$$

(b) $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if there is a number k such that $l_1 = kl_2$, $m_1 = km_2$ and $n_1 = kn_2$.

14.4 The Vector Product

In this section we introduce the *vector product* (or *cross product*) $\mathbf{a} \times \mathbf{b}$ of two vectors $\mathbf{a}$ and $\mathbf{b}$. Unlike the dot product, which is a scalar, this new operation produces another vector. The vector product was first used as a tool for physical problems involving moments of forces. It is possible to define $\mathbf{a} \times \mathbf{b}$ geometrically and then obtain an algebraic form by introducing a rectangular coordinate system. We shall reverse this process and begin with an algebraic definition. This approach disguises the geometric nature of $\mathbf{a} \times \mathbf{b}$; however, it leads to simpler proofs of properties.

It is convenient to use *determinants* when working with vector products. A **determinant of order 2** is defined by

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1$$

where all letters represent real numbers. For example,

$$\begin{vmatrix} 2 & -3 \\ 4 & 5 \end{vmatrix} = (2)(5) - (-3)(4) = 10 + 12 = 22.$$

A **determinant of order 3** is given by

$$\begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} c_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} c_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} c_3.$$

This is sometimes called the *expansion of the determinant by the first row*. The numerical value can be found by evaluating the second-order determinants on the right side of the equation.

Example 1 Find the value of $\begin{vmatrix} 2 & -1 & 3 \\ -2 & 5 & 1 \\ 1 & 2 & -4 \end{vmatrix}$.

Solution By definition,

$$\begin{aligned} \begin{vmatrix} 2 & -1 & 3 \\ -2 & 5 & 1 \\ 1 & 2 & -4 \end{vmatrix} &= \begin{vmatrix} 5 & 1 \\ 2 & -4 \end{vmatrix}(2) - \begin{vmatrix} -2 & 1 \\ 1 & -4 \end{vmatrix}(-1) + \begin{vmatrix} -2 & 5 \\ 1 & 2 \end{vmatrix}(3) \\ &= (-20 - 2)(2) - (8 - 1)(-1) + (-4 - 5)(3) \\ &= -44 + 7 - 27 = -64. \end{aligned}$$

Definition (14.30)

The **vector product** (or **cross product**) $\mathbf{a} \times \mathbf{b}$ of $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ is

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}.$$

The symbol $\mathbf{a} \times \mathbf{b}$ is read “**a cross b**”. Note that the formula for $\mathbf{a} \times \mathbf{b}$ can be obtained by replacing c_1, c_2, c_3 in our definition of a determinant of order 3 by the unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$. This suggests the following notation for the formula in Definition (14.30):

$$(14.31) \quad \mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The symbol on the right side of this equation is not a determinant, since the first row contains vectors instead of scalars. Consequently, properties of determinants may not be applied to (14.31). The determinant notation is used as a mnemonic device for remembering the more cumbersome formula (14.30). With this warning in mind we shall use (14.31) to find vector products, as in the following example.

Example 2 Find $\mathbf{a} \times \mathbf{b}$ if $\mathbf{a} = \langle 2, -1, 6 \rangle$ and $\mathbf{b} = \langle -3, 5, 1 \rangle$.

Solution Writing

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 6 \\ -3 & 5 & 1 \end{vmatrix}$$

we obtain

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} -1 & 6 \\ 5 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & 6 \\ -3 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & -1 \\ -3 & 5 \end{vmatrix} \mathbf{k} \\ &= (-1 - 30)\mathbf{i} - (2 + 18)\mathbf{j} + (10 - 3)\mathbf{k} \\ &= -31\mathbf{i} - 20\mathbf{j} + 7\mathbf{k}. \end{aligned}$$

If $\mathbf{a}$ is any vector in V_3 , then

$$\mathbf{a} \times \mathbf{0} = \mathbf{0} = \mathbf{0} \times \mathbf{a}$$

for if one of the vectors in Definition (14.30) is $\mathbf{0}$, then each determinant has a row of zeros. Also, $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ for every $\mathbf{a}$, since in this case each determinant in (14.30) has equal rows.

The next theorem brings out an important property of vector products.

Theorem (14.32)

The vector $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$.

Proof By Theorem (14.24) it is sufficient to show that

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0 \quad \text{and} \quad (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$$

If we apply the definition of dot product to (14.30) and if $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, then

$$\begin{aligned} (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} a_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} a_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} a_3 \\ &= (a_2 b_3 - a_3 b_2) a_1 - (a_1 b_3 - a_3 b_1) a_2 + (a_1 b_2 - a_2 b_1) a_3 \\ &= a_2 b_3 a_1 - a_3 b_2 a_1 - a_1 b_3 a_2 + a_3 b_1 a_2 + a_1 b_2 a_3 - a_2 b_1 a_3 \\ &= 0. \end{aligned}$$

Hence $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{a}$. The verification that $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$ is left to the reader. $\square$

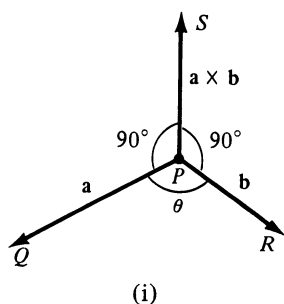


FIGURE 14.24

In geometric terms, Theorem (14.32) implies that if nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are represented by vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ with the same initial point P , then $\mathbf{a} \times \mathbf{b}$ may be represented by a vector $\overrightarrow{PS}$ which is perpendicular to the plane determined by P , Q , and R , as illustrated in (i) of Figure 14.24. We shall write

$$\overrightarrow{PS} = \overrightarrow{PQ} \times \overrightarrow{PR}.$$

It can be shown that the direction of $\overrightarrow{PS}$ may be obtained using the right-hand rule illustrated in (ii) of Figure 14.24. Specifically, if θ denotes the angle between $\overrightarrow{PQ}$ and $\overrightarrow{PR}$, and if the fingers of the right hand are curled such that a rotation through θ will transform $\overrightarrow{PQ}$ into a vector having the same direction as $\overrightarrow{PR}$, then the extended thumb points in the direction of $\overrightarrow{PQ} \times \overrightarrow{PR}$.

The following result provides information about the magnitude of $\mathbf{a} \times \mathbf{b}$.

Theorem (14.33)

If θ is the angle between nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, then

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta.$$

Proof Applying Definitions (14.2) and (14.30),

$$\begin{aligned}
 |\mathbf{a} \times \mathbf{b}|^2 &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}^2 + \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}^2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}^2 \\
 &= (a_2b_3 - a_3b_2)^2 + (a_1b_3 - a_3b_1)^2 + (a_1b_2 - a_2b_1)^2 \\
 &= a_2^2b_3^2 - 2a_2a_3b_2b_3 + a_3^2b_2^2 + a_1^2b_3^2 - 2a_1a_3b_1b_3 \\
 &\quad + a_3^2b_1^2 + a_1^2b_2^2 - 2a_1a_2b_1b_2 + a_2^2b_1^2 \\
 &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2.
 \end{aligned}$$

The last equality may be verified by multiplying the indicated expressions. The vector form of this identity is

$$|\mathbf{a} \times \mathbf{b}|^2 = (|\mathbf{a}||\mathbf{b}|)^2 - (\mathbf{a} \cdot \mathbf{b})^2$$

or, since $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$,

$$\begin{aligned}
 |\mathbf{a} \times \mathbf{b}|^2 &= (|\mathbf{a}||\mathbf{b}|)^2 - (|\mathbf{a}||\mathbf{b}|)^2 \cos^2 \theta \\
 &= (|\mathbf{a}||\mathbf{b}|)^2(1 - \cos^2 \theta) = (|\mathbf{a}||\mathbf{b}|)^2 \sin^2 \theta.
 \end{aligned}$$

Finally, taking square roots, we obtain

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta. \quad \square$$

Corollary (14.34)

Two vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

Proof Suppose $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors. If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then the vectors are parallel if and only if $\theta = 0$ or $\theta = \pi$, or equivalently $\sin \theta = 0$. By Theorem (14.33) the last statement is equivalent to $\mathbf{a} \times \mathbf{b} = \mathbf{0}$. If either $\mathbf{a}$ or $\mathbf{b}$ is the zero vector the proof is trivial. $\square$

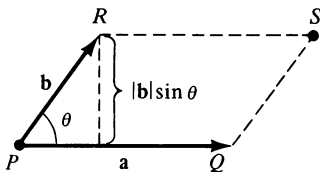


FIGURE 14.25

To interpret $|\mathbf{a} \times \mathbf{b}|$ geometrically let us represent $\mathbf{a}$ and $\mathbf{b}$ by vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ having the same initial point P . Let S be the point such that segments PQ and PR are adjacent sides of a parallelogram with vertices P, Q, R , and S , as illustrated in Figure 14.25. An altitude of the parallelogram is $|b| \sin \theta$ and hence its area is $|a| |b| \sin \theta$. Comparing this with Corollary (14.33) we see that the magnitude of the vector product $\mathbf{a} \times \mathbf{b}$ equals the area of the parallelogram determined by $\mathbf{a}$ and $\mathbf{b}$. Physical interpretations of the vector product will be given in the next chapter.

Example 3 Find the area of the triangle determined by $P(4, -3, 1)$, $Q(6, -4, 7)$, and $R(1, 2, 2)$.

Solution Applying Theorem (14.18), the vectors in V_3 corresponding to $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are $\mathbf{a} = \langle 2, -1, 6 \rangle$ and $\mathbf{b} = \langle -3, 5, 1 \rangle$, respectively. From

Example 2, $\mathbf{a} \times \mathbf{b} = -31\mathbf{i} - 20\mathbf{j} + 7\mathbf{k}$. Hence the area of the parallelogram with adjacent sides PQ and PR is

$$|\mathbf{a} \times \mathbf{b}| = \sqrt{961 + 400 + 49} = \sqrt{1410}.$$

It follows that the area of the triangle is $\frac{1}{2}\sqrt{1410}$. ■

The vector products of the special unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are of interest. For example, using Definition (14.30) with $\mathbf{a} = \mathbf{i} = \langle 1, 0, 0 \rangle$ and $\mathbf{b} = \mathbf{j} = \langle 0, 1, 0 \rangle$,

$$\mathbf{i} \times \mathbf{j} = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \mathbf{k} = \mathbf{k}.$$

In general, each of the following is true:

$$(14.35) \quad \begin{aligned} \mathbf{i} \times \mathbf{j} &= \mathbf{k}, & \mathbf{j} \times \mathbf{k} &= \mathbf{i}, & \mathbf{k} \times \mathbf{i} &= \mathbf{j} \\ \mathbf{j} \times \mathbf{i} &= -\mathbf{k}, & \mathbf{k} \times \mathbf{j} &= -\mathbf{i}, & \mathbf{i} \times \mathbf{k} &= -\mathbf{j} \\ \mathbf{i} \times \mathbf{i} &= \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0}. \end{aligned}$$

The fact that $\mathbf{i} \times \mathbf{j} \neq \mathbf{j} \times \mathbf{i}$ shows that the vector product is not commutative. The associative law does not hold either since, for example,

$$\mathbf{i} \times (\mathbf{j} \times \mathbf{j}) = \mathbf{i} \times \mathbf{0} = \mathbf{0},$$

$$\text{whereas} \quad (\mathbf{i} \times \mathbf{j}) \times \mathbf{j} = \mathbf{k} \times \mathbf{j} = -\mathbf{i}.$$

The following theorem lists some important properties of the vector product, where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are any vectors, and m is a scalar.

Theorem (14.36)

- (i) $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
- (ii) $(m\mathbf{a}) \times \mathbf{b} = m(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (m\mathbf{b})$
- (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$
- (iv) $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{c})$
- (v) $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$
- (vi) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

Proof All of the properties may be established by straightforward (but sometimes lengthy) applications of Definition (14.30). For example, if $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, then

$$\mathbf{b} \times \mathbf{a} = \begin{vmatrix} b_2 & b_3 \\ a_2 & a_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ a_1 & a_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ a_1 & a_2 \end{vmatrix} \mathbf{k}.$$

Since interchanging two rows of a determinant changes its sign, we have

$$\begin{aligned}\mathbf{b} \times \mathbf{a} &= - \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} + \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} - \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \\ &= -\mathbf{a} \times \mathbf{b}.\end{aligned}$$

This proves property (i).

If $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$, then the $\mathbf{i}$ component of $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$ is

$$\begin{aligned}\begin{vmatrix} a_2 & a_3 \\ b_2 + c_2 & b_3 + c_3 \end{vmatrix} &= a_2(b_3 + c_3) - a_3(b_2 + c_2) \\ &= (a_2 b_3 - a_3 b_2) + (a_2 c_3 - a_3 c_2) \\ &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} + \begin{vmatrix} a_2 & a_3 \\ c_2 & c_3 \end{vmatrix}\end{aligned}$$

which is equal to the $\mathbf{i}$ component of $(\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$. A similar calculation can be used to prove that the $\mathbf{j}$ and $\mathbf{k}$ components of $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$ are the same as those of $(\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$. This establishes (iii). The proofs of the remaining properties are left to the reader. $\square$

We shall conclude this section with two geometric applications of the vector product.

Example 4 Find a formula for the distance d from a point R to a line l .

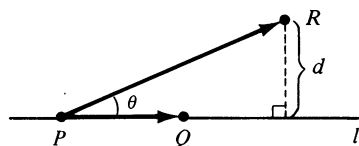


FIGURE 14.26

Solution As illustrated in Figure 14.26, let P and Q be points on l , and let θ be the angle between $\overrightarrow{PQ}$ and $\overrightarrow{PR}$. Using the fact that $d = |\overrightarrow{PR}| \sin \theta$ and $|\overrightarrow{PQ} \times \overrightarrow{PR}| = |\overrightarrow{PQ}| |\overrightarrow{PR}| \sin \theta$ gives us

$$d = \frac{|\overrightarrow{PQ} \times \overrightarrow{PR}|}{|\overrightarrow{PQ}|}.$$

Example 5 Suppose $\overrightarrow{PQ}$, $\overrightarrow{PR}$, and $\overrightarrow{PS}$ represent adjacent sides of a rectangular parallelepiped. If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are the corresponding vectors in V_3 , show that $|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|$ is the volume of the parallelepiped.

Solution One illustration of the parallelepiped is given in Figure 14.27.

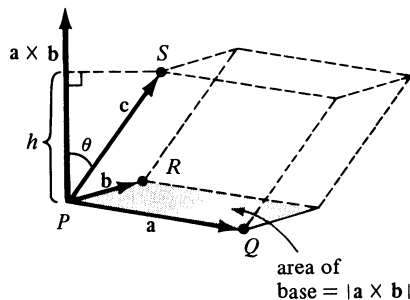


FIGURE 14.27

The area of the base is $|\mathbf{a} \times \mathbf{b}|$. Let θ be the angle between $\mathbf{c}$ and $\mathbf{a} \times \mathbf{b}$. Since the vector corresponding to $\mathbf{a} \times \mathbf{b}$ is perpendicular to the base, it follows that the altitude h of the parallelepiped is given by $h = |\mathbf{c}| |\cos \theta|$. It is necessary to use the absolute value $|\cos \theta|$ since θ may be an obtuse angle. Hence the volume V of the parallelepiped is given by

$$\begin{aligned} V &= (\text{area of base}) (\text{altitude}) \\ &= |\mathbf{a} \times \mathbf{b}| |\mathbf{c}| |\cos \theta| \\ &= |(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|. \end{aligned}$$

14.4 Exercises

In Exercises 1–10 find $\mathbf{a} \times \mathbf{b}$.

- 1 $\mathbf{a} = \langle 1, -2, 3 \rangle$, $\mathbf{b} = \langle 2, 1, -4 \rangle$
- 2 $\mathbf{a} = \langle -5, 1, -1 \rangle$, $\mathbf{b} = \langle 3, 6, -2 \rangle$
- 3 $\mathbf{a} = 5\mathbf{i} - 6\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$
- 4 $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{b} = -5\mathbf{j} + 2\mathbf{k}$
- 5 $\mathbf{a} = \langle 0, 1, 2 \rangle$, $\mathbf{b} = \langle 1, 2, 0 \rangle$
- 6 $\mathbf{a} = -3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = 9\mathbf{i} - 3\mathbf{j} - 6\mathbf{k}$
- 7 $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 8\mathbf{k}$, $\mathbf{b} = 5\mathbf{j}$
- 8 $\mathbf{a} = \langle 0, 0, 4 \rangle$, $\mathbf{b} = \langle -7, 1, 0 \rangle$
- 9 $\mathbf{a} = 4\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$
- 10 $\mathbf{a} = 3\mathbf{i}$, $\mathbf{b} = 4\mathbf{k}$
- 11 If $\mathbf{a} = \langle 2, 0, -1 \rangle$, $\mathbf{b} = \langle -3, 1, 0 \rangle$, and $\mathbf{c} = \langle 1, -2, 4 \rangle$, find $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ and $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$.
- 12 If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are the vectors of Exercise 11, find $\mathbf{a} \times (\mathbf{b} - \mathbf{c})$ and $(\mathbf{a} \times \mathbf{b}) - (\mathbf{a} \times \mathbf{c})$.
- 13 Prove that $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0$ for all vectors $\mathbf{a}$ and $\mathbf{b}$.
- 14 Prove (14.35).
- 15 Complete the proof of Theorem (14.36).
- 16 (a) If $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$ and $\mathbf{a} \neq \mathbf{0}$, does it follow that $\mathbf{b} = \mathbf{c}$? Explain.
(b) Let $\mathbf{a} \neq \mathbf{0}$. If $\mathbf{a} \times \mathbf{b} = \mathbf{a} \times \mathbf{c}$ and $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$, prove that $\mathbf{b} = \mathbf{c}$.

In Exercises 17–20 find (a) a vector perpendicular to the plane determined by the points P , Q , and R ; (b) the area of the triangle determined by P , Q , and R .

- 17 $P(1, -1, 2)$, $Q(0, 3, -1)$, $R(3, -4, 1)$
- 18 $P(-3, 0, 5)$, $Q(2, -1, -3)$, $R(4, 1, -1)$

$$19 \quad P(4, 0, 0), Q(0, 5, 0), R(0, 0, 2)$$

$$20 \quad P(-1, 2, 0), Q(0, 2, -3), R(5, 0, 1)$$

- 21 If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$, prove that

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

(This number is called the **triple scalar product** of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.)

- 22 If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are represented by vectors with a common initial point, show that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$ if and only if the vectors are coplanar.
- 23 Given $P(1, -1, 2)$, $Q(0, 3, -1)$ and $R(3, -4, 1)$, use Exercise 21 to find the volume of the parallelepiped having adjacent sides OP , OQ , and OR .
- 24 Given $A(2, 1, -1)$, $B(3, 0, 2)$, $C(4, -2, 1)$, and $D(5, -3, 0)$, use Exercise 21 to find the volume of the parallelepiped having adjacent sides AB , AC , and AD .

Without using components, verify the identities in Exercises 25–30, where $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ are arbitrary vectors.

- 25 $(\mathbf{a} + \mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = 2\mathbf{b} \times \mathbf{a}$
- 26 $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$
(Hint: Use (vi) of Theorem (14.36).)
- 27 $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$
- 28 $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = \begin{vmatrix} \mathbf{a} \cdot \mathbf{c} & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{a} \cdot \mathbf{d} & \mathbf{b} \cdot \mathbf{d} \end{vmatrix}$
- 29 $(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \times \mathbf{b} \cdot \mathbf{d})\mathbf{c} - (\mathbf{a} \times \mathbf{b} \cdot \mathbf{c})\mathbf{d}$
- 30 $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) = (\mathbf{a} \cdot \mathbf{b} \times \mathbf{c})^2$

14.5 Lines in Space

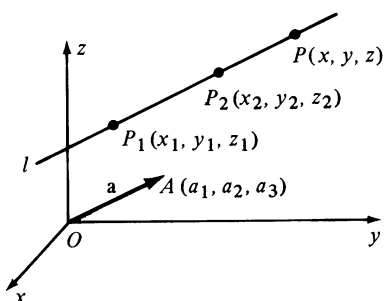


FIGURE 14.28

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are distinct points, then by Theorem (14.18), the vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ in V_3 with geometric representation $\overrightarrow{P_1P_2}$ is given by $\mathbf{a} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$. To find equations for the line l through P_1 and P_2 we observe that a point $P(x, y, z)$ is on l if and only if $\overrightarrow{P_1P}$ and $\overrightarrow{P_1P_2}$ are parallel. This is illustrated in Figure 14.28, where the position vector $\overrightarrow{OA}$ corresponding to $\mathbf{a}$ is also shown.

We know from our work in Section 14.3 that $\overrightarrow{P_1P}$ and $\overrightarrow{P_1P_2}$ are parallel if and only if

$$\overrightarrow{P_1P} = t\overrightarrow{P_1P_2} \quad \text{for some scalar } t.$$

This condition may be stated algebraically as follows:

$$\langle x - x_1, y - y_1, z - z_1 \rangle = t\langle a_1, a_2, a_3 \rangle = \langle a_1t, a_2t, a_3t \rangle.$$

Equating components and solving for x , y , and z gives us

$$x = x_1 + a_1t, \quad y = y_1 + a_2t, \quad z = z_1 + a_3t$$

where t is a real number. These equations are called **parametric equations for the line l** , and the variable t is called a **parameter**. The points $P(x, y, z)$ on l are obtained by letting t take on all real values. For example, P_1 corresponds to $t = 0$, the midpoint of P_1P_2 to $t = \frac{1}{2}$, and P_2 to $t = 1$.

We may summarize the preceding discussion as follows.

Theorem (14.37)

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are distinct points, and if $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ is the vector corresponding to $\overrightarrow{P_1P_2}$, then parametric equations for the line through P_1 and P_2 are

$$x = x_1 + a_1t, \quad y = y_1 + a_2t, \quad z = z_1 + a_3t.$$

Example 1 Find parametric equations for the line l through $P_1(3, 1, -2)$ and $P_2(-2, 7, -4)$. At what point does l intersect the xy -plane?

Solution The vector $\mathbf{a}$ in V_3 corresponding to $\overrightarrow{P_1P_2}$ is

$$\mathbf{a} = \langle -2 - 3, 7 - 1, -4 + 2 \rangle = \langle -5, 6, -2 \rangle.$$

Applying Theorem (14.37), parametric equations for l are

$$x = 3 - 5t, \quad y = 1 + 6t, \quad z = -2 - 2t.$$

The line intersects the xy -plane at the point $R(x, y, z)$ if $z = -2 - 2t = 0$, that is, if $t = -1$. Thus R is the point with coordinates $(8, -5, 0)$. ■

If, in Figure 14.28, we let

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad \text{and} \quad \mathbf{r}_1 = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$$

then $\overrightarrow{OP}$ and $\overrightarrow{OP_1}$ are position vectors of $\mathbf{r}$ and $\mathbf{r}_1$, respectively, and $\overrightarrow{P_1P}$ represents the vector $\mathbf{r} - \mathbf{r}_1$. Thus we may write $\mathbf{r} - \mathbf{r}_1 = t\mathbf{a}$. This leads to the following vector equation for the line in Theorem (14.37):

$$\mathbf{r} = \mathbf{r}_1 + t\mathbf{a}.$$

If $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ is a nonzero vector in V_3 and $P_1(x_1, y_1, z_1)$ is any point, then there is a unique line l through P_1 parallel to the position vector $\overrightarrow{OA}$ corresponding to $\mathbf{a}$. We shall refer to l as **the line through P_1 parallel to $\mathbf{a}$** . Parametric equations for l are given in Theorem (14.37). If $\mathbf{b}$ is any nonzero vector that is parallel to $\mathbf{a}$, then the same line is determined, for in this case $\mathbf{b} = c\mathbf{a} = \langle ca_1, ca_2, ca_3 \rangle$ and parametric equations for the line through P_1 determined by $\mathbf{b}$ are

$$x = x_1 + (ca_1)v, \quad y = y_1 + (ca_2)v, \quad z = z_1 + (ca_3)v$$

where the parameter v ranges through all real numbers. These equations determine the same line, since the point given by t can be obtained by letting $v = t/c$.

Example 2 Find parametric equations for the line l through $P(5, -6, 2)$ parallel to $\mathbf{a} = \langle \frac{1}{2}, 2, -\frac{4}{3} \rangle$.

Solution To avoid fractions we shall use the vector $\mathbf{b} = 6\mathbf{a} = \langle 3, 12, -8 \rangle$ instead of $\mathbf{a}$. Applying Theorem (14.37), parametric equations for l are

$$x = 5 + 3t, \quad y = -6 + 12t, \quad z = 2 - 8t. \quad \blacksquare$$

If l_1 and l_2 are lines parallel to vectors $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, respectively, then the angle between l_1 and l_2 is defined as the angle θ between $\mathbf{a}$ and $\mathbf{b}$, whether or not the lines intersect. The lines are **orthogonal** if $\mathbf{a} \cdot \mathbf{b} = 0$ or, equivalently, if $a_1b_1 + a_2b_2 + a_3b_3 = 0$. The lines are **parallel** if $\mathbf{b} = c\mathbf{a}$ for some scalar c , that is, if $b_1 = ca_1, b_2 = ca_2, b_3 = ca_3$.

In the next section we shall discuss representations of lines as intersections of planes.

14.5 Exercises

In Exercises 1–4 find parametric equations for the line through P_1 and P_2 . Determine (if possible) the points at which the line intersects each of the coordinate planes.

1 $P_1(5, -2, 4), P_2(2, 6, 1)$

2 $P_1(-3, 1, -1), P_2(7, 11, -8)$

3 $P_1(2, 0, 5), P_2(-6, 0, 3)$

4 $P_1(2, -2, 4), P_2(2, -2, -3)$

In Exercises 5–8 find parametric equations for the line through P parallel to $\mathbf{a}$.

5 $P(4, 2, -3), \mathbf{a} = \langle \frac{1}{3}, 2, \frac{1}{2} \rangle$

- 6 $P(5, 0, -2)$, $\mathbf{a} = \langle -1, -4, 1 \rangle$
- 7 $P(0, 0, 0)$, $\mathbf{a} = \mathbf{j}$
- 8 $P(1, 2, 3)$, $\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$
- 9 If a line l has parametric equations $x = 5 - 3t$, $y = -2 + t$, $z = 1 + 9t$, find parametric equations for a line through $P(-6, 4, -3)$ that is parallel to l .
- 10 If l_1 is the line through $P(5, -2, 4)$ and $Q(2, 6, 1)$, and if l_2 is the line through $R(-3, 1, -1)$ and $S(7, 11, -8)$, find the angles between l_1 and l_2 .
- 11 Find the angles between the two lines having parametric equations $x = 7 - 2t$, $y = 4 + 3t$, $z = 5t$ and $x = -1 + t$, $y = 3 + 4t$, $z = 1 + t$, respectively.
- 12 Find parametric equations for the line through $P(4, -1, 0)$ that is parallel to the line through $P_1(-3, 9, -2)$ and $P_2(5, 7, -3)$.

In Exercises 13–16 determine whether the two lines intersect and, if so, find the point of intersection.

- 13 $x = 1 + 2t$, $y = 1 - 4t$, $z = 5 - t$;
 $x = 4 - v$, $y = -1 + 6v$, $z = 4 + v$
- 14 $x = 1 - 6t$, $y = 3 + 2t$, $z = 1 - 2t$;
 $x = 2 + 2v$, $y = 6 + v$, $z = 2 + v$
- 15 $x = 3 + t$, $y = 2 - 4t$, $z = t$;
 $x = 4 - v$, $y = 3 + v$, $z = -2 + 3v$
- 16 $x = 2 - 5t$, $y = 6 + 2t$, $z = -3 - 2t$;
 $x = 4 - 3v$, $y = 7 + 5v$, $z = 1 + 4v$
- In Exercises 17 and 18 show that the two lines intersect orthogonally.
- 17 $x = 2 + 3t$, $y = -4 - 2t$, $z = -1 + 4t$;
 $x = 6 + 4v$, $y = -2 + 2v$, $z = -3 - 2v$
- 18 $x = 4 - t$, $y = -1 - t$, $z = 4 + 3t$;
 $x = 3 + 2v$, $y = -1 + v$, $z = v$
- 19 Use the dot product to find the distance d from the point $A(2, -6, 1)$ to the line l through $B(3, 4, -2)$ and $C(7, -1, 5)$.
- 20 Work Exercise 19 for the points $A(1, 5, 0)$, $B(-2, 1, -4)$, and $C(0, -3, 2)$.
- 21 Find the distance from the point $C(2, 1, -2)$ to the line having parametric equations $x = 3 - 2t$, $y = -4 + 3t$, $z = 1 + 2t$.
- 22 A line l has parametric equations $x = 2t + 1$, $y = -t + 3$, $z = 5t$. Find the distance from $A(3, 1, -1)$ to l .
- 23 Find parametric equations for the line through $P(3, 1, -2)$ that is orthogonal to, and intersects, the line $x = -1 + t$, $y = -2 + t$, $z = -1 + t$.
- 24 Find the point on the line through $A(2, 1, -2)$ and $B(1, -3, 2)$ that is equidistant from $C(0, 1, 1)$ and $D(1, 2, 3)$.

14.6 Planes

If P_1 and P_2 are distinct points, then all points P such that $\overrightarrow{P_1P}$ is orthogonal to $\overrightarrow{P_1P_2}$ lie on a plane Γ through P_1 , as illustrated in Figure 14.29.

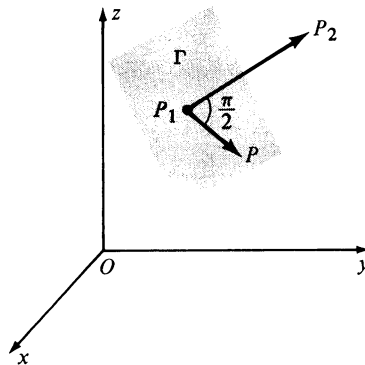


FIGURE 14.29

We shall call $\overrightarrow{P_1P_2}$, or the vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ with geometric representation $\overrightarrow{P_1P_2}$, a **normal vector** to Γ . It follows from Theorem (14.24) that P is on Γ if and only if $\overrightarrow{P_1P_2} \cdot \overrightarrow{P_1P} = 0$ or, equivalently,

$$\langle a_1, a_2, a_3 \rangle \cdot \langle x - x_1, y - y_1, z - z_1 \rangle = 0.$$

Applying the definition of dot product gives us the next result.

Theorem (14.38)

An equation of the plane through $P_1(x_1, y_1, z_1)$ with normal vector $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ is

$$a_1(x - x_1) + a_2(y - y_1) + a_3(z - z_1) = 0.$$

If we let

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad \text{and} \quad \mathbf{r}_1 = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$$

then $\overrightarrow{OP}$ and $\overrightarrow{OP_1}$ are position vectors of $\mathbf{r}$ and $\mathbf{r}_1$, respectively, and $\overrightarrow{P_1P}$ represents the vector $\mathbf{r} - \mathbf{r}_1$. This leads to the following **vector equation for a plane**:

$$\mathbf{a} \cdot (\mathbf{r} - \mathbf{r}_1) = 0.$$

Example 1 Find an equation of the plane through the point $(5, -2, 4)$ with normal vector $\mathbf{a} = \langle 1, 2, 3 \rangle$.

Solution Applying Theorem (14.38), we obtain

$$1(x - 5) + 2(y + 2) + 3(z - 4) = 0$$

which reduces to

$$x + 2y + 3z - 13 = 0. \quad \blacksquare$$

Example 2 Find an equation of the plane determined by the points $P(4, -3, 1)$, $Q(6, -4, 7)$, and $R(1, 2, 2)$.

Solution Vectors $\mathbf{a}$ and $\mathbf{b}$ corresponding to $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ are

$$\mathbf{a} = \langle 2, -1, 6 \rangle \quad \text{and} \quad \mathbf{b} = \langle -3, 5, 1 \rangle.$$

As illustrated in Figure 14.24, the vector $\mathbf{a} \times \mathbf{b}$ is normal to the plane determined by P , Q , and R . From Example 2 of Section 14.4,

$$\mathbf{a} \times \mathbf{b} = -31\mathbf{i} - 20\mathbf{j} + 7\mathbf{k}.$$

Using Theorem (14.38) with $P_1 = P$ gives us the equation

$$-31(x - 4) - 20(y + 3) + 7(z - 1) = 0,$$

or

$$-31x - 20y + 7z + 57 = 0. \quad \blacksquare$$

The equation of the plane in Theorem (14.38) may be written in the form

$$ax + by + cz + d = 0$$

where $a = a_1$, $b = b_1$, $c = c_1$, and $d = -a_1x_1 - a_2y_1 - a_3z_1$. Conversely, given $ax + by + cz + d = 0$, where a, b , and c are not all zero, we may choose numbers x_1, y_1 , and z_1 such that $ax_1 + by_1 + cz_1 + d = 0$. Consequently, $d = -ax_1 - by_1 - cz_1$, and hence

$$ax + by + cz - ax_1 - by_1 - cz_1 = 0,$$

or
$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0.$$

According to Theorem (14.38), the graph of the last equation is a plane through $P(x_1, y_1, z_1)$ with normal vector $\langle a, b, c \rangle$. An equation of the form $ax + by + cz + d = 0$, where a, b , and c are not all zero, is called a **linear equation in three variables** x, y , and z . We have proved the following theorem.

Theorem (14.39)

The graph of every linear equation $ax + by + cz + d = 0$ is a plane with normal vector $\langle a, b, c \rangle$.

For simplicity we often use the phrase “the plane $ax + by + cz + d = 0$ ” instead of the more accurate statement “the plane that has equation $ax + by + cz + d = 0$.”

To sketch the graph of a linear equation we often find, if possible, the **trace** of the graph in each coordinate plane, that is, the line in which the graph intersects the coordinate plane. To find the trace in the xy -plane we substitute 0 for z , since this will lead to all points of the graph that lie on the xy -plane. Similarly, to find the trace in the yz -plane or the xz -plane we let $x = 0$ or $y = 0$, respectively, in the equation $ax + by + cz + d = 0$.

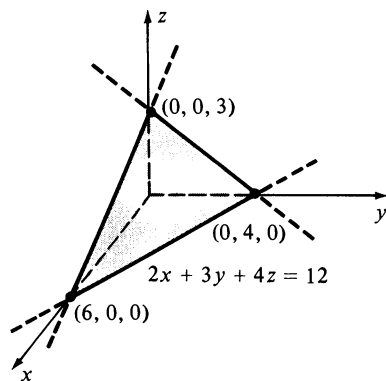


FIGURE 14.30

Example 3 Sketch the graph of the equation $2x + 3y + 4z = 12$.

Solution There are three points on the plane that are easily found, namely the points of intersection of the plane with the coordinate axes. Substituting 0 for both y and z in the equation, we obtain $2x = 12$, or $x = 6$. Thus the point $(6, 0, 0)$ is on the graph. As in two dimensions, 6 is called the *x-intercept* of the graph. Similarly, substitution of 0 for x and z gives us the *y-intercept* 4, and hence the point $(0, 4, 0)$ is on the graph. The point $(0, 0, 3)$ (or *z-intercept* 3) is obtained in like manner. The trace in the xy -plane is found by substituting 0 for z in the given equation. This leads to $2x + 3y = 12$, which has as its graph in the xy -plane a straight line with x -intercept 6 and y -intercept 4. This trace and the traces of the graph in the xz - and yz -planes are illustrated in Figure 14.30. ■

Definition (14.40)

Two planes with normal vectors $\mathbf{a}$ and $\mathbf{b}$ are

- (i) **parallel** if $\mathbf{a}$ and $\mathbf{b}$ are parallel.
- (ii) **orthogonal** if $\mathbf{a}$ and $\mathbf{b}$ are orthogonal.

Example 4 Prove that the graphs of the equations $2x - 3y - z - 5 = 0$ and $-6x + 9y + 3z + 2 = 0$ are parallel planes.

Solution By Theorem (14.39), the graphs are planes with normal vectors $\mathbf{a} = \langle 2, -3, -1 \rangle$ and $\mathbf{b} = \langle -6, 9, 3 \rangle$. Since $\mathbf{b} = -3\mathbf{a}$, the vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel and hence, by Definition (14.40), so are the planes. ■

Example 5 Find an equation of the plane Γ through $P(5, -2, 4)$ that is parallel to the plane $3x + y - 6z + 8 = 0$.

Solution The vector $\mathbf{a} = \langle 3, 1, -6 \rangle$ may be used as a normal vector for Γ and hence the desired equation may be written $3x + y - 6z + d = 0$ for some number d . If $P(5, -2, 4)$ is on Γ , then its coordinates must satisfy this equation, that is, $3(5) + (-2) - 6(4) + d = 0$ or $d = 11$. Hence an equation for Γ is $3x + y - 6z + 11 = 0$. ■

The vector $\mathbf{i} = \langle 1, 0, 0 \rangle$ is a normal vector for the yz -plane. A plane that has an equation of the form $x - x_1 = 0$ (or $x = x_1$) also has normal vector $\mathbf{i}$ and hence is parallel to the yz -plane (and orthogonal to both the xy - and xz -planes). A portion of the graph of $x = a$ is sketched in (i) of Figure 14.31. Similarly, the graph of $y = b$ is a plane parallel to the xz -plane with y -intercept b , whereas the graph of $z = c$ is a plane parallel to the xy -plane with z -intercept c (see (ii) and (iii) of Figure 14.31).

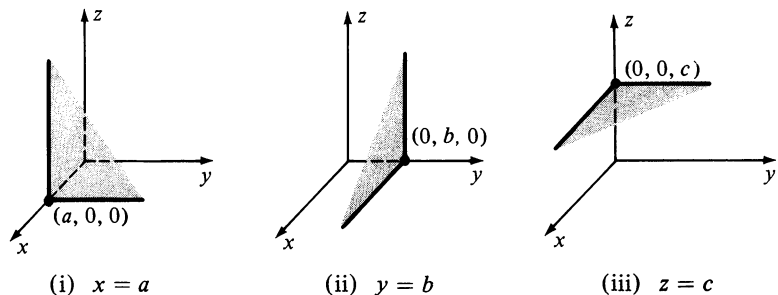


FIGURE 14.31

A plane with an equation of the form $by + cz + d = 0$ has normal vector $\mathbf{a} = \langle 0, b, c \rangle$ and is orthogonal to the yz -plane since $\mathbf{a} \cdot \mathbf{i} = 0$. Similarly, graphs of $ax + by + d = 0$ and $ax + cz + d = 0$ are planes that are orthogonal to the xy -plane and xz -plane, respectively.

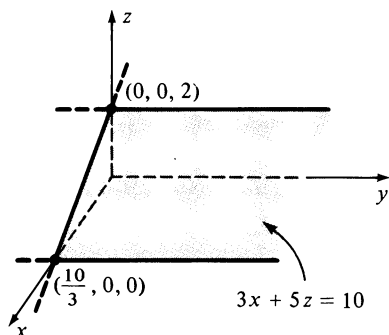


FIGURE 14.32

Example 6 Sketch the graph of the equation $3x + 5z = 10$.

Solution The graph is a plane orthogonal to the xz -plane with x -intercept $10/3$ and z -intercept 2 . Note that the trace in the yz -plane has equation $5z = 10$ and hence is a line parallel to the y -axis with z -intercept 2 . Similarly, the trace in the xy -plane has equation $3x = 10$ and is a line parallel to the y -axis with x -intercept $10/3$. A portion of the graph showing traces in the three coordinate planes is sketched in Figure 14.32. ■

Lines may be described as intersections of planes. If a line l is given parametrically as in Theorem (14.37), and if a_1, a_2, a_3 are different from zero, we may solve each equation for t , obtaining

$$t = \frac{x - x_1}{a_1}, \quad t = \frac{y - y_1}{a_2}, \quad t = \frac{z - z_1}{a_3}.$$

It follows that a point $P(x, y, z)$ is on l if and only if

$$(14.41) \quad \frac{x - x_1}{a_1} = \frac{y - y_1}{a_2} = \frac{z - z_1}{a_3}.$$

The equations in (14.41) are called a **symmetric form** for the line l . A symmetric form is not unique, since in (14.41) we may use any three numbers b_1, b_2, b_3 that are proportional to a_1, a_2, a_3 , or any point on l other than (x_1, y_1, z_1) .

If, in (14.41), we take the indicated expressions in pairs, say

$$\frac{x - x_1}{a_1} = \frac{y - y_1}{a_2} \quad \text{and} \quad \frac{x - x_1}{a_1} = \frac{z - z_1}{a_3},$$

we obtain a description of l as an intersection of two planes, the first orthogonal to the xy -plane and the second orthogonal to the xz -plane. If one of the numbers a_1, a_2 , or a_3 is zero, we cannot solve each equation in (14.37) for t . For example, if $a_3 = 0$ and $a_1 a_2 \neq 0$, then the third equation reduces to $z = z_1$, and a symmetric form may be written as

$$\frac{x - x_1}{a_1} = \frac{y - y_1}{a_2}, \quad z = z_1$$

which again expresses l as an intersection of two planes. A similar situation exists if $a_1 = 0$ or $a_2 = 0$.

Example 7 Find a symmetric form for the line through $P_1(3, 1, -2)$ and $P_2(-2, 7, -4)$.

Solution As in Example 1 of the preceding section, a vector $\mathbf{a}$ corresponding to $\overrightarrow{P_1 P_2}$ is

$$\mathbf{a} = \langle -2 - 3, 7 - 1, -4 + 2 \rangle = \langle -5, 6, -2 \rangle$$

and by Theorem (14.37) a parametric representation for the line is

$$x = 3 - 5t, \quad y = 1 + 6t, \quad z = -2 - 2t.$$

Solving each equation for t and equating the results we obtain the symmetric form

$$\frac{x - 3}{-5} = \frac{y - 1}{6} = \frac{z + 2}{-2}.$$

Example 8 Find a formula for the distance h from a point $P(x_0, y_0, z_0)$ to the plane $ax + by + cz + d = 0$.

Solution Let $R(x_1, y_1, z_1)$ be any point on the plane and let $\mathbf{n}$ be a normal vector to the plane. The vector $\mathbf{p} = \langle x_0 - x_1, y_0 - y_1, z_0 - z_1 \rangle$ has a geometric representation $\overline{RP}$. As illustrated in Figure 14.33, the distance h is given by

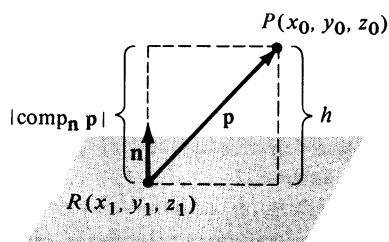


FIGURE 14.33

$$h = |\text{comp}_{\mathbf{n}} \mathbf{p}| = \left| \mathbf{p} \cdot \frac{1}{|\mathbf{n}|} \mathbf{n} \right|.$$

Since $\langle a, b, c \rangle$ is a normal vector to the plane we may let

$$\frac{1}{|\mathbf{n}|} \mathbf{n} = \frac{1}{\sqrt{a^2 + b^2 + c^2}} \langle a, b, c \rangle.$$

Consequently,

$$\begin{aligned} h &= \left| \mathbf{p} \cdot \frac{1}{|\mathbf{n}|} \mathbf{n} \right| = \left| \frac{a(x_0 - x_1) + b(y_0 - y_1) + c(z_0 - z_1)}{\sqrt{a^2 + b^2 + c^2}} \right| \\ &= \frac{|(ax_0 + by_0 + cz_0) + (-ax_1 - by_1 - cz_1)|}{\sqrt{a^2 + b^2 + c^2}}. \end{aligned}$$

Since R is on the plane, $ax_1 + by_1 + cz_1 + d = 0$ and, therefore, $d = -ax_1 - by_1 - cz_1$. Hence the preceding formula may be written

$$h = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Example 9 Find the distance from the point $P(-6, 2, 3)$ to the plane $4x - 5y + 8z - 7 = 0$.

Solution Using the formula obtained in Example 8,

$$h = \frac{|4(-6) - 5(2) + 8(3) - 7|}{\sqrt{16 + 25 + 64}} = \frac{17}{\sqrt{105}}.$$

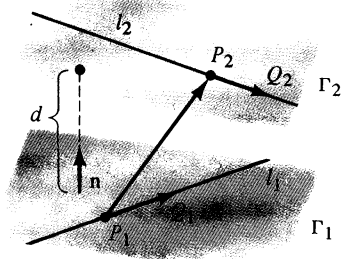


FIGURE 14.34

Example 10 Find a formula for the shortest distance d between two skew lines l_1 and l_2 .

Solution Two lines are skew if they are not parallel and do not intersect. Choose points P_1, Q_1 on l_1 and P_2, Q_2 on l_2 , as illustrated in Figure 14.34. It follows that the vector

$$\mathbf{n} = \frac{1}{|\overrightarrow{P_1Q_1} \times \overrightarrow{P_2Q_2}|} (\overrightarrow{P_1Q_1} \times \overrightarrow{P_2Q_2})$$

is a unit vector orthogonal to both $\overrightarrow{P_1Q_1}$ and $\overrightarrow{P_2Q_2}$. (Why?)

Let us consider planes Γ_1 and Γ_2 through P_1 and P_2 , respectively, each having normal vector $\mathbf{n}$. Thus Γ_1 and Γ_2 are parallel planes containing l_1 and l_2 , respectively. The distance d between the planes is measured along a line parallel to the common normal $\mathbf{n}$, as shown in the figure. It follows that d is the shortest distance between l_1 and l_2 . Moreover,

$$\begin{aligned} d &= |\text{comp}_{\mathbf{n}} \overrightarrow{P_1P_2}| = |\mathbf{n} \cdot \overrightarrow{P_1P_2}| \\ &= \frac{1}{|\overrightarrow{P_1Q_1} \times \overrightarrow{P_2Q_2}|} |(\overrightarrow{P_1Q_1} \times \overrightarrow{P_2Q_2}) \cdot \overrightarrow{P_1P_2}|. \quad \blacksquare \end{aligned}$$

14.6 Exercises

Sketch the graphs of the equations in Exercises 1–8.

- | | |
|--------------------------|--------------------------|
| 1 (a) $x = 3$ | 2 (a) $x = -4$ |
| (b) $y = -2$ | (b) $y = 0$ |
| (c) $z = 5$ | (c) $z = -2/3$ |
| 3 $2x + y - 6 = 0$ | 4 $3x - 2z - 24 = 0$ |
| 5 $4y - 2z - 15 = 0$ | 6 $5x + y - 4z + 20 = 0$ |
| 7 $2x - y + 5z + 10 = 0$ | 8 $x + y + z = 0$ |

In Exercises 9–18 find an equation of the plane that satisfies the stated conditions.

- 9 Through $P(6, -7, 4)$, parallel to (a) the xy -plane; (b) the yz -plane; (c) the xz -plane
- 10 Through $P(-2, 5, -8)$ with normal vector (a) $\mathbf{i}$; (b) $\mathbf{j}$; (c) $\mathbf{k}$
- 11 Through $P(-11, 4, -2)$ with normal vector $\mathbf{a} = 6\mathbf{i} - 5\mathbf{j} - \mathbf{k}$
- 12 Through $P(4, 2, -9)$ with normal vector $\overrightarrow{OP}$
- 13 Through $P(2, 5, -6)$ and parallel to the plane $3x - y + 2z = 10$
- 14 Through the origin and parallel to the plane $x - 6y + 4z = 7$

- 15 Through the origin and the points $P(0, 2, 5)$ and $Q(1, 4, 0)$
- 16 Through the points $P(3, 2, 1)$, $Q(-1, 1, -2)$, and $R(3, -4, 1)$
- 17 Through $P(5, 2, -3)$ and orthogonal to the planes $2x - y + 4z = 7$ and $3y - z = 8$
- 18 Through $P(-4, 1, 6)$ and having the same trace in the xz -plane as $x + 4y - 5z = 8$

In Exercises 19–22, find a symmetric form for the line through P_1 and P_2 .

- 19 $P_1(5, -2, 4)$, $P_2(2, 6, 1)$
- 20 $P_1(-3, 1, -1)$, $P_2(7, 11, -8)$
- 21 $P_1(4, 2, -3)$, $P_2(-3, 2, 5)$
- 22 $P_1(5, -7, 4)$, $P_2(-2, -1, 4)$

In Exercises 23–26 determine whether the given lines (a) are parallel; (b) are orthogonal; (c) intersect.

- 23 $(x + 1)/4 = (y + 7)/(-2) = (z - 5)/3$;
 $(x - 3)/3 = (y + 2)/3 = z/(-2)$
- 24 $(2x - 3)/6 = (y - 4)/(-1) = (z + 1)/2$;
 $(x + 3)/(-6) = (y + 1)/2 = (z - 5)/(-4)$

25 $(x + 1)/3 = (y - 3)/(-1) = (z + 5)/2$;
 $x/(-4) = (y - 4)/2 = (z - 9)/4$

26 $(x - 3)/5 = (y + 6)/(-2) = (z - 2)$;
 $x/7 = (y - 4)/3 = (1 - z)/5$.

In Exercises 27–30 find the distance from the point P to the given plane.

27 $3x - 7y + z - 5 = 0$, $P(1, -1, 2)$

28 $2x + 4y - 5z + 1 = 0$, $P(3, 1, -2)$

29 $4x - 3z = 2$, $P(5, -8, 1)$

30 $6y - z = 10$, $P(4, 1, -3)$

In Exercises 31 and 32 show that the two planes are parallel and find the distance between the planes.

31 $4x - 2y + 6z = 3$, $-6x + 3y - 9z = 4$

32 $3x + 12y - 6z = -2$, $5x + 20y - 10z = 7$

33 If a line l has parametric equations $x = 3t + 1$, $y = -2t + 4$, $z = t - 3$, find an equation of the plane that contains l and the point $P(5, 0, 2)$.

34 Find parametric equations for the line of intersection of the planes $2x + y + 4z = 8$ and $x + 3y - z = -1$.

35 Find an equation of the plane through the point $A(1, 2, -3)$ and the line of intersection of the planes $x + 3y - 5z = 10$ and $6x - y + z = 0$.

36 Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be vectors in V_3 . Prove that their position vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$, $\overrightarrow{OC}$ are coplanar if and only if

$$\mathbf{c} = m\mathbf{a} + n\mathbf{b}$$

for some scalars m and n .

In Exercises 37 and 38 let l_1 be the line through A , B and let l_2 be the line through C , D . Find the distance between l_1 and l_2 .

37 $A(1, -2, 3)$, $B(2, 0, 5)$, $C(4, 1, -1)$, $D(-2, 3, 4)$

38 $A(1, 3, 0)$, $B(0, 4, 5)$, $C(-2, -1, 2)$, $D(5, 1, 0)$

39 If a plane has nonzero x -, y - and z -intercepts a , b and c , respectively, show that an equation of the plane may be written $(x/a) + (y/b) + (z/c) = 1$. (This is called the **intercept form** for the equation of a plane.)

40 Prove that the planes with equation $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2$ are

(a) orthogonal if and only if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

(b) parallel if and only if there is a real number m such that $a_1 = ma_2$, $b_1 = mb_2$, and $c_1 = mc_2$.

The following concept is used in Exercises 41–44. If S_1 and S_2 are two spheres with equations

$$x^2 + y^2 + z^2 + a_1x + b_1y + c_1z + d_1 = 0,$$

and $x^2 + y^2 + z^2 + a_2x + b_2y + c_2z + d_2 = 0,$

respectively, then their **radical plane** is the plane whose equation is obtained by subtracting the equations of the spheres:

$$(a_1 - a_2)x + (b_1 - b_2)y + (c_1 - c_2)z + (d_1 - d_2) = 0.$$

41 If P_1 and P_2 are the centers of the spheres S_1 and S_2 , respectively, show that $\overrightarrow{P_1P_2}$ is normal to the radical plane of the spheres.

42 Show that if two spheres are tangent to each other, then their radical plane is tangent to both spheres.

43 Show that if two spheres intersect in a circle C ; then their radical plane contains C .

44 Show that the three radical planes obtained from three intersecting spheres are either parallel or intersect in a common line.

14.7 Cylinders and Surfaces of Revolution

In this section and the next we shall consider equations in x , y , and z whose graphs are fundamental in the study of analytic geometry. We previously defined the concept of *trace* in a coordinate plane. More generally, the trace of a surface in *any* plane is the intersection of the surface and the plane. To find the shape of a surface from its equation, we often make considerable use of traces in planes that are parallel to coordinate planes.

Example 1 Find traces, in various planes, of the surface having equation $z = x^2 + y^2$, and sketch the graph of the equation.

Solution Substitution of 0 for x in the equation gives us $z = y^2$, and hence the trace of the surface in the yz -plane is a parabola with vertex at the origin and opening upward, as shown in Figure 14.35.

Similarly, substituting 0 for y leads to $z = x^2$, and hence the trace in the xz -plane is also a parabola. It is instructive to find traces in planes parallel to the xy -plane, that is, planes with equations of the form $z = c$. Substituting c for z in the given equation produces $x^2 + y^2 = c$. Thus, if $c > 0$, then the trace in the plane $z = c$ is a circle of radius $\sqrt{c}$. Three such circles are sketched in Figure 14.35. If $c < 0$, then $x^2 + y^2 = c$ has no graph and consequently none of the points beneath the xy -plane is on the surface. The trace in the xy -plane has equation $x^2 + y^2 = 0$ and, therefore, consists of only one point, the origin. Although traces in planes parallel to the xz - or yz -planes could be determined, those we have obtained are sufficient for an accurate description of the graph. It will follow from the discussion at the end of this section that the surface in this example may be regarded as having been generated by revolving the graph of the parabola $z = y^2$ in the yz -plane about the z -axis. This surface is called a **circular paraboloid**, or **paraboloid of revolution**. ■

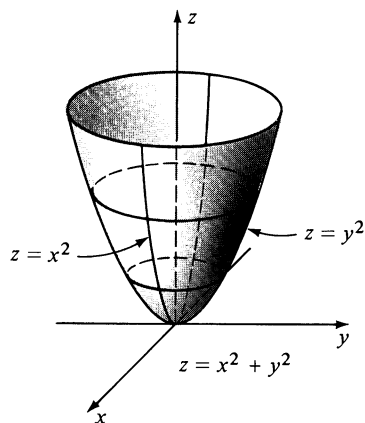


FIGURE 14.35

Definition (14.42)

If C is a curve in a plane and l is a line not in the plane, then the set of points on all lines that intersect C and are parallel to l is called a **cylinder**.

The curve C in Definition (14.42) is called a **directrix** for the cylinder, and each line through C parallel to l is a **ruling** of the cylinder. The most familiar type of cylinder is a **right circular cylinder**, obtained by letting C be a circle in a plane and l a line perpendicular to the plane, as illustrated in (i) of Figure 14.36. Although we have cut off the cylinder in the figure, it is to be understood that the rulings extend indefinitely. It is not required that the directrix C in (14.42) be a closed curve. This is illustrated in (ii) of Figure 14.36, where C is a parabola.

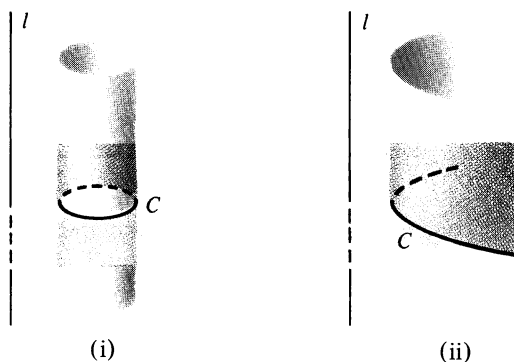


FIGURE 14.36

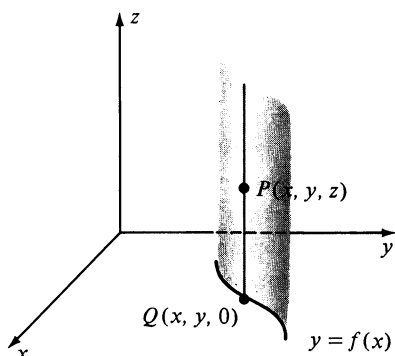


FIGURE 14.37

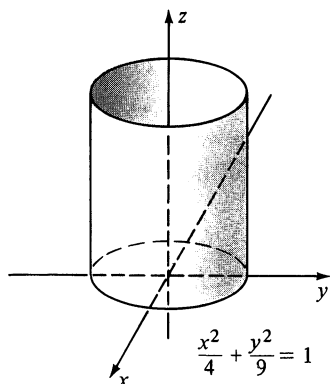


FIGURE 14.38

In the discussion to follow we shall only consider the case in which the directrix C is on a coordinate plane and the line l is parallel to the coordinate axis that is not on the plane. Suppose that C is on the xy -plane and has equation $y = f(x)$, where f is a function, and that the rulings are parallel to the z -axis. As illustrated in Figure 14.37, a point $P(x, y, z)$ is on the cylinder if and only if $Q(x, y, 0)$ is on C ; that is, if and only if the first two coordinates x and y of P satisfy the equation $y = f(x)$. It follows that the equation of the cylinder is $y = f(x)$, and hence is the same as the equation of the directrix in the xy -plane.

Example 2 Sketch the graph of $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

Solution From our previous remarks, the graph is a cylinder with rulings parallel to the z -axis. We begin by sketching the graph of $x^2/4 + y^2/9 = 1$ in the xy -plane. This ellipse is a directrix for the cylinder. All traces in planes parallel to the xy -plane are ellipses congruent to this directrix. A portion of the graph is shown in Figure 14.38. This surface is called an **elliptic cylinder**. ■

It can be shown that the graph of an equation that contains only the variables y and z is a cylinder with rulings parallel to the x -axis and whose trace (directrix) in the yz -plane is the graph of the given equation. Similarly, the graph of an equation that does not contain the variable y is a cylinder with rulings parallel to the y -axis and whose directrix is the graph of the given equation in the xz -plane.

Example 3 Sketch the graphs of the following equations.

- (a) $y^2 = 9 - z$ (b) $z = \sin x$

Solution (a) The graph is a cylinder with rulings parallel to the x -axis. A directrix in the yz -plane is the graph of $y^2 = 9 - z$. Part of the graph is sketched in (i) of Figure 14.39. This surface is called a **parabolic cylinder**.

(b) The graph is a cylinder with rulings parallel to the y -axis and whose directrix in the xz -plane is the graph of the equation $z = \sin x$. A portion of the graph is sketched in (ii) of Figure 14.39.

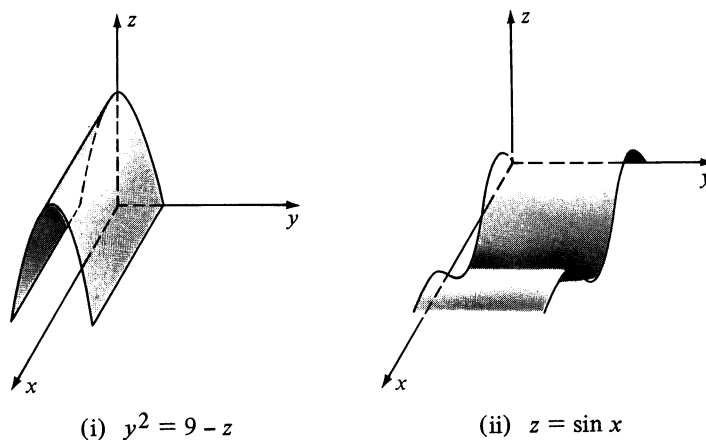


FIGURE 14.39

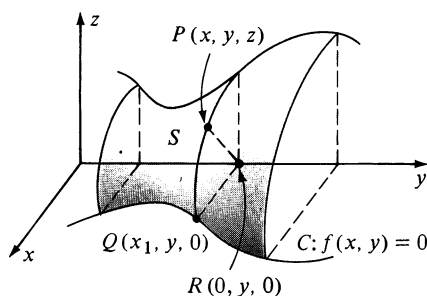


FIGURE 14.40

A surface of revolution was defined in Chapter 13 as the surface generated by revolving a plane curve C about a line in the plane, called the axis of revolution. In the discussion to follow, C will always lie in a coordinate plane and the axis of revolution will be one of the coordinate axes. It will be convenient to use the symbol $f(x, y)$ to denote an expression in the variables x and y . In this case $f(a, b)$ will denote the number obtained by substituting a for x and b for y . This notation will be discussed further in Chapter 16.

The graph of the equation $f(x, y) = 0$ in the xy -plane is a curve C . (We are interested here only in the graph in the xy -plane and not in the three-dimensional graph, which is a cylinder.) Suppose, for simplicity, that x and y are nonnegative for all points (x, y) on C and let S denote the surface obtained by revolving C about the y -axis, as illustrated in Figure 14.40. A point $P(x, y, z)$ is on S if and only if $Q(x_1, y, 0)$ is on C , where $x_1 = \sqrt{x^2 + z^2}$. Consequently, $P(x, y, z)$ is on S if and only if $f(\sqrt{x^2 + z^2}, y) = 0$. Thus, to find an equation for S we replace the variable x in the equation for C by $\sqrt{x^2 + z^2}$. Similarly, if the graph of $f(x, y) = 0$ is revolved about the x -axis, then an equation for the resulting surface may be found by replacing y by $\sqrt{y^2 + z^2}$. For some curves that contain points (x, y) where x or y is negative, the preceding discussion can be extended by substituting $\pm\sqrt{x^2 + z^2}$ for x or $\pm\sqrt{y^2 + z^2}$ for y . If, as in the next example, x or y appear only to even powers, then there is no need to make this distinction since the radical disappears when the equation is simplified.

Example 4 The graph of $9x^2 + 4y^2 = 36$ is revolved about the y -axis. Find an equation for the resulting surface.

Solution The desired equation may be found by substituting $x^2 + z^2$ for x^2 . This gives us

$$9(x^2 + z^2) + 4y^2 = 36.$$

The surface is called an **ellipsoid of revolution**. ■

A similar discussion can be given for curves that lie in the yz -plane or the xz -plane. For example, if a suitable curve C in the xz -plane is revolved about the z -axis, then an equation for the resulting surface may be found by replacing x by $\sqrt{x^2 + y^2}$. If C is revolved about the x -axis we replace z by $\sqrt{y^2 + z^2}$.

Finally, note that equations for surfaces of revolution are characterized by the fact that two of the variables appear in combinations such as $x^2 + y^2$, $y^2 + z^2$, or $x^2 + z^2$.

Example 5 Describe the graph of the equation $y^2 + z^2 - 4x^2 = 16$ as a surface of revolution.

Solution The given equation may be obtained from the equation $y^2 - 4x^2 = 16$ by substituting $\sqrt{y^2 + z^2}$ for y . Hence the graph is a surface

generated by revolving the hyperbola in the xy -plane having equation $y^2 - 4x^2 = 16$ about the x -axis. We could also describe the surface as that generated by revolving the hyperbola in the xz -plane that has the equation $z^2 - 4x^2 = 16$ about the x -axis. ■

14.7 Exercises

In Exercises 1–16 sketch the graph of the equation in three dimensions.

- | | |
|----------------------|---------------------|
| 1 $x^2 + y^2 = 9$ | 2 $y^2 + z^2 = 16$ |
| 3 $4y^2 + 9z^2 = 36$ | 4 $x^2 + 5z^2 = 25$ |
| 5 $x^2 = 9z$ | 6 $x^2 - 4y = 0$ |
| 7 $y^2 - x^2 = 16$ | 8 $xz = 1$ |
| 9 $4x - 3y = 12$ | 10 $2z + y = 5$ |
| 11 $z = x^3 - 4x$ | 12 $z = \cos(y/2)$ |
| 13 $yz = 0$ | 14 $ x + y = 2$ |
| 15 $z = e^y$ | 16 $z = \log x$ |

In Exercises 17–22 find an equation of the surface obtained by revolving the graph of the given equation about the indicated axis. Sketch the graph of the surface.

- | | |
|----------------------------------|-------------------------------|
| 17 $x^2 + 4y^2 = 16$; y -axis | 18 $y^2 = 4x$; x -axis |
| 19 $z = 4 - y^2$; z -axis | 20 $z = e^{-y^2}$; y -axis |
| 21 $z^2 - x^2 = 1$; x -axis | 22 $xz = 1$; z -axis |

In Exercises 23–26 sketch the graph of the given equation.

- | | |
|----------------------------|----------------------------|
| 23 $x^2 + 4y^2 + z^2 = 16$ | 24 $x^2 - 4y^2 + z^2 = 16$ |
| 25 $y - 4z^2 = 4x^2$ | 26 $x^2 - y^2 + z^2 = 1$ |

14.8 Quadric Surfaces

In Chapter 12 it was shown that, in two dimensions, the graph of a second-degree equation in x and y is a conic section. In three-dimensional analytic geometry, the graph of a second-degree equation in x , y , and z is referred to as a **quadric surface**. In this section we shall investigate equations for such surfaces. Let us begin with the following, where a , b , and c are positive real numbers. The graph of the equation is given the name listed in the margin.

Ellipsoid (14.43)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

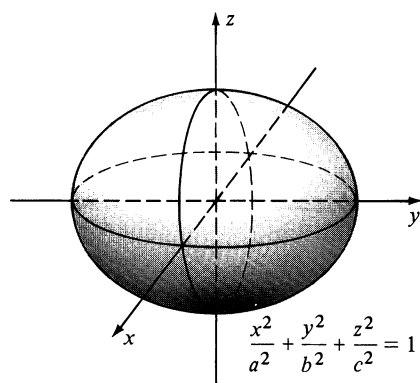


FIGURE 14.41

A typical graph of an ellipsoid is sketched in Figure 14.41. Note that traces of this surface in planes parallel to coordinate planes are ellipses. For example, the trace in the xy -plane is the ellipse $x^2/a^2 + y^2/b^2 = 1$. Similarly, the traces in the yz - and xz -planes are ellipses, as indicated in the figure. Let us find the trace in an arbitrary plane parallel to the xy -plane, that is, in a plane having an equation of the form $z = k$. Substituting k for z in Definition (14.43) gives us the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 - \frac{k^2}{c^2}.$$

If $|k| > c$, then $1 - k^2/c^2 < 0$ and there is no graph. Thus the graph of (14.43) lies between the planes $z = -c$ and $z = c$. If $|k| < c$, then $1 - k^2/c^2 > 0$ and hence the trace in the plane $z = k$ is an ellipse. Similarly, traces in planes parallel to the other two coordinate planes are ellipses, provided they do not

intersect the x -axis outside of the open interval $(-a, a)$ or the y -axis outside of $(-b, b)$. If $a = b = c$ then the graph is a sphere of radius a with center at the origin.

Another quadric surface is given by the following equation.

Hyperboloid of One Sheet (14.44)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Observe that the traces in the xz - and yz -planes are hyperbolas with equations

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 \quad \text{and} \quad \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1,$$

respectively. Traces on planes parallel to the xy -plane have equations of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 + \frac{k^2}{c^2},$$

where k is a real number and, therefore, are ellipses. The graph of (14.44) is sketched in Figure 14.42. The z -axis is called the **axis of the hyperboloid**.

The graphs of

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{and} \quad -\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

are also hyperboloids of one sheet; however, in the first case the axis of the hyperboloid is the y -axis, whereas in the second the axis of the hyperboloid coincides with the x -axis.

Hyperboloid of Two Sheets (14.45)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Traces in the xy - and xz -planes are hyperbolas, whereas traces in planes with equations of the form $x = k$ where $|k| > a$ are ellipses. We leave it to the reader to show that the graph of (14.45) has the appearance shown in Figure 14.43. The x -axis is the axis of the hyperboloid.

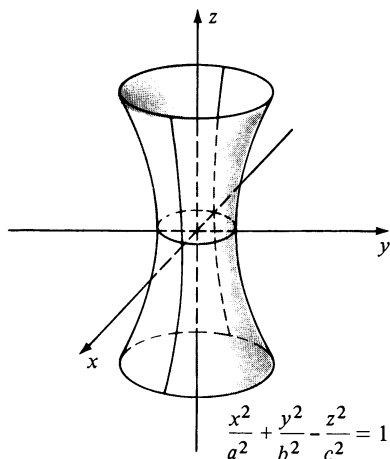


FIGURE 14.42

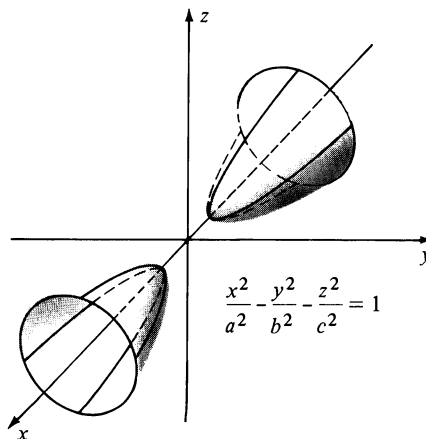


FIGURE 14.43

By using minus signs on different terms, we can obtain a hyperboloid of two sheets whose axis is the y -axis or the z -axis.

Cone (14.46)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

A typical graph of a cone is sketched in Figure 14.44. The trace in the yz -plane has equation

$$\frac{y^2}{b^2} - \frac{z^2}{c^2} = 0.$$

Solving for y we obtain $y = \pm(b/c)z$, which gives us the equations of two straight lines through the origin. Similarly, the trace in the xz -plane is a pair of straight lines that intersect at the origin. Traces in planes parallel to the xy -plane are ellipses. (Why?). The z -axis is the **axis of the cone**. By changing signs of the terms in (14.46), we obtain a cone whose axis is either the x - or the y -axis.

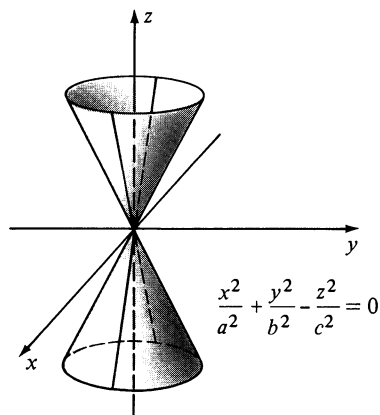


FIGURE 14.44

Paraboloid (14.47)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = cz$$

Example 1 in the previous section is the special case of (14.47) in which $a = b = c = 1$. If $c > 0$, then the graph of (14.47) is similar to that shown in Figure 14.35, except that if $a \neq b$, then traces in planes parallel to the xy -plane are ellipses instead of circles. If $c < 0$, then the paraboloid opens downward. The z -axis is called the **axis of the paraboloid**. The graphs of the equations.

$$\frac{x^2}{a^2} + \frac{z^2}{b^2} = cy \quad \text{and} \quad \frac{y^2}{a^2} + \frac{z^2}{b^2} = cx$$

are paraboloids whose axes are the y -axis and x -axis, respectively.

Hyperbolic Paraboloid (14.48)

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = cz$$

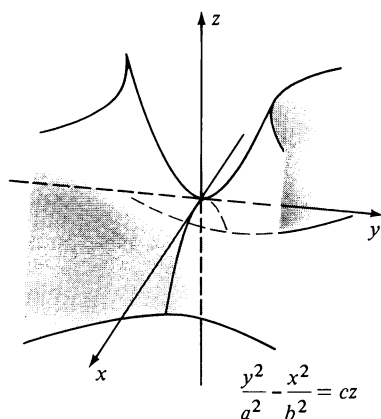


FIGURE 14.45

A typical sketch of this saddle-shaped surface for the case $c > 0$ is shown in Figure 14.45. Variations are obtained by interchanging x , y , and z in (14.48).

It is possible to obtain formulas for translation or rotation of axes in three dimensions that are analogous to those in two dimensions. These can be used to show that the graph of an equation of degree two in x , y , and z , is one of the surfaces discussed in this chapter, except for degenerate cases.

Example 1 Sketch the graph of $16x^2 - 9y^2 + 36z^2 = 144$ and identify the surface.

Solution Dividing both sides of the equation by 144 leads to

$$\frac{x^2}{9} - \frac{y^2}{16} + \frac{z^2}{4} = 1.$$

The trace in the xy -plane is the hyperbola $x^2/9 - y^2/16 = 1$, with vertices at $(\pm 3, 0, 0)$. The trace in the yz -plane is another hyperbola $z^2/4 - y^2/16 = 1$, with vertices at $(0, 0, \pm 2)$. The trace in the xz -plane is the ellipse with the equation $x^2/9 + z^2/4 = 1$. These traces are illustrated in Figure 14.46. To find traces on planes that are parallel to the xz -plane we substitute $y = k$ in the given equation, obtaining

$$\frac{x^2}{9} + \frac{z^2}{4} = 1 + \frac{k^2}{16}.$$

This shows that every cross section in a plane parallel to the xz -plane is an ellipse. For example, if $k = \pm 8$ we obtain $x^2/9 + z^2/4 = 5$ or, equivalently, $x^2/45 + z^2/20 = 1$. Several of these traces are illustrated in Figure 14.46. Traces on planes parallel to the xy -plane or the yz -plane may be found by substituting $z = k$ or $x = k$, respectively, in the given equation. We leave it to the reader to verify that in either case the trace is a hyperbola. The surface is a hyperboloid of one sheet with the y -axis as its axis.

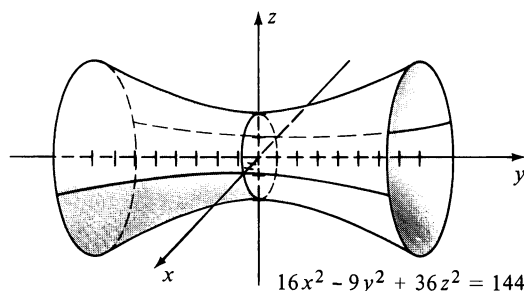


FIGURE 14.46

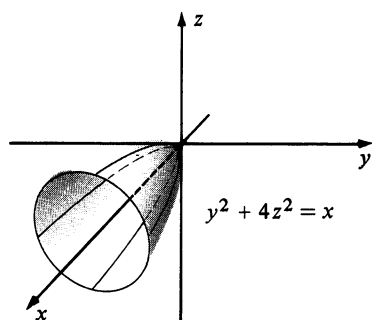


FIGURE 14.47

Example 2 Sketch the graph of $y^2 + 4z^2 = x$ and identify the surface.

Solution The trace in the xy -plane is the parabola $y^2 = x$. The trace in the xz -plane is the parabola $4z^2 = x$. These traces are illustrated in Figure 14.47. The trace in the yz -plane is the graph of $y^2 + 4z^2 = 0$ and hence consists of only one point, the origin. To find traces in planes parallel to the yz -plane we let $x = k$ in the given equation, obtaining

$$y^2 + 4z^2 = k.$$

If $k < 0$ there is no graph. (Why?) If $k > 0$ the trace is an ellipse. For example, letting $k = 9$, we obtain

$$\frac{y^2}{9} + \frac{z^2}{\frac{9}{4}} = 1$$

which is an ellipse with semi-axes of lengths 3 and $\frac{3}{2}$, as illustrated in Figure 14.47. The reader should verify that traces on planes parallel to either the xz - or xy -planes are parabolas. The surface is a paraboloid having the x -axis as its axis. ■

14.8 Exercises

Sketch the graphs of the equations in Exercises 1–20 and identify the surfaces.

- 1 $4x^2 + 9y^2 = 36z$
- 2 $8x^2 + 4y^2 + z^2 = 16$
- 3 $16x^2 + 100y^2 - 25z^2 = 400$
- 4 $25x^2 - 225y^2 + 9z^2 = 225$
- 5 $3x^2 - 4y^2 - z^2 = 12$
- 6 $4x^2 + y^2 = 9z^2$
- 7 $x^2 - 16y^2 = 4z^2$
- 8 $4y^2 - 25z^2 = 100x$
- 9 $9x^2 + 4y^2 + z^2 = 36$
- 10 $16x^2 - 25y^2 + 100z^2 = 200$
- 11 $16x^2 - 4y^2 - z^2 + 1 = 0$
- 12 $36x = 9y^2 + z^2$
- 13 $y^2 - 9x^2 - z^2 - 9 = 0$
- 14 $z^2 - x^2 - y^2 = 1$
- 15 $4y^2 + 25z^2 + 100x = 0$

- 16 $16y = x^2 + 4z^2$
- 17 $36x^2 - 16y^2 + 9z^2 = 0$
- 18 $4y^2 + 9z^2 = 9x^2$
- 19 $4y = x^2 - z^2$
- 20 $4x^2 + 16y = z^2$

In Exercises 21–24, find an equation for the set of all points $P(x, y, z)$ satisfying the given condition, and identify the graph of the equation.

- 21 Equally distant from the point $(0, 0, -1)$ and the plane $z = 1$
- 22 Equally distant from the z -axis and the xy -plane
- 23 Half as far from the xy -plane as from the origin
- 24 The sum of the squares of the distances from P to the origin and to the xy -plane is 4
- 25 Show that all cross sections of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ by planes parallel to the xy -plane are ellipses with the same eccentricity.
- 26 Identify the surface $z = xy$ by making a suitable rotation of axes in the xy -plane.

14.9 Cylindrical and Spherical Coordinate Systems

The system of polar coordinates can be extended to three dimensions. We merely represent a point P by an ordered triple (r, θ, z) where z is the usual (third) rectangular coordinate of P and (r, θ) are polar coordinates for the projection P' of P onto the xy -plane (see Figure 14.48). This system of coordinates is called the **cylindrical coordinate system**.

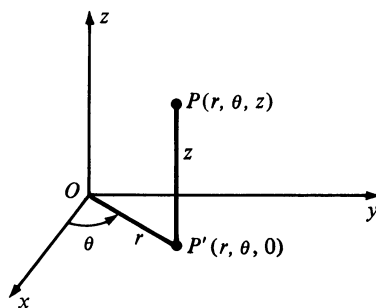


FIGURE 14.48

Since we have changed to polar coordinates in the xy -plane, the formulas in Theorem (13.4) give relationships between the rectangular coordinates and the cylindrical coordinates of P . Let us restate these formulas as follows.

Theorem (14.49)

The rectangular coordinates (x, y, z) and cylindrical coordinates (r, θ, z) of a point P are related as follows:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad \tan \theta = \frac{y}{x}, \quad r^2 = x^2 + y^2$$

The reason for using the term *cylindrical* for this system is that if $k > 0$, then the graph of the equation $r = k$, or equivalently of $x^2 + y^2 = k^2$, is a circular cylinder of radius k with axis along the z -axis. Note that the graph of $\theta = k$ is a plane containing the z -axis, whereas the graph of $z = k$ is a plane perpendicular to the z -axis.

Example 1 Find a rectangular equation for, and describe the graph of, each of the following equations.

(a) $z = r^2$ (b) $r = 4 \sin \theta$

Solution

(a) By Theorem (14.49), a rectangular equation for the graph of $z = r^2$ is $z = x^2 + y^2$. The graph is the paraboloid sketched in Figure 14.35.

(b) Multiplying both sides of $r = 4 \sin \theta$ by r gives us $r^2 = 4r \sin \theta$. Using (14.49) we obtain the rectangular equation $x^2 + y^2 = 4y$, or equivalently,

$$x^2 + (y - 2)^2 = 4.$$

The graph is a cylinder with rulings parallel to the z -axis. The directrix of the cylinder is a circle of radius 2 in the xy -plane whose center in rectangular coordinates is $(0, 2, 0)$. ■

Example 2 For each of the following equations, find an equation in cylindrical coordinates and describe the graph.

(a) $z^2 = x^2 - y^2$ (b) $z^2 = x^2 + y^2$

Solution

(a) Applying the first two equations in Theorem (14.49),

$$\begin{aligned} z^2 &= r^2 \cos^2 \theta - r^2 \sin^2 \theta \\ &= r^2(\cos^2 \theta - \sin^2 \theta), \end{aligned}$$

or

$$z^2 = r^2 \cos 2\theta.$$

The graph is a circular cone with axis along the x -axis.

(b) By the last equation in (14.49),

$$z^2 = r^2 \quad \text{or} \quad z = r.$$

The graph is a circular cone with axis along the z -axis. ■

A system called **spherical coordinates** may also be introduced in three dimensions. In this system a point P different from the origin is represented by an ordered triple (ρ, ϕ, θ) , where $\rho = |\overline{OP}|$, ϕ is the angle between the positive z -axis and $\overline{OP}$, and θ is a polar angle associated with the projection P' of P onto the xy -plane (see Figure 14.49). The origin is represented by any triple of the form $(0, \phi, \theta)$. The term *spherical* arises from the fact that the graph of the equation $\rho = k$, where $k > 0$, is a sphere with center at O . As in cylindrical coordinates, the graph of $\theta = k$ is a half-plane containing the z -axis. The graph of $\phi = k$ is usually a half-cone with vertex at O .

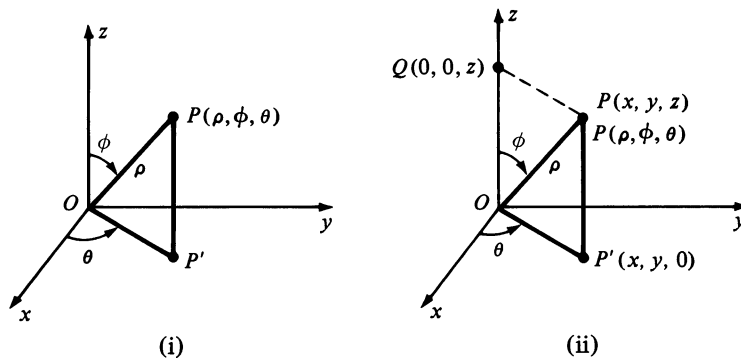


FIGURE 14.49

The relationship between spherical coordinates (ρ, ϕ, θ) and rectangular coordinates (x, y, z) of a point P can be found by referring to (ii) of Figure 14.49. Using the fact that

$$x = |\overline{OP'}| \cos \theta \quad \text{and} \quad y = |\overline{OP'}| \sin \theta$$

together with

$$|\overline{OP'}| = |\overline{QP}| = \rho \sin \phi \quad \text{and} \quad |\overline{OQ}| = \rho \cos \phi$$

gives us (i) of the next theorem. Part (ii) is a consequence of the Distance Formula.

Theorem (14.50)

The rectangular coordinates (x, y, z) and spherical coordinates (ρ, ϕ, θ) of a point P are related as follows:

- (i) $x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$
- (ii) $\rho^2 = x^2 + y^2 + z^2$

Example 3 Find a rectangular equation for $\rho = 2 \sin \phi \cos \theta$ and describe the graph of this equation.

Solution Multiplying both sides of the given equation by ρ we obtain $\rho^2 = 2\rho \sin \phi \cos \theta$. Applying (14.50) gives us $x^2 + y^2 + z^2 = 2x$ or, equivalently, $(x - 1)^2 + y^2 + z^2 = 1$. By (14.15), the graph is a sphere of radius 1 whose center in rectangular coordinates is $(1, 0, 0)$. ■

Example 4 If a point P has spherical coordinates $(4, \pi/6, \pi/3)$, find rectangular and cylindrical coordinates for P .

Solution Using (i) of Theorem (14.50) with $\rho = 4$, $\phi = \pi/6$, and $\theta = \pi/3$, we obtain the rectangular coordinates.

$$x = 4 \sin \frac{\pi}{6} \cos \frac{\pi}{3} = 4 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = 1$$

$$y = 4 \sin \frac{\pi}{6} \sin \frac{\pi}{3} = 4 \left(\frac{1}{2} \right) \left(\frac{\sqrt{3}}{2} \right) = \sqrt{3}$$

$$z = 4 \cos \frac{\pi}{6} = 4 \left(\frac{\sqrt{3}}{2} \right) = 2\sqrt{3}.$$

To find the cylindrical coordinates we note that $r^2 = x^2 + y^2 = 1 + 3 = 4$. Hence cylindrical coordinates for P are (r, θ, z) or $(2, \pi/3, 2\sqrt{3})$. ■

14.9 Exercises

- 1 Change the following from cylindrical coordinates to rectangular coordinates.
(a) $(5, \pi/2, 3)$ (b) $(6, \pi/3, -5)$
- 2 Change the following from spherical coordinates to rectangular coordinates.
(a) $(4, \pi/6, \pi/2)$ (b) $(1, 3\pi/4, 2\pi/3)$
- 3 Change the following from rectangular coordinates to spherical coordinates.
(a) $(1, 1, -2\sqrt{2})$ (b) $(1, \sqrt{3}, 0)$
- 4 Change the following from rectangular coordinates to cylindrical coordinates.
(a) $(2\sqrt{3}, 2, -2)$ (b) $(\sqrt{2}, -\sqrt{2}, 1)$
- 5 Change the following from spherical coordinates to cylindrical coordinates.
(a) $(4, \pi/3, \pi/3)$ (b) $(2, 5\pi/6, \pi/4)$
- 6 Change the following from cylindrical coordinates to spherical coordinates.
(a) $(\sqrt{2}, \pi/4, 1)$ (b) $(3, \pi/3, 1)$

Describe the graphs of the equations in Exercises 7–20.

- | | |
|-----------------------------|---|
| 7 $r = 4$ | 8 $\theta = \pi/4$ |
| 9 $\phi = \pi/6$ | 10 $\rho = 3$ |
| 11 $r = 4 \cos \theta$ | 12 $r = \sin 2\theta$ |
| 13 $\rho = 4 \cos \phi$ | 14 $\rho = \sec \phi$ |
| 15 $\phi = 0$ | 16 $\phi = \pi/2$ |
| 17 $r^2 = \cos 2\theta$ | 18 $z = 1 - r^2$ |
| 19 $\rho^2 - 3\rho + 2 = 0$ | 20 $\phi^2 - \pi\phi + (3\pi^2/16) = 0$ |

Find equations in cylindrical coordinates and in spherical coordinates for the graphs of the equations in Exercises 21–30.

- | | |
|--------------------------|---------------------------|
| 21 $x^2 + y^2 + z^2 = 4$ | 22 $x^2 + y^2 = 4z$ |
| 23 $3x + y - 4z = 12$ | 24 $x^2 - y^2 - z^2 = 1$ |
| 25 $y^2 + z^2 = 9$ | 26 $x^2 + z^2 = 9$ |
| 27 $6x = x^2 + y^2$ | 28 $x^2 - 4z^2 + y^2 = 0$ |
| 29 $y = xz$ | 30 $y = x$ |

14.10 Review: Concepts

Define or discuss each of the following.

- | | |
|--|--|
| 1 Vectors in V_2 and V_3 | 15 Equation of a sphere |
| 2 Scalar | 16 The dot product and its properties |
| 3 Scalar multiples of vectors | 17 The angle between two vectors |
| 4 Addition of vectors | 18 Orthogonal vectors |
| 5 Components of a vector | 19 Cauchy–Schwarz Inequality |
| 6 Zero vector | 20 Triangle Inequality |
| 7 Magnitude of a vector | 21 Work done by a force |
| 8 Position vector | 22 The vector product and its properties |
| 9 Subtraction of vectors | 23 Equations of a line |
| 10 Parallel vectors | 24 Equations of a plane |
| 11 Unit vector | 25 Cylinders |
| 12 Rectangular coordinate system in three dimensions | 26 Quadric surfaces |
| 13 Distance Formula | 27 Cylindrical coordinates |
| 14 Midpoint Formula | 28 Spherical coordinates |

Review: Exercises

Find the vectors or scalars indicated in Exercises 1–10 if $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 4\mathbf{i} - \mathbf{j}$.

- 1 $4\mathbf{a} + \mathbf{b}$
- 2 $2\mathbf{a} - 3\mathbf{b}$
- 3 $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} - 2\mathbf{b})$
- 4 $3\mathbf{a} \cdot (5\mathbf{b} + \mathbf{i})$
- 5 $|\mathbf{a}| - |\mathbf{b}|$
- 6 $|\mathbf{a} - \mathbf{b}|$
- 7 The angle between $\mathbf{a}$ and $\mathbf{i}$
- 8 A unit vector having the same direction as $\mathbf{b}$
- 9 A unit vector orthogonal to $\mathbf{a}$
- 10 The cosine of the angle between $\mathbf{a}$ and $\mathbf{b}$
- 11 Given the points $A(5, -3, 2)$ and $B(-1, -4, 3)$, find
 - (a) $d(A, B)$.
 - (b) the coordinates of the midpoint of the line segment AB .
 - (c) an equation of the sphere with center B and tangent to the xz -plane.
 - (d) an equation of the plane through B parallel to the xz -plane.
 - (e) parametric equations for the line through A and B .
 - (f) an equation of the plane through A with normal vector $\overrightarrow{AB}$.
- 12 Find an equation for the plane through $A(0, 4, 9)$ and $B(0, -3, 7)$ which is perpendicular to the yz -plane.
- 13 Find an equation for the plane that has x -intercept 5, y -intercept -2 , and z -intercept 6.
- 14 Find an equation for the cylinder which is perpendicular to the xy -plane and has, for its directrix, the circle in the xy -plane with center $C(4, -3, 0)$ and radius 5.
- 15 Find an equation for an ellipsoid with center O and having x -intercept 8, y -intercept 3, and z -intercept 1.
- 16 Find an equation for the surface obtained by revolving the graph of the equation $z = x$ about the z -axis.

In Exercises 17–27 sketch the graph of the given equation.

- 17 $x^2 + y^2 + z^2 - 14x + 6y - 8z + 10 = 0$
- 18 $4y - 3z - 15 = 0$
- 19 $3x - 5y + 2z = 10$
- 20 $y = z^2 + 1$
- 21 $9x^2 + 4z^2 = 36$
- 22 $x^2 + 4y + 9z^2 = 0$
- 23 $z^2 - 4x^2 = 9 - 4y^2$

- 24 $2x^2 + 4z^2 - y^2 = 0$
- 25 $z^2 - 4x^2 - y^2 = 4$
- 26 $x^2 + 2y^2 + 4z^2 = 16$
- 27 $x^2 - 4y^2 = 4z$

If $\mathbf{a} = 3\mathbf{i} - \mathbf{j} - 4\mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$, and $\mathbf{c} = -\mathbf{i} + 6\mathbf{k}$, find the vectors or scalars in Exercises 28–44.

- 28 $3\mathbf{a} - 2\mathbf{b}$
- 29 $\mathbf{a} \cdot (\mathbf{b} - \mathbf{c})$
- 30 $|\mathbf{b} + \mathbf{c}|$
- 31 $|\mathbf{b}| + |\mathbf{c}|$
- 32 A unit vector having the same direction as $\mathbf{a}$
- 33 The direction cosines of $\mathbf{a}$
- 34 The cosine of the angle between $\mathbf{a}$ and $\mathbf{c}$
- 35 $\mathbf{a} \times \mathbf{b}$
- 36 $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$
- 37 $\text{comp}_{\mathbf{b}} \mathbf{a}$
- 38 $\text{comp}_{\mathbf{a}} (\mathbf{b} \times \mathbf{c})$
- 39 $\mathbf{a} \cdot \mathbf{a}$
- 40 $\mathbf{a} \times \mathbf{a}$
- 41 $(\mathbf{a} \times \mathbf{c}) + (\mathbf{c} \times \mathbf{a})$
- 42 $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$
- 43 A vector having the opposite direction of $\mathbf{b}$ and twice the magnitude of $\mathbf{b}$
- 44 Two unit vectors orthogonal to both $\mathbf{b}$ and $\mathbf{c}$
- 45 Given the points $P(2, -1, 1)$, $Q(-3, 2, 0)$, and $R(4, -5, 3)$, find
 - (a) a unit normal vector orthogonal to the plane determined by P , Q , and R .
 - (b) an equation for the plane determined by P , Q , and R .
 - (c) parametric equations for a line through P that is parallel to the line through Q and R .
 - (d) $\overrightarrow{QP} \cdot \overrightarrow{QR}$.
 - (e) the angle between $\overrightarrow{QP}$ and $\overrightarrow{QR}$.
 - (f) the area of the triangle with vertices P , Q , and R .
- 46 Find the angle between the two lines

$$(x - 3)/2 = (y + 1)/(-4) = (z - 5)/8$$
 and

$$(x + 1)/7 = (6 - y)/2 = (2z + 7)/(-4).$$

47 Find parametric equations for each of the lines in Exercise 46.

48 Determine whether the following lines intersect and, if so, find the point of intersection:

$$\begin{aligned}x &= 2 + t, y = 1 + t, z = 4 + 7t; \\x &= -4 + 5t, y = 2 - 2t, z = 1 - 4t.\end{aligned}$$

49 Find the angle between the lines in Exercise 48.

50 The position of a particle at time t is $(t, t \sin t, t \cos t)$. Find the distance the particle moves during the time interval $[0, 5]$.

51 If the rectangular coordinates of a point P are $(2, -2, 1)$, find cylindrical and spherical coordinates for P .

52 If spherical coordinates of a point P are $(12, \pi/6, 3\pi/4)$, find cylindrical and rectangular coordinates for P .

Find a rectangular equation and describe the graph of each equation in Exercises 53–56.

53 $\phi = 3\pi/4$

54 $r = \cos 2\theta$

55 $\rho \sin \phi \cos \theta = 1$

56 $\rho^2 - 3\rho = 0$

Find equations in cylindrical coordinates and in spherical coordinates for the graphs of the equations in Exercises 57–60.

57 $x^2 + y^2 = 1$

58 $z = x^2 - y^2$

59 $x^2 + y^2 + z^2 - 2z = 0$

60 $2x + y - 3z = 4$

15

Vector-Valued Functions

In this chapter we shall extend the theory of limits, derivatives, and integrals to functions whose ranges consist of vectors instead of real numbers. Our principal

application of these concepts will be to the study of motion. Further applications will be given in later chapters.

15.1 Definitions and Graphs

A function was defined in (1.21) as a correspondence that associates with each element of a set X a unique element of a set Y . If X is a set of real numbers and Y is a set of vectors, then the function will be called a **vector-valued function** and denoted by symbols such as $\mathbf{r}$ or $\vec{r}$. Thus, if $\mathbf{r}$ is a vector-valued function, then for each real number t in the domain X there exists a unique vector

$$\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

If, with each t , we associate the first component of $\mathbf{r}(t)$, we obtain a function f from X to $\mathbb{R}$, where $x = f(t)$. Similarly, there are functions g and h such that $y = g(t)$ and $z = h(t)$, respectively. Hence for every t in X we have the following:

$$(15.1) \qquad \mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

Conversely, if f , g , and h are functions from X to $\mathbb{R}$, then we may define a vector-valued function $\mathbf{r}$ by means of (15.1). Consequently, $\mathbf{r}$ is a *vector-valued function if and only if $\mathbf{r}(t)$ is expressible in the form (15.1)*. If it is not stated explicitly, the domain of $\mathbf{r}$ is assumed to be the intersection of the domains of f , g , and h .

Example 1 If $\mathbf{r}(t) = 3t^2\mathbf{i} + 2t^3\mathbf{j} + (t - 2)\mathbf{k}$, find $\mathbf{r}(1)$, $\mathbf{r}(2)$, and $\mathbf{r}(0)$.

Solution The specified vectors may be found by substituting 1, 2 and 0 for t . Thus

$$\mathbf{r}(1) = 3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

$$\mathbf{r}(2) = 12\mathbf{i} + 16\mathbf{j} + 0\mathbf{k} = 12\mathbf{i} + 16\mathbf{j}$$

$$\mathbf{r}(0) = 0\mathbf{i} + 0\mathbf{j} - 2\mathbf{k} = -2\mathbf{k}. \quad \blacksquare$$

There is a close connection between vector-valued functions and curves in space. As in Definition (13.1) for plane curves, we define a **space curve**, or a **curve in three dimensions**, as a set C of ordered triples of the form

$$(f(t), g(t), h(t))$$

where the functions f , g , and h are continuous on an interval I . The graph of C in a rectangular coordinate system consists of all points $P(f(t), g(t), h(t))$ that correspond to the ordered triples in C . The equations

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

where t is in I , are called **parametric equations** for C . The notions of **endpoints**, **closed curve**, **simple closed curve**, and **length of a curve** are defined exactly as in the two-dimensional case. A curve C is **smooth** if it has a parametric representation $x = f(t)$, $y = g(t)$, $z = h(t)$ on an interval I such that f' , g' and h' are continuous and not simultaneously zero, except possibly at endpoints. In this case the parametric equations are called a **smooth parametric representation** of C .

The next result is analogous to Theorem (13.11).

Theorem (15.2)

If a smooth curve C is given parametrically by $x = f(t)$, $y = g(t)$, $z = h(t)$, where $a \leq t \leq b$, and if C does not intersect itself, except possibly at the endpoints of $[a, b]$, then the length L of C is given by

$$\begin{aligned} L &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt. \end{aligned}$$

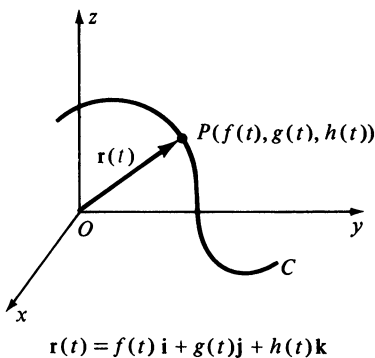


FIGURE 15.1

Consider a vector function $\mathbf{r}$, where $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$. As illustrated in Figure 15.1, if $\overrightarrow{OP}$ is the position vector corresponding to $\mathbf{r}(t)$, then as t varies, the terminal point P traces the curve having parametric equations $x = f(t)$, $y = g(t)$, $z = h(t)$. This geometric representation of $\mathbf{r}(t)$ is especially useful if t is time and P is the position of a particle in motion.

Example 2 Let $\mathbf{r}(t) = a \cos t \mathbf{i} + a \sin t \mathbf{j} + b t \mathbf{k}$ and let $\overrightarrow{OP}$ be the position vector of $\mathbf{r}(t)$.

- (a) Sketch the graph of the curve C traced by P as t varies.
 (b) Find the length of C between the points corresponding to $t = 0$ and $t = 2\pi$.

Solution

(a) Parametric equations for C are $x = a \cos t$, $y = a \sin t$, $z = bt$. If the point $P(x, y, z)$ is on C , then P is on the circular cylinder having equation $x^2 + y^2 = a^2$. (Why?) As t varies from 0 to 2π , the point P starts at $(a, 0, 0)$ and winds around this cylinder one time. Other intervals of length 2π lead to similar loops (see Figure 15.2). This curve is called a **circular helix**.

(b) The length of C between the points corresponding to $t = 0$ and $t = 2\pi$ can be found by means of Theorem (15.2). Thus

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + b^2} dt \\ &= \int_0^{2\pi} \sqrt{a^2 + b^2} dt = \sqrt{a^2 + b^2} t \Big|_0^{2\pi} = \sqrt{a^2 + b^2} 2\pi. \quad \blacksquare \end{aligned}$$

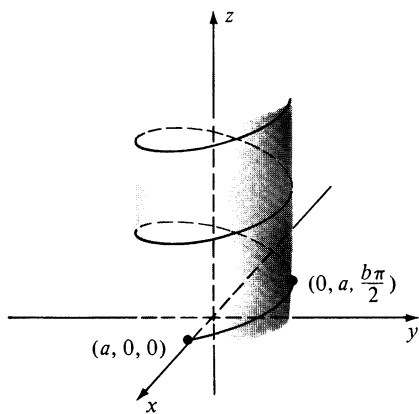


FIGURE 15.2

Example 3 If $\mathbf{r}(t) = (2t + 1)\mathbf{i} + (3t - 4)\mathbf{j} - 5t\mathbf{k}$, and if $\overrightarrow{OP}$ is the position vector of $\mathbf{r}(t)$, describe the graph of the curve C traced by P as t varies.

Solution Parametric equations for C are

$$x = 2t + 1, \quad y = 3t - 4, \quad z = -5t.$$

By Theorem (14.37) the graph is the line through the point $(1, -4, 0)$ that is parallel to the vector $\langle 2, 3, -5 \rangle$. $\blacksquare$

As in the preceding chapter, we may regard V_2 as the subset of V_3 consisting of all vectors whose third component is 0. In this case (15.1) takes on the form

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$$

and as t varies, the curve traced by the terminal point of the position vector $\overrightarrow{OP}$ lies in the xy -plane.

Example 4 Let $\mathbf{r}(t) = (\frac{1}{2}t \cos t)\mathbf{i} + (\frac{1}{2}t \sin t)\mathbf{j}$, and let $\overrightarrow{OP}$ be the position vector of $\mathbf{r}(t)$.

- (a) Sketch the graph of the curve traced by P as t varies from 0 to 3π .
 (b) Find $\mathbf{r}(6)$ and sketch its position vector.

Solution

(a) The curve is in the xy -plane and has parametric equations

$$x = \frac{1}{2}t \cos t, \quad y = \frac{1}{2}t \sin t, \quad \text{where } 0 \leq t \leq 3\pi.$$

There is no loss of generality in assuming that t is the radian measure of an angle θ . Indeed, we may write

$$x = \frac{1}{2}\theta \cos \theta, \quad y = \frac{1}{2}\theta \sin \theta, \quad \text{where } 0 \leq \theta \leq 3\pi.$$

Moreover, since

$$\frac{y}{x} = \frac{\frac{1}{2}\theta \sin \theta}{\frac{1}{2}\theta \cos \theta} = \tan \theta$$

it follows from Theorem (13.4) that θ is an angle that can be used to express (x, y) in polar coordinates. We next observe that

$$\begin{aligned} x^2 + y^2 &= \frac{1}{4}\theta^2 \cos^2 \theta + \frac{1}{4}\theta^2 \sin^2 \theta \\ &= \frac{1}{4}\theta^2 (\cos^2 \theta + \sin^2 \theta) = \frac{1}{4}\theta^2. \end{aligned}$$

Changing $x^2 + y^2$ to polar coordinates gives us

$$r^2 = \frac{1}{4}\theta^2 \quad \text{or} \quad r = \frac{1}{2}\theta.$$

The graph of the polar equation $r = \frac{1}{2}\theta$, where $0 \leq \theta \leq 3\pi$, is the portion of the spiral illustrated in Figure 15.3.

(b) Substitution of 6 for t gives us

$$\mathbf{r}(6) = (3 \cos 6)\mathbf{i} + (3 \sin 6)\mathbf{j} \approx 2.9\mathbf{i} - 0.8\mathbf{j}$$

and we sketch $\mathbf{r}(6)$ as indicated in Figure 15.3. ■

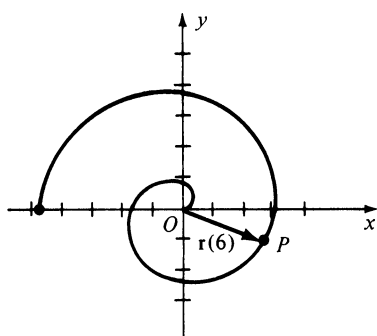


FIGURE 15.3

15.1 Exercises

In Exercises 1–14 sketch the graph of the curve C traced by the terminal point of the position vector of $\mathbf{r}(t)$ if t varies as indicated.

- 1 $\mathbf{r}(t) = 3t\mathbf{i} + (1 - 9t^2)\mathbf{j}; t \text{ in } \mathbb{R}$
- 2 $\mathbf{r}(t) = (1 - t^3)\mathbf{i} + t\mathbf{j}; t \geq 0$
- 3 $\mathbf{r}(t) = (t^3 - 1)\mathbf{i} + (t^2 + 2)\mathbf{j}; -2 \leq t \leq 2$
- 4 $\mathbf{r}(t) = (2 + \cos t)\mathbf{i} - (3 - \sin t)\mathbf{j}; 0 \leq t \leq 2\pi$
- 5 $\mathbf{r}(t) = e^t \cos t\mathbf{i} + e^t \sin t\mathbf{j}; 0 \leq t \leq \pi$
- 6 $\mathbf{r}(t) = 2 \cosh t\mathbf{i} + 3 \sinh t\mathbf{j}; t \text{ in } \mathbb{R}$
- 7 $\mathbf{r}(t) = t\mathbf{i} + 4 \cos t\mathbf{j} + 9 \sin t\mathbf{k}; t \geq 0$
- 8 $\mathbf{r}(t) = \tan t\mathbf{i} + \sec t\mathbf{j} + 2\mathbf{k}; -\pi/2 < t < \pi/2$
- 9 $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}; t \text{ in } \mathbb{R}$
- 10 $\mathbf{r}(t) = t^3\mathbf{i} + t^2\mathbf{j} + t\mathbf{k}; 0 \leq t \leq 4$

- 11 $\mathbf{r}(t) = (t^2 + 1)\mathbf{i} + t\mathbf{j} + 3\mathbf{k}; t \text{ in } \mathbb{R}$
- 12 $\mathbf{r}(t) = 6 \sin t\mathbf{i} + 4\mathbf{j} + 25 \cos t\mathbf{k}; -2\pi \leq t \leq 2\pi$
- 13 $\mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + \sin t\mathbf{k}; t \text{ in } \mathbb{R}$
- 14 $\mathbf{r}(t) = t\mathbf{i} + 2t\mathbf{j} + e^t\mathbf{k}; t \text{ in } \mathbb{R}$

In Exercises 15–20 find the length of the curve.

- 15 $C = \{(5t, 4t^2, 3t^2): 0 \leq t \leq 2\}$
- 16 $C = \{(t^2, t \sin t, t \cos t): 0 \leq t \leq 1\}$
- 17 $x = e^t \cos t, y = e^t, z = e^t \sin t; 0 \leq t \leq 2\pi$
- 18 $x = 2t, y = 4 \sin 3t, z = 4 \cos 3t; 0 \leq t \leq 2\pi$
- 19 $x = 3t^2, y = t^3, z = 6t; 0 \leq t \leq 1$
- 20 $x = 1 - 2t^2, y = 4t, z = 3 + 2t^2; 0 \leq t \leq 2$

15.2 Limits, Derivatives, and Integrals

The limit of a vector-valued function may be defined as follows.

Definition (15.3)

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left[\lim_{t \rightarrow a} f(t) \right] \mathbf{i} + \left[\lim_{t \rightarrow a} g(t) \right] \mathbf{j} + \left[\lim_{t \rightarrow a} h(t) \right] \mathbf{k}.$$

provided f , g , and h have limits as t approaches a .

The preceding definition may be extended to one-sided limits as was done in Chapter 2. If, in Definition (15.3),

$$\lim_{t \rightarrow a} f(t) = a_1, \quad \lim_{t \rightarrow a} g(t) = a_2, \quad \text{and} \quad \lim_{t \rightarrow a} h(t) = a_3$$

then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k} = \mathbf{a}.$$

Intuitively, we have a situation of the type illustrated in Figure 15.4, where if $\overrightarrow{OP}$ and $\overrightarrow{OA}$ are position vectors corresponding to $\mathbf{r}(t)$ and $\mathbf{a}$, respectively, then as t approaches a , $\overrightarrow{OP}$ approaches $\overrightarrow{OA}$, in the sense that P gets closer and closer to A .

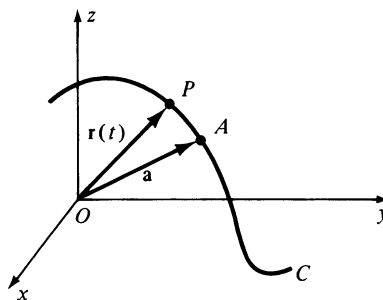


FIGURE 15.4

Theorems on limits of vector-valued functions similar to those in Chapter 2 can be established and are left as exercises. Limits of vector-valued functions may also be formulated using an ε - δ approach similar to that in Definition (2.3) (see Exercise 35).

Definition (15.4)

A vector-valued function $\mathbf{r}$ is **continuous** at a if

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a).$$

It follows that if $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then $\mathbf{r}$ is continuous at a if and only if f , g , and h are continuous at a . Continuity on an interval is defined in the usual way.

Definition (15.5)

If $\mathbf{r}$ is a vector-valued function, then the **derivative** of $\mathbf{r}$ is the vector-valued function $\mathbf{r}'$ defined by

$$\mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} [\mathbf{r}(t + \Delta t) - \mathbf{r}(t)]$$

for all t such that the limit exists.

If we adopt the convention that

$$\frac{1}{\Delta t} [\mathbf{r}(t + \Delta t) - \mathbf{r}(t)] = \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

then Definition (15.5) takes on the familiar form for derivatives introduced in Chapter 3. We shall use this notation in the proof of the next theorem, which states that to find $\mathbf{r}'(t)$ we may differentiate each component of $\mathbf{r}(t)$.

Theorem (15.6)

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$

where f , g , and h are differentiable, then

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}.$$

Proof

$$\begin{aligned} \mathbf{r}'(t) &= \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{[f(t + \Delta t)\mathbf{i} + g(t + \Delta t)\mathbf{j} + h(t + \Delta t)\mathbf{k}] - [f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}]}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \left[\frac{f(t + \Delta t) - f(t)}{\Delta t} \mathbf{i} + \frac{g(t + \Delta t) - g(t)}{\Delta t} \mathbf{j} + \frac{h(t + \Delta t) - h(t)}{\Delta t} \mathbf{k} \right]. \end{aligned}$$

Taking the limit of each component gives us the desired result. $\square$

If $\mathbf{r}'(t)$ exists we say that the function $\mathbf{r}$ is **differentiable** at t . We also denote derivatives as follows:

$$\mathbf{r}'(t) = D_t \mathbf{r}(t) = \frac{d}{dt} \mathbf{r}(t).$$

Higher derivatives may be obtained in like manner. For example,

$$\mathbf{r}''(t) = f''(t)\mathbf{i} + g''(t)\mathbf{j} + h''(t)\mathbf{k}.$$

Example 1 If $\mathbf{r}(t) = (\ln t)\mathbf{i} + e^{-3t}\mathbf{j} + t^2\mathbf{k}$,

- (a) find the domain of $\mathbf{r}$ and determine where $\mathbf{r}$ is continuous.
- (b) find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.

Solution

(a) Since $\ln t$ is undefined if $t \leq 0$, the domain of $\mathbf{r}$ is the set of positive real numbers. Moreover, $\mathbf{r}$ is continuous throughout its domain since each component determines a continuous function.

(b) By Theorem (15.6),

$$\mathbf{r}'(t) = (1/t)\mathbf{i} - 3e^{-3t}\mathbf{j} + 2t\mathbf{k},$$

and

$$\mathbf{r}''(t) = (-1/t^2)\mathbf{i} + 9e^{-3t}\mathbf{j} + 2\mathbf{k}. \quad \blacksquare$$

Since the definition of $\mathbf{r}'(t)$ in (15.5) has the same form as that given for $f'(x)$ in Chapter 3, it is reasonable to expect that differentiation formulas for vector-valued functions are similar to those for real-valued functions. The next theorem brings out this fact. Other differentiation formulas may be found in the Exercises.

Theorem (15.7)

If $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector-valued functions and c is a scalar, then

- (i) $D_t[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t).$
- (ii) $D_t[c\mathbf{u}(t)] = c\mathbf{u}'(t).$
- (iii) $D_t[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}(t) \cdot \mathbf{v}'(t) + \mathbf{u}'(t) \cdot \mathbf{v}(t).$
- (iv) $D_t[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}(t) \times \mathbf{v}'(t) + \mathbf{u}'(t) \times \mathbf{v}(t).$

Proof
we write

We shall prove (iii) and leave the other parts as exercises. If

$$\mathbf{u}(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$$

$$\mathbf{v}(t) = g_1(t)\mathbf{i} + g_2(t)\mathbf{j} + g_3(t)\mathbf{k}$$

where each f_i and g_i is a differentiable function of t , then

$$\mathbf{u}(t) \cdot \mathbf{v}(t) = \sum_{i=1}^3 f_i(t)g_i(t).$$

Consequently,

$$\begin{aligned}
 D_t[\mathbf{u}(t) \cdot \mathbf{v}(t)] &= D_t \sum_{i=1}^3 f_i(t)g_i(t) = \sum_{i=1}^3 D_t[f_i(t)g_i(t)] \\
 &= \sum_{i=1}^3 [f_i(t)g'_i(t) + f'_i(t)g_i(t)] \\
 &= \sum_{i=1}^3 f_i(t)g'_i(t) + \sum_{i=1}^3 f'_i(t)g_i(t) \\
 &= \mathbf{u}(t) \cdot \mathbf{v}'(t) + \mathbf{u}'(t) \cdot \mathbf{v}(t). \quad \square
 \end{aligned}$$

There is an interesting geometric interpretation for the derivative $\mathbf{r}'(t)$. Given $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ where f , g , and h are differentiable, let C denote the curve with parametric equations $x = f(t)$, $y = g(t)$, $z = h(t)$. As in (i) of Figure 15.5, if $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are the position vectors corresponding to $\mathbf{r}(t)$ and $\mathbf{r}(t + \Delta t)$, respectively, then

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

is a geometric vector corresponding to $\mathbf{r}(t + \Delta t) - \mathbf{r}(t)$. It follows that $(1/\Delta t)\overrightarrow{PQ}$ is a vector corresponding to $(1/\Delta t)[\mathbf{r}(t + \Delta t) - \mathbf{r}(t)]$. If $\Delta t > 0$, then $(1/\Delta t)\overrightarrow{PQ}$ is a vector $\overrightarrow{PR}$ having the same direction as $\overrightarrow{PQ}$. Moreover, if $0 < \Delta t < 1$, then $(1/\Delta t) > 1$ and $|\overrightarrow{PR}| > |\overrightarrow{PQ}|$ as illustrated in (ii) of Figure 15.5.

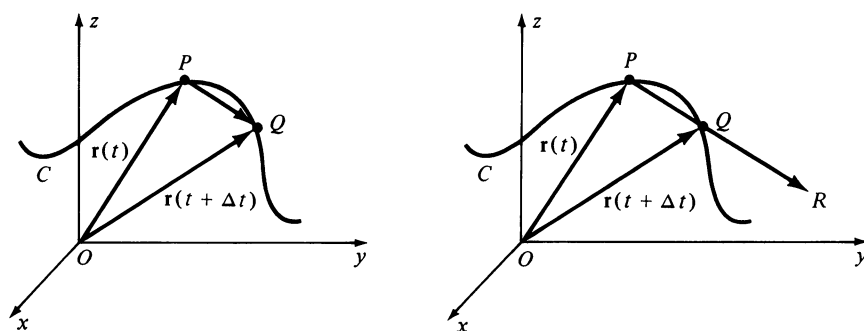


FIGURE 15.5

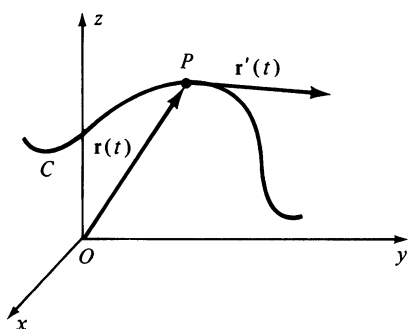


FIGURE 15.6

If we let $\Delta t \rightarrow 0^+$, then Q approaches P and it appears that the vector $\overrightarrow{PR}$ approaches a vector that lies on the tangent line to C at P , as illustrated in Figure 15.6. A similar discussion may be given if $\Delta t < 0$. For this reason we refer to $\mathbf{r}'(t)$, or any of its geometric representations, as a **tangent vector** to C at P . The vector $\mathbf{r}'(t)$ points in the direction of increasing values of t . By definition, the tangent line to C at P is the line through P parallel to $\mathbf{r}'(t)$.

For the two-dimensional case $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$ we have

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j}.$$

If $f'(t) \neq 0$, then the slope of the line parallel to $\mathbf{r}'(t)$ is $g'(t)/f'(t)$. This is in agreement with the formula in Theorem (13.3) obtained for tangent lines to plane curves. These remarks are illustrated in the next example.

Example 2 If $\mathbf{r}(t) = 2t\mathbf{i} + (4 - t^2)\mathbf{j}$, where $-2 \leq t \leq 2$, find $\mathbf{r}'(t)$. Sketch the curve C with parametric equations $x = 2t$, $y = 4 - t^2$, where $-2 \leq t \leq 2$. Illustrate $\mathbf{r}(1)$ and $\mathbf{r}'(1)$ geometrically.

Solution Since we are dealing with V_2 , it is sufficient to use an xy -plane to represent vectors. Eliminating the parameter from $x = 2t$, $y = 4 - t^2$, we obtain $y = 4 - (x/2)^2$, which is an equation of a parabola. Coordinates of points on C that correspond to several values of t are listed below.

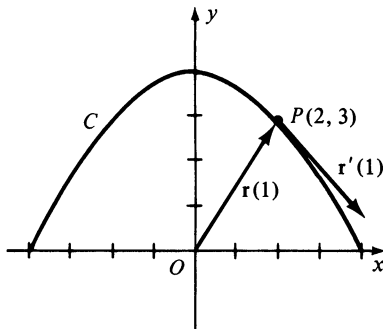


FIGURE 15.7

t	-2	-1	0	1	2
x	-4	-2	0	2	4
y	0	3	4	3	0

The curve C is that part of the parabola illustrated in Figure 15.7. Since $\mathbf{r}(1) = 2\mathbf{i} + 3\mathbf{j}$, the position vector corresponding to $\mathbf{r}(1)$ is $\overrightarrow{OP}$, where P is the point on C with coordinates $(2, 3)$. By differentiation we obtain $\mathbf{r}'(t) = 2\mathbf{i} - 2t\mathbf{j}$. In the figure we have represented $\mathbf{r}'(1) = 2\mathbf{i} - 2\mathbf{j}$ by a vector with initial point P to illustrate that it is a tangent vector to C . ■

Example 3 Let C be the curve with parametric equations

$$x = t^2 - 5, \quad y = t^3, \quad z = 3t + 1$$

where t is in $\mathbb{R}$. Find parametric equations for the tangent line to C at the point corresponding to $t = 2$.

Solution The point $P(t)$ on C corresponding to t is $(t^2 - 5, t^3, 3t + 1)$. If we let

$$\mathbf{r}(t) = (t^2 - 5)\mathbf{i} + t^3\mathbf{j} + (3t + 1)\mathbf{k}$$

then from the previous discussion, a tangent vector to C at $P(t)$ is

$$\mathbf{r}'(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + 3\mathbf{k}.$$

In particular, the point corresponding to $t = 2$ is $P(2) = (-1, 8, 7)$ and a tangent vector is

$$\mathbf{r}'(2) = 4\mathbf{i} + 12\mathbf{j} + 3\mathbf{k}.$$

Applying Theorem (14.37), parametric equations of the tangent line are

$$x = 4t - 1, \quad y = 12t + 8, \quad z = 3t + 7. \quad \blacksquare$$

Example 4 Prove that if $|\mathbf{r}(t)|$ is constant, then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for every t .

Solution By hypothesis

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = |\mathbf{r}(t)|^2 = c$$

for some real number c and, therefore,

$$D_t [\mathbf{r}(t) \cdot \mathbf{r}(t)] = D_t c = 0.$$

Using (iii) of Theorem (15.7) gives us

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) + \mathbf{r}'(t) \cdot \mathbf{r}(t) = 0,$$

or

$$2\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0.$$

Hence, by Theorem (14.24), $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal. $\blacksquare$

The result stated in Example 4 is also geometrically evident, for if $|\mathbf{r}(t)|$ is constant, then as t varies, the endpoint of the position vector corresponding to $\mathbf{r}(t)$ moves on a curve C that lies on the surface of a sphere with center at O . The tangent vector $\mathbf{r}'(t)$ to C is always orthogonal to the position vector and hence to $\mathbf{r}(t)$.

Definite integrals of vector-valued functions may also be considered. Specifically, if

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

where f , g , and h are integrable on $[a, b]$, then by definition

$$\int_a^b \mathbf{r}(t) dt = \left[\int_a^b f(t) dt \right] \mathbf{i} + \left[\int_a^b g(t) dt \right] \mathbf{j} + \left[\int_a^b h(t) dt \right] \mathbf{k}.$$

In this event we say that $\mathbf{r}$ is **integrable** on $[a, b]$.

If $\mathbf{R}'(t) = \mathbf{r}(t)$, then $\mathbf{R}(t)$ is called an **antiderivative** of $\mathbf{r}(t)$. The next result is analogous to the Fundamental Theorem of Calculus.

Theorem (15.8)

If $\mathbf{R}(t)$ is an antiderivative of $\mathbf{r}(t)$ on $[a, b]$, then

$$\int_a^b \mathbf{r}(t) dt = \mathbf{R}(t) \Big|_a^b = \mathbf{R}(b) - \mathbf{R}(a).$$

The proof is left as an exercise.

Example 5 Find $\int_0^2 \mathbf{r}(t) dt$ if $\mathbf{r}(t) = 12t^3\mathbf{i} + 4e^{2t}\mathbf{j} + (t+1)^{-1}\mathbf{k}$.

Solution By finding an antiderivative of each component of $\mathbf{r}(t)$, we obtain

$$\mathbf{R}(t) = 3t^4\mathbf{i} + 2e^{2t}\mathbf{j} + \ln(t+1)\mathbf{k}.$$

Since $\mathbf{R}'(t) = \mathbf{r}(t)$ it follows from Theorem (15.8) that

$$\begin{aligned} \int_0^2 \mathbf{r}(t) dt &= \mathbf{R}(2) - \mathbf{R}(0) \\ &= (48\mathbf{i} + 2e^4\mathbf{j} + \ln 3\mathbf{k}) - (0\mathbf{i} + 2\mathbf{j} + 0\mathbf{k}) \\ &= 48\mathbf{i} + 2(e^4 - 1)\mathbf{j} + \ln 3\mathbf{k}. \end{aligned}$$

Indefinite integrals of vector-valued functions have a theory similar to that developed for real-valued functions. The proofs of theorems require only minor modifications of those given earlier and are omitted. If $\mathbf{R}(t)$ is an antiderivative of $\mathbf{r}(t)$, then every antiderivative has the form $\mathbf{R}(t) + \mathbf{c}$ for some (constant) vector $\mathbf{c}$ and we write

$$\int \mathbf{r}(t) dt = \mathbf{R}(t) + \mathbf{c}, \quad \text{where } \mathbf{R}'(t) = \mathbf{r}(t).$$

If boundary conditions are known, then $\mathbf{c}$ may be determined (see Exercises 27–30).

15.2 Exercises

In Exercises 1–4 find (a) the domain of $\mathbf{r}$ and determine where $\mathbf{r}$ is continuous; (b) $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.

1 $\mathbf{r}(t) = \sqrt{t-1}\mathbf{i} + \sqrt{2-t}\mathbf{j}$

2 $\mathbf{r}(t) = (1/t)\mathbf{i} + \sin 3t\mathbf{j}$

3 $\mathbf{r}(t) = \tan t\mathbf{i} + (t^2 + 8t)\mathbf{j}$

4 $\mathbf{r}(t) = e^{t^2}\mathbf{i} + \sin^{-1} t\mathbf{j}$

In Exercises 5–12 (a) sketch the curve in the xy -plane determined by the components of $\mathbf{r}(t)$; (b) find $\mathbf{r}'(t)$ and sketch vectors corresponding to $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ at the indicated value of t .

5 $\mathbf{r}(t) = (-t^4/4)\mathbf{i} + t^2\mathbf{j}; t = 2$

6 $\mathbf{r}(t) = e^{2t}\mathbf{i} + e^{-4t}\mathbf{j}; t = 0$

7 $\mathbf{r}(t) = 4 \cos t\mathbf{i} + 2 \sin t\mathbf{j}; t = 3\pi/4$

8 $\mathbf{r}(t) = 2 \sec t\mathbf{i} + 3 \tan t\mathbf{j}; t = \pi/4$

9 $\mathbf{r}(t) = t^3\mathbf{i} + t^{-3}\mathbf{j}; t = 1$

10 $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j}; t = -1$

11 $\mathbf{r}(t) = (2t-1)\mathbf{i} + (4-t)\mathbf{j}; t = 3$

12 $\mathbf{r}(t) = 5\mathbf{i} + t^3\mathbf{j}; t = 2$

In Exercises 13–16 (a) find the domain of $\mathbf{r}$ and determine where $\mathbf{r}$ is continuous; (b) find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.

13 $\mathbf{r}(t) = t^2\mathbf{i} + \tan t\mathbf{j} + 3\mathbf{k}$

14 $\mathbf{r}(t) = \sqrt[3]{t}\mathbf{i} + (1/t)\mathbf{j} + e^{-t}\mathbf{k}$

15 $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + e^{2t}\mathbf{j} + t\mathbf{k}$

16 $\mathbf{r}(t) = \ln(1-t)\mathbf{i} + \sin t\mathbf{j} + t^2\mathbf{k}$

Parametric equations of a curve C are given in each of Exercises 17–20. Find parametric equations for the tangent line to C at the point P .

17 $x = 2t^3 - 1, y = -5t^2 + 3, z = 8t + 2; P(1, -2, 10)$

18 $x = 4\sqrt{t}, y = t^2 - 10, z = 4/t; P(8, 6, 1)$

19 $x = e^t, y = te^t, z = t^2 + 4; P(1, 0, 4)$

20 $x = t \sin t, y = t \cos t, z = t; P(\pi/2, 0, \pi/2)$

In Exercises 21 and 22 find two different unit tangent vectors to the given curve at the indicated point P .

21 $x = e^{2t}$, $y = e^{-t}$, $z = t^2 + 4$; $P(1, 1, 4)$

22 $x = 2 + \sin t$, $y = \cos t$, $z = t$; $P(2, 1, 0)$

Evaluate the integrals in Exercises 23–26.

23 $\int_0^2 (6t^2\mathbf{i} - 4t\mathbf{j} + 3\mathbf{k}) dt$

24 $\int_{-1}^1 (-5t\mathbf{i} + 8t^3\mathbf{j} - 3t^2\mathbf{k}) dt$

25 $\int_0^{\pi/4} (\sin t\mathbf{i} - \cos t\mathbf{j} + \tan t\mathbf{k}) dt$

26 $\int_0^1 [te^{t^2}\mathbf{i} + \sqrt{t}\mathbf{j} + (t^2 + 1)^{-1}\mathbf{k}] dt$

27 If $\mathbf{u}'(t) = t^2\mathbf{i} + (6t + 1)\mathbf{j} + 8t^3\mathbf{k}$ and $\mathbf{u}(0) = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, find $\mathbf{u}(t)$.

28 If $\mathbf{r}'(t) = 2\mathbf{i} - 4t^3\mathbf{j} + 6\sqrt{t}\mathbf{k}$ and $\mathbf{r}(0) = \mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$, find $\mathbf{r}(t)$.

29 If $\mathbf{u}''(t) = 6t\mathbf{i} - 12t^2\mathbf{j} + \mathbf{k}$, $\mathbf{u}'(0) = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, and $\mathbf{u}(0) = 7\mathbf{i} + \mathbf{k}$, find $\mathbf{u}(t)$.

30 If $\mathbf{r}''(t) = 6t\mathbf{i} + 3\mathbf{j}$, $\mathbf{r}'(0) = 4\mathbf{i} - \mathbf{j} + \mathbf{k}$, and $\mathbf{r}(0) = 5\mathbf{j}$, find $\mathbf{r}(t)$.

31 If a curve C has a tangent vector $\mathbf{a}$ at a point P , then the **normal plane** to C at P is the plane through P with normal vector $\mathbf{a}$. Find an equation of the normal plane to the curve of Exercise 19 at the point $P(1, 0, 4)$.

32 Refer to Exercise 31. Find an equation for the normal plane to the curve of Exercise 20 at the point $P(\pi/2, 0, \pi/2)$.

33 If $\mathbf{r}$ and $\mathbf{s}$ are vector-valued functions that have limits as $t \rightarrow a$, prove the following.

(a) $\lim_{t \rightarrow a} [\mathbf{r}(t) + \mathbf{s}(t)] = \lim_{t \rightarrow a} \mathbf{r}(t) + \lim_{t \rightarrow a} \mathbf{s}(t)$

(b) $\lim_{t \rightarrow a} [\mathbf{r}(t) \cdot \mathbf{s}(t)] = \lim_{t \rightarrow a} \mathbf{r}(t) \cdot \lim_{t \rightarrow a} \mathbf{s}(t)$

(c) $\lim_{t \rightarrow a} c\mathbf{r}(t) = c \lim_{t \rightarrow a} \mathbf{r}(t)$ where c is a scalar

34 If a function f and a vector-valued function $\mathbf{u}$ have limits as $t \rightarrow a$, prove that

$$\lim_{t \rightarrow a} f(t)\mathbf{u}(t) = \left[\lim_{t \rightarrow a} f(t) \right] \left[\lim_{t \rightarrow a} \mathbf{u}(t) \right].$$

35 Prove that $\lim_{t \rightarrow a} \mathbf{u}(t) = \mathbf{b}$ if and only if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|\mathbf{u}(t) - \mathbf{b}| < \varepsilon$ whenever $0 < |t - a| < \delta$. Interpret this fact geometrically.

36 If $\mathbf{u}$ and $\mathbf{v}$ are vector-valued functions that have limits as $t \rightarrow a$, prove that

$$\lim_{t \rightarrow a} [\mathbf{u}(t) \times \mathbf{v}(t)] = \lim_{t \rightarrow a} \mathbf{u}(t) \times \lim_{t \rightarrow a} \mathbf{v}(t).$$

37 Complete the proof of Theorem (15.7).

38 Prove Theorem (15.8).

39 Prove that if f and $\mathbf{u}$ are differentiable functions of t , then

$$D_t[f(t)\mathbf{u}(t)] = f(t)\mathbf{u}'(t) + f'(t)\mathbf{u}(t).$$

40 Prove the Chain Rule for vector-valued functions:

$$D_t\mathbf{u}(f(t)) = f'(t)\mathbf{u}'(f(t))$$

provided $\mathbf{u}$ and f are differentiable and the domains are suitably restricted.

41 Prove that if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are differentiable vector-valued functions, then

$$D_t[\mathbf{u}(t) \cdot \mathbf{v}(t) \times \mathbf{w}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) \times \mathbf{w}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t) \times \mathbf{w}(t) + \mathbf{u}(t) \cdot \mathbf{v}(t) \times \mathbf{w}'(t).$$

42 Prove that if $\mathbf{u}'(t)$ and $\mathbf{u}''(t)$ exist, then

$$D_t[\mathbf{u}(t) \times \mathbf{u}'(t)] = \mathbf{u}(t) \times \mathbf{u}''(t).$$

43 If $\mathbf{u}$ and $\mathbf{v}$ are integrable on $[a, b]$ and if c is a scalar, prove the following.

(a) $\int_a^b [\mathbf{u}(t) + \mathbf{v}(t)] dt = \int_a^b \mathbf{u}(t) dt + \int_a^b \mathbf{v}(t) dt$

(b) $\int_a^b c\mathbf{u}(t) dt = c \int_a^b \mathbf{u}(t) dt$

44 If $\mathbf{u}$ is integrable on $[a, b]$ and $\mathbf{c}$ is any vector in V_3 , prove that

$$\int_a^b \mathbf{c} \cdot \mathbf{u}(t) dt = \mathbf{c} \cdot \int_a^b \mathbf{u}(t) dt.$$

In Exercises 45 and 46, find $D_t[\mathbf{u}(t) \cdot \mathbf{v}(t)]$ and $D_t[\mathbf{u}(t) \times \mathbf{v}(t)]$.

45 $\mathbf{u}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$
 $\mathbf{v}(t) = \sin t\mathbf{i} + \cos t\mathbf{j} + 2 \sin t\mathbf{k}$

46 $\mathbf{u}(t) = 2t\mathbf{i} + 6t\mathbf{j} + t^2\mathbf{k}$
 $\mathbf{v}(t) = e^{-t}\mathbf{i} - e^{-t}\mathbf{j} + \mathbf{k}$

15.3 Motion

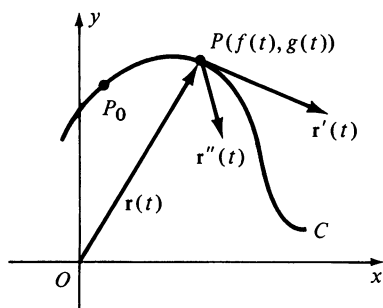


FIGURE 15.8

In this section we introduce concepts that are needed for the study of motion. If we assume that the mass of an object is concentrated at its center of gravity, then we can reduce our study to the motion of a point (or a particle). Motion often takes place in a plane. For example, although the earth moves through space, its orbit lies in a plane. (This will be proved in Section 15.6.) Let us begin, therefore, by studying a particle moving in a coordinate plane. To analyze the behavior of the particle, it is essential to know its position (x, y) at every instant. Since at time t the particle has a unique position, it follows that the coordinates x and y are functions of t . Thus, we shall assume that at time t the particle is at the point $P(x, y)$, where $x = f(t)$ and $y = g(t)$ for certain functions f and g . If we let

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$$

then as t varies, the endpoint of the position vector $\overline{OP}$ corresponding to $\mathbf{r}(t)$ traces the path C of the particle. From the discussion in the preceding section, the tangent line to C at P is parallel to

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j}.$$

(See Figure 15.8.) The magnitude of this vector is

$$|\mathbf{r}'(t)| = \sqrt{[f'(t)]^2 + [g'(t)]^2}.$$

If P_0 is the point corresponding to $t = t_0$, and if C is smooth, then by Theorem (13.11), the arc length $s(t)$ of C from P_0 to P is given by

$$s(t) = \int_{t_0}^t \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_{t_0}^t |\mathbf{r}'(t)| dt.$$

Applying Theorem (5.25) we obtain $s'(t) = |\mathbf{r}'(t)|$; that is, $|\mathbf{r}'(t)|$ is the rate of change of arc length with respect to time. For this reason we refer to $|\mathbf{r}'(t)|$ as the **speed** of the particle. The tangent vector $\mathbf{r}'(t)$, whose magnitude is the speed, is defined as the **velocity** of the particle at time t . In analogy with Theorem (4.28) we call the vector $\mathbf{r}''(t)$ the **acceleration** of the particle at time t and usually represent it by a vector with initial point P . It can be shown that in most cases $\mathbf{r}''(t)$ is directed toward the concave side of C , as illustrated in Figure 15.8.

Example 1 The position of a particle moving in a plane is given by

$$\mathbf{r}(t) = (t^2 + t)\mathbf{i} + t^3\mathbf{j}$$

where $0 \leq t \leq 2$. Find the velocity and acceleration at time t . Sketch the path C of the particle and represent $\mathbf{r}'(1)$ and $\mathbf{r}''(1)$ geometrically.

Solution The velocity and acceleration are

$$\mathbf{r}'(t) = (2t + 1)\mathbf{i} + 3t^2\mathbf{j} \quad \text{and} \quad \mathbf{r}''(t) = 2\mathbf{i} + 6t\mathbf{j}.$$

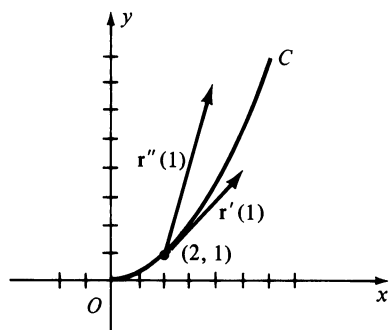


FIGURE 15.9

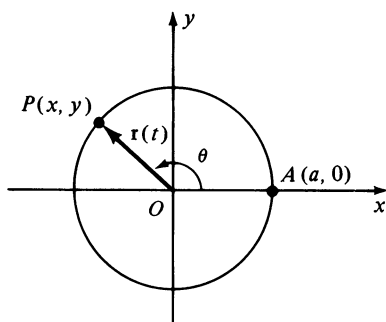


FIGURE 15.10

In particular, at $t = 1$

$$\mathbf{r}'(1) = 3\mathbf{i} + 3\mathbf{j} \quad \text{and} \quad \mathbf{r}''(1) = 2\mathbf{i} + 6\mathbf{j}.$$

These vectors and the path of the particle are sketched in Figure 15.9. ■

Example 2 Show that if a point P moves around a circle of radius a at a constant speed v , then the acceleration vector has a constant magnitude v^2/a and is directed from P toward the center of the circle. (This vector is called the **centripetal acceleration vector**.)

Solution There is no loss of generality in assuming that the center of the circle is at the origin O of a rectangular coordinate system. Suppose that the radius of the circle is a , the initial position of P is $A(a, 0)$, and θ is the angle generated by $\overrightarrow{OP}$ at the end of time t (see Figure 15.10).

The fact that P moves around the circle at a constant speed is equivalent to stating that the rate at which the angle θ changes per unit time (called the **angular speed**) is a constant ω . This fact may be expressed by means of the differential equation $d\theta/dt = \omega$, or equivalently by $\theta = \omega t + c$ for some number c . Since $\theta = 0$ when $t = 0$, the number c is 0; that is $\theta = \omega t$. It follows that the coordinates (x, y) of P are

$$x = a \cos \omega t, \quad y = a \sin \omega t$$

and the motion of P is given by

$$\mathbf{r}(t) = a \cos \omega t \mathbf{i} + a \sin \omega t \mathbf{j}.$$

Consequently,

$$\mathbf{r}'(t) = -\omega a \sin \omega t \mathbf{i} + \omega a \cos \omega t \mathbf{j}$$

$$\mathbf{r}''(t) = -\omega^2 a \cos \omega t \mathbf{i} - \omega^2 a \sin \omega t \mathbf{j}$$

and hence

$$\mathbf{r}''(t) = -\omega^2 \mathbf{r}(t).$$

This shows that the direction of the acceleration vector is opposite that of $\mathbf{r}(t)$, and hence is directed from P toward O .

The magnitude of $\mathbf{r}''(t)$ is

$$|\mathbf{r}''(t)| = |-\omega^2 \mathbf{r}(t)| = |-\omega^2| |\mathbf{r}(t)| = \omega^2 a$$

which is a constant. Since

$$v = |\mathbf{r}'(t)| = \sqrt{(\omega a)^2 \sin^2 \omega t + (\omega a)^2 \cos^2 \omega t} = \sqrt{\omega^2 a^2} = \omega a$$

we see that $\omega = v/a$. Substitution in the formula $|\mathbf{r}''(t)| = \omega^2 a$ gives us

$$|\mathbf{r}''(t)| = \frac{v^2}{a}.$$

Note that $|\mathbf{r}''(t)|$ will increase if we either increase v or decrease a , a fact that is evident to anyone who has attached an object to a string or rope and twirled it about in a circular path.

For an application of Example 2, see Exercises 23 and 24. ■

Our discussion of motion may be extended to a particle moving in a three-dimensional coordinate system. Thus, suppose the position of a particle at time t is given by $P(t) = (f(t), g(t), h(t))$ where the functions f , g , and h are defined on an interval I . If we let

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

then as t varies, the endpoint of the position vector $\overline{OP}$ traces the path of the particle. As before, the derivative $\mathbf{r}'(t)$, if it exists, is called the **velocity** of the particle at time t and is a tangent vector to C at $P(t)$. The vector $\mathbf{r}''(t)$ is called the **acceleration** of the particle at time t . The **speed** of the particle is the magnitude $|\mathbf{r}'(t)|$ of the velocity and, as in two dimensions, equals the rate of change of arc length with respect to time.

Example 3 The position of a particle is given by $\mathbf{r}(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k}$, where $0 \leq t \leq 2$. Find the velocity and acceleration at time t . Sketch the path of the particle and illustrate $\mathbf{r}'(1)$ and $\mathbf{r}''(1)$ geometrically.

Solution The velocity and acceleration are

$$\mathbf{r}'(t) = 2\mathbf{i} + 6t\mathbf{j} + 3t^2\mathbf{k} \quad \text{and} \quad \mathbf{r}''(t) = 6\mathbf{j} + 6t\mathbf{k}.$$

In particular, at $t = 1$ the particle is at point $P(2, 3, 1)$ and

$$\mathbf{r}'(1) = 2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k} \quad \text{and} \quad \mathbf{r}''(1) = 6\mathbf{j} + 6\mathbf{k}.$$

These vectors and the path of the particle are sketched in Figure 15.11. ■

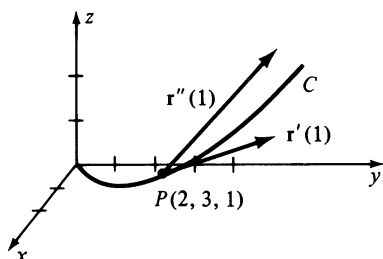


FIGURE 15.11

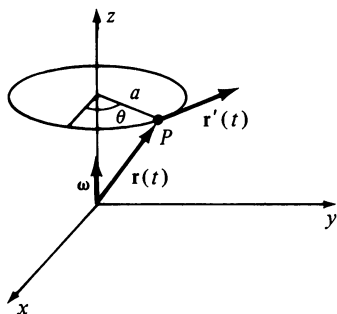


FIGURE 15.12

Suppose a particle P is rotating about the z -axis on a circle of radius a that lies in a plane parallel to the xy -plane, as illustrated in Figure 15.12. In addition, suppose the angular speed $d\theta/dt$ is a constant ω . The vector $\boldsymbol{\omega} = \omega\mathbf{k}$ directed along the z -axis and having magnitude ω is called the **angular velocity** of P . As in Example 2, the motion of P is given by

$$\mathbf{r}(t) = a \cos \omega t \mathbf{i} + a \sin \omega t \mathbf{j} + h\mathbf{k},$$

where h is the distance from the xy -plane to the particle. A direct computation shows that

$$\boldsymbol{\omega} \times \mathbf{r}(t) = -\omega a \sin \omega t \mathbf{i} + \omega a \cos \omega t \mathbf{j} = \mathbf{r}'(t).$$

Thus the velocity vector is the cross product of the angular velocity and the position vector of P . This fact can be generalized to the rotation of a particle P about any line.

As a final illustration of how vectors may be applied to problems involving motion, suppose a projectile is fired from a cannon, with an angle of elevation α . If the initial speed is v_0 , then we may introduce a coordinate plane as in Figure 15.13, where the muzzle of the cannon is at the origin O and the initial velocity $\mathbf{v}_0$ has direction α and magnitude v_0 . Let $P(x, y)$ denote the position of the projectile after time t and let $\mathbf{r}(t)$ correspond to $\overline{OP}$. Our objective is to find an explicit form for $\mathbf{r}(t)$.

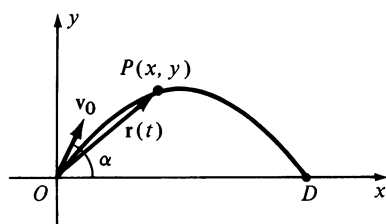


FIGURE 15.13

We shall assume that air resistance is negligible and that the only force acting on the projectile is the force $\mathbf{g}$ of gravitational acceleration. Since $\mathbf{g}$ acts in the downward direction we may write $\mathbf{g} = -g\mathbf{j}$, where $|\mathbf{g}| = g \approx 32 \text{ ft/sec}^2$. According to Newton's Second Law of Motion, the force $\mathbf{F}$ acting on an object of mass m is related to its acceleration $\mathbf{a}$ by the formula $\mathbf{F} = m\mathbf{a}$. In the present situation, $m\mathbf{a} = m\mathbf{g}$, or $\mathbf{a} = \mathbf{g}$, which leads to the vector differential equation

$$\mathbf{r}''(t) = \mathbf{g}.$$

Indefinite integration gives us

$$\mathbf{r}'(t) = t\mathbf{g} + \mathbf{c}$$

where $\mathbf{c}$ is some vector. Since $\mathbf{r}'(t)$ is the velocity at time t ,

$$\mathbf{v}_0 = \mathbf{r}'(0) = \mathbf{c}$$

and hence

$$\mathbf{r}'(t) = t\mathbf{g} + \mathbf{v}_0.$$

Integrating both sides of the last equation we obtain

$$\mathbf{r}(t) = \frac{1}{2}t^2\mathbf{g} + t\mathbf{v}_0 + \mathbf{d}$$

for some vector $\mathbf{d}$. Since $\mathbf{r}(0) = \mathbf{0}$, it follows that $\mathbf{d} = \mathbf{0}$. Consequently,

$$\mathbf{r}(t) = \frac{1}{2}t^2\mathbf{g} + t\mathbf{v}_0$$

which may also be written

$$x\mathbf{i} + y\mathbf{j} = -\frac{1}{2}t^2g\mathbf{j} + t(v_0 \cos \alpha \mathbf{i} + v_0 \sin \alpha \mathbf{j}).$$

Equating components we see that

$$(*) \quad x = (v_0 \cos \alpha)t, \quad y = -\frac{1}{2}gt^2 + (v_0 \sin \alpha)t.$$

These are parametric equations for the path of the projectile. Eliminating the parameter gives us the rectangular equation

$$y = \frac{-g}{2v_0^2 \cos^2 \alpha} x^2 + (\tan \alpha)x.$$

This shows that the path of the projectile is parabolic. To find the range, that is, the distance $d(O, D)$ from O to D in Figure 15.13, we let $y = 0$ in (*), obtaining

$$t(-\frac{1}{2}gt + v_0 \sin \alpha) = 0.$$

Thus the point D corresponds to $t = (2v_0 \sin \alpha)/g$. Using the first equation in (*) gives us

$$d(O, D) = (v_0 \cos \alpha) \left(\frac{2v_0 \sin \alpha}{g} \right) = \frac{v_0^2 \sin 2\alpha}{g}.$$

In particular, the range will have its maximum value v_0^2/g if $\sin 2\alpha = 1$, or $\alpha = 45^\circ$. The maximum altitude h occurs when $t = (v_0 \sin \alpha)/g$. (Why?)

Substituting in the second equation of (*), we obtain

$$h = -\frac{1}{2}g\left(\frac{v_0 \sin \alpha}{g}\right)^2 + (v_0 \sin \alpha)\left(\frac{v_0 \sin \alpha}{g}\right) = \frac{v_0^2 \sin^2 \alpha}{2g}.$$

15.3 Exercises

In Exercises 1–8 the position of a particle moving in a plane is given by $\mathbf{r}(t)$. Find its velocity, acceleration, and speed at time t . Sketch the path of the particle together with vectors corresponding to the velocity and acceleration at the indicated time t .

- 1 $\mathbf{r}(t) = 2t\mathbf{i} + (4t^2 + 1)\mathbf{j}$, $t = 1$
- 2 $\mathbf{r}(t) = (4 - 9t^2)\mathbf{i} + 3t\mathbf{j}$, $t = 1$
- 3 $\mathbf{r}(t) = (2/t)\mathbf{i} + [3/(t + 1)]\mathbf{j}$, $t = 2$
- 4 $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + (1 + \sqrt{t})\mathbf{j}$, $t = 4$
- 5 $\mathbf{r}(t) = \sin t\mathbf{i} + 4 \cos 2t\mathbf{j}$, $t = \pi/6$
- 6 $\mathbf{r}(t) = \cos^2 t\mathbf{i} + 2 \sin t\mathbf{j}$, $t = 3\pi/4$
- 7 $\mathbf{r}(t) = e^{2t}\mathbf{i} + e^{-t}\mathbf{j}$, $t = 0$
- 8 $\mathbf{r}(t) = 2t\mathbf{i} + e^{-t^2}\mathbf{j}$, $t = 1$

In Exercises 9–16 $\mathbf{r}(t)$ is the position vector of a particle moving in space. Find the velocity, acceleration, and speed at time t . Sketch the path of the particle together with vectors corresponding to $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$ for the indicated values of t .

- 9 $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$; $t = 0, \pi/4, \pi/2$
- 10 $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j} + t\mathbf{k}$; $t = 0, 1, 2$
- 11 $\mathbf{r}(t) = t^2\mathbf{i} + 2\sqrt{t}\mathbf{j} + 4\sqrt{t^3}\mathbf{k}$; $t = 1, 4, 9$
- 12 $\mathbf{r}(t) = 4 \sin t\mathbf{i} + 2t\mathbf{j} + 9 \cos t\mathbf{k}$; $t = 0, \pi/2, 3\pi/4$
- 13 $\mathbf{r}(t) = e^t(\cos t\mathbf{i} + \sin t\mathbf{j} + \mathbf{k})$, $t = 0, \pi/4, \pi/2$
- 14 $\mathbf{r}(t) = t(\cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k})$, $t = 0, \pi/4, \pi/2$

- 15 $\mathbf{r}(t) = (1 + t)\mathbf{i} + 2t\mathbf{j} + (2 + 3t)\mathbf{k}$, $t = 0, 1, 2$
- 16 $\mathbf{r}(t) = 2t\mathbf{i} + \mathbf{j} + 9t^2\mathbf{k}$, $t = 0, 1, 2$
- 17 Prove that if a particle moves at a constant speed, then the velocity and acceleration vectors are orthogonal.
- 18 Prove that if the acceleration of a moving particle is always $\mathbf{0}$, then the motion is rectilinear.
- 19 A projectile is fired with an initial speed of 1500 ft/sec and angle of elevation 30° . Find (a) the velocity at time t ; (b) the maximum altitude; (c) the range; and (d) the speed at which the projectile strikes level ground.
- 20 Work Exercise 19 if the angle of elevation is 60° .
- 21 An outfielder on a baseball team threw a ball a distance of 250 ft. If he released the ball at an angle of 45° to the ground, what was the initial speed?
- 22 A projectile is fired horizontally with a velocity of 1800 ft/sec from an altitude of 1000 ft above level ground. When and where will it strike the ground?

Use the results of Example 2 to solve Exercises 23 and 24. Take the radius of the earth as 4,000 miles.

- 23 If a space shuttle is in a circular orbit 150 mi above the surface of the earth approximate (a) its speed; (b) the length of time for one complete revolution around the earth.
- 24 A satellite that is in a circular orbit makes one revolution around the earth every 88 minutes. Approximate its altitude.

15.4 Curvature

For many applications involving vector-valued functions it is convenient to employ unit tangent vectors to curves. Let us begin by considering a vector-valued function $\mathbf{r}$ in two dimensions such that $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$. Let C be the curve determined by the endpoint of the position vector corresponding to $\mathbf{r}(t)$. From our work in Section 15.3 we know that $\mathbf{r}'(t)$ is a tangent vector

to C . If $\mathbf{r}'(t) \neq \mathbf{0}$, then by Theorem (14.12), a **unit tangent vector** $\mathbf{T}(t)$ to C is given by the following formula.

$$(15.9) \quad \mathbf{T}(t) = \frac{1}{|\mathbf{r}'(t)|} \mathbf{r}'(t)$$

Since $|\mathbf{T}(t)| = 1$, it follows from Example 4 of Section 15.2 that $\mathbf{T}'(t)$ is orthogonal to $\mathbf{T}(t)$ for every t . Letting $\mathbf{N}'(t) = (1/|\mathbf{T}'(t)|)\mathbf{T}'(t)$ gives us a **unit vector** orthogonal to $\mathbf{T}(t)$. We shall refer to $\mathbf{N}(t)$ as a **unit normal vector** to C . Let us state this for reference as follows.

$$(15.10) \quad \mathbf{N}(t) = \frac{1}{|\mathbf{T}'(t)|} \mathbf{T}'(t)$$

The next example illustrates these concepts.

Example 1 Let C be the curve determined by $\mathbf{r}(t) = t^2\mathbf{i} + t\mathbf{j}$.

- Find the unit tangent and normal vectors $\mathbf{T}(t)$ and $\mathbf{N}(t)$.
- Sketch C , together with $\mathbf{T}(1)$ and $\mathbf{N}(1)$.

Solution Using (15.9) with $\mathbf{r}'(t) = 2t\mathbf{i} + \mathbf{j}$, we obtain

$$\mathbf{T}(t) = \frac{1}{(4t^2 + 1)^{1/2}} (2t\mathbf{i} + \mathbf{j}) = \frac{2t}{(4t^2 + 1)^{1/2}} \mathbf{i} + \frac{1}{(4t^2 + 1)^{1/2}} \mathbf{j}.$$

Differentiating the components of $\mathbf{T}(t)$ gives us

$$\mathbf{T}'(t) = \frac{2}{(4t^2 + 1)^{3/2}} \mathbf{i} - \frac{4t}{(4t^2 + 1)^{3/2}} \mathbf{j} = \frac{2}{(4t^2 + 1)^{3/2}} (\mathbf{i} - 2t\mathbf{j}).$$

It is easy to verify that $|\mathbf{T}'(t)| = 2/(4t^2 + 1)$. Applying (15.10) and simplifying leads to

$$\mathbf{N}(t) = \frac{1}{(4t^2 + 1)^{1/2}} (\mathbf{i} - 2t\mathbf{j}).$$

Note that

$$\mathbf{T}(t) \cdot \mathbf{N}(t) = \frac{1}{4t^2 + 1} (2t - 2t) = 0$$

and hence $\mathbf{T}(t)$ and $\mathbf{N}(t)$ are, indeed, orthogonal.

At $t = 1$ we have

$$\mathbf{T}(1) = \frac{1}{\sqrt{5}} (2\mathbf{i} + \mathbf{j}) \quad \text{and} \quad \mathbf{N}(1) = \frac{1}{\sqrt{5}} (\mathbf{i} - 2\mathbf{j}).$$

These vectors and the curve (a parabola) are sketched in Figure 15.14. ■

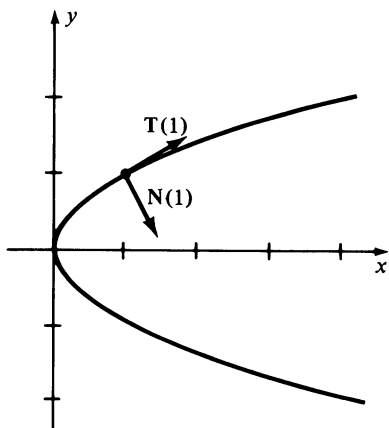


FIGURE 15.14

Note that the unit normal vector $\mathbf{N}(t)$ in Example 1 points toward the concave side of the curve. It can be shown that this is always the case.

A plane curve may be represented parametrically in many different ways. Sometimes it is convenient to use arc length as a parameter. Thus, suppose a curve C in the xy -plane is given by

$$x = f(s), \quad y = g(s)$$

where f' and g' are continuous on an interval I and where the parameter s is arc length measured along C from some fixed point A . Geometrically we have a situation similar to that illustrated in Figure 15.15, where for each s in I there corresponds a unique point $P(s) = (f(s), g(s))$ which is s units from A (measured along C). The positive direction along C is determined by increasing values of s . We shall call s the **arc length parameter** for the curve C . The position of the fixed point A is irrelevant.

If, as illustrated in Figure 15.15, $\mathbf{r}(s) = f(s)\mathbf{i} + g(s)\mathbf{j}$ is the position vector of $P(s)$, then we know that $\mathbf{r}'(s)$ is a tangent vector to C . However, if s is the arc length parameter for C , it can be shown that $\mathbf{r}'(s)$ is a *unit* tangent vector. To establish this fact first note that by Theorem (13.11)

$$s = \int_0^s \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

Differentiating with respect to s gives us

$$1 = \sqrt{[f'(s)]^2 + [g'(s)]^2} = |\mathbf{r}'(s)|.$$

We shall denote the unit tangent vector $\mathbf{r}'(s)$ by $\mathbf{T}(s)$ and study its variation as $P(s)$ moves along C . For each s let θ be the angle between $\mathbf{T}(s)$ and $\mathbf{i}$, as illustrated in Figure 15.16. Observe that θ is a function of s , because for each s there corresponds a point $P(s)$ on C that, in turn, determines a value of θ . The rate of change $d\theta/ds$ of θ with respect to s is the basis for the next definition.

Definition (15.11)

Let a smooth curve C be given by $x = f(s)$, $y = g(s)$, where s is the arc length parameter. The **curvature** K of C at the point $P(s) = P(x, y)$ is

$$K = \left| \frac{d\theta}{ds} \right|.$$

Thus, curvature is the absolute value of the rate at which θ changes with respect to arc length s . The curvature is small for points such as R and S in Figure 15.16, since θ changes slowly as $P(s)$ moves along C . The curvature is large for the points Q and V , since θ changes rapidly near those points. Thus, roughly speaking, K provides information about the sharpness of the curve at various points.

The next two examples give important illustrations of Definition 15.11.

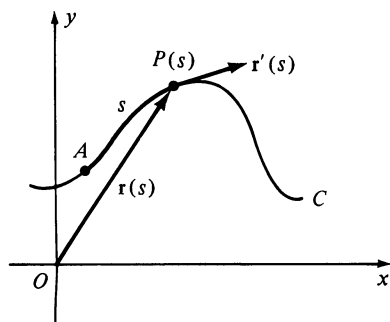


FIGURE 15.15

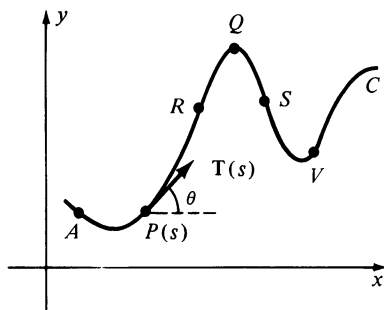


FIGURE 15.16

Example 2 Prove that the curvature of a line is 0 at every point on the line.

Solution If C is a line, then the angle θ is the same for every point $P(s)$ on the line, that is, θ is constant. Hence

$$K = |d\theta/ds| = |0| = 0. \quad \blacksquare$$

Example 3 Prove that the curvature at any point on a circle of radius a is $1/a$.

Solution There is no loss of generality in assuming, as in Figure 15.17, that the circle has its center at O , and that the point P is in the first quadrant. If we let $A(a, 0)$ be fixed, and let s be the length of arc AP , then from Theorem (1.26),

$$s = a\alpha \quad \text{or} \quad \alpha = s/a.$$

where α is angle POA . Referring to Figure 15.17 we see that

$$\theta = \alpha + (\pi/2) = (s/a) + (\pi/2)$$

and, therefore,
$$\frac{d\theta}{ds} = \frac{1}{a} \quad \text{and} \quad K = \frac{1}{a}. \quad \blacksquare$$

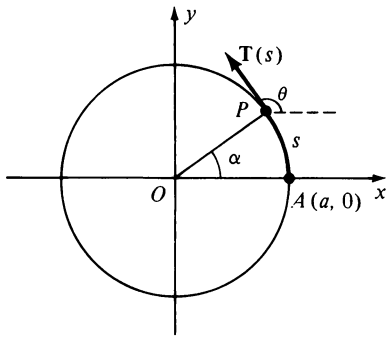


FIGURE 15.17

Note, from Example 3, that as the radius a of a circle increases, its curvature $K = 1/a$ decreases. Indeed, if we let a increase without bound, K approaches 0, the curvature of a line. As the radius a approaches 0, the curvature K increases without bound.

Suppose a curve C is the graph of a rectangular equation $y = h(x)$, where h' is continuous on some interval. Since y' is the slope of the tangent line at P we see, as in Figure 15.16, that

$$(a) \quad \tan \theta = y' \quad \text{or} \quad \theta = \tan^{-1} y'.$$

From Definition (6.17), the arc length function s may be defined by

$$(b) \quad s(x) = \int_a^x \sqrt{1 + (y')^2} \, dx$$

where a is the x -coordinate of the fixed point A on C . If y'' exists, then by the Chain Rule

$$\frac{d\theta}{dx} = \frac{d\theta}{ds} \frac{ds}{dx}.$$

Consequently,

$$(c) \quad K = \left| \frac{d\theta}{ds} \right| = \left| \frac{d\theta/dx}{ds/dx} \right|.$$

Referring to (a) and (b),

$$\frac{d\theta}{dx} = \frac{y''}{1 + (y')^2} \quad \text{and} \quad \frac{ds}{dx} = \sqrt{1 + (y')^2}.$$

Substitution in (c) gives us the following theorem.

Theorem (15.12)

If C is the graph of $y = f(x)$, then the curvature K at $P(x, y)$ is

$$K = \frac{|y''|}{[1 + (y')^2]^{3/2}}.$$

Example 4 Sketch the graph of $y = 1 - x^2$ and find the curvature at the points (x, y) , $(0, 1)$, $(1, 0)$, and $(2, -3)$.

Solution The graph (a parabola) is sketched in Figure 15.18.

Since $y' = -2x$ and $y'' = -2$ we have, from Theorem (15.12),

$$K = \frac{2}{(1 + 4x^2)^{3/2}}.$$

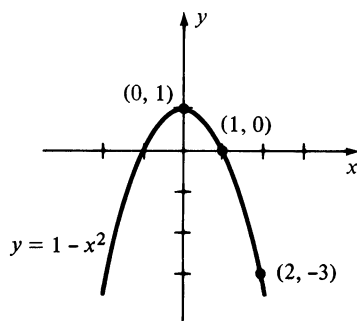


FIGURE 15.18

In particular, at $(0, 1)$ we see that $K = 2$; that is, the direction of the tangent vector changes at the rate of 2 radians per unit change in arc length. At the point $(1, 0)$, $K = 2/(5)^{3/2} \approx 0.18$, which shows that the curve is less sharp here than at $(0, 1)$. This can also be seen from the graph. Finally, at $(2, -3)$, $K = 2/(17)^{3/2} \approx 0.03$. Observe that as x increases without bound the curvature approaches 0. ■

In the next example we shall employ Theorem (15.12) to work Example 3.

Example 5 Prove that the curvature at any point on a circle of radius a is $1/a$.

Solution Let us assume, as in Figure 15.17, that the circle has center at the origin and equation $x^2 + y^2 = a^2$. Implicit differentiation yields

$$2x + 2yy' = 0, \quad \text{or} \quad y' = -x/y.$$

Consequently,

$$\begin{aligned} y'' &= -\frac{y - xy'}{y^2} = -\frac{y + (x^2/y)}{y^2} \\ &= -\frac{y^2 + x^2}{y^3} = -\frac{a^2}{y^3}. \end{aligned}$$

Substituting in (15.12), and using the fact that $x^2 + y^2 = a^2$,

$$K = \frac{|a^2/y^3|}{[1 + x^2/y^2]^{3/2}} = \frac{|a^2/y^3|}{[(y^2 + x^2)/y^2]^{3/2}} = \frac{|a^2/y^3|}{(a^2/y^2)^{3/2}} = \frac{1}{a}. \quad \blacksquare$$

Let us next derive a formula for finding K if C is described in terms of any parameter t . Suppose C is given parametrically by

$$x = f(t), \quad y = g(t)$$

and that f'' and g'' exist for all t under consideration. As in the proof of Theorem (15.12)

$$K = \left| \frac{d\theta}{ds} \right| = \left| \frac{d\theta/dt}{ds/dt} \right|.$$

Since the slope of the tangent line is given by $g'(t)/f'(t)$, we have

$$\tan \theta = \frac{g'(t)}{f'(t)}, \quad \text{or} \quad \theta = \tan^{-1} \frac{g'(t)}{f'(t)}$$

provided $f'(t) \neq 0$. Hence

$$\frac{d\theta}{dt} = \frac{1}{1 + [g'(t)/f'(t)]^2} \frac{f'(t)g''(t) - g'(t)f''(t)}{[f'(t)]^2} = \frac{f'(t)g''(t) - g'(t)f''(t)}{[f'(t)]^2 + [g'(t)]^2}.$$

Moreover,

$$\left| \frac{ds}{dt} \right| = \sqrt{[f'(t)]^2 + [g'(t)]^2}.$$

Substituting in the formula $K = |(d\theta/dt)/(ds/dt)|$ gives us the following.

Theorem (15.13)

If a curve C is given parametrically by $x = f(t)$, $y = g(t)$, where f'' and g'' exist, then the curvature K at $P(x, y)$ is

$$K = \frac{|f'(t)g''(t) - g'(t)f''(t)|}{[(f'(t))^2 + (g'(t))^2]^{3/2}}.$$

If the curvature at a point P on a curve C is not zero, then the circle of radius $\rho = 1/K$ whose center lies on the concave side of C and which has the same tangent line at P as C is called the **circle of curvature** for P . Its radius ρ and center are called the **radius of curvature** and **center of curvature**, respectively, for P . According to Examples 3 and 5 the curvature of the circle of curvature is $1/\rho$, or K , and hence is the same as the curvature of C . For this reason the circle of curvature may be thought of as the circle that best coincides with C at P .

Example 6 Let C be the curve with parametric equations $x = t^2$, $y = t^3$. Find the curvature at the point P corresponding to $t = \frac{1}{2}$. Sketch the graph of C and the circle of curvature for P .

Solution Letting $f(t) = t^2$ and $g(t) = t^3$, we obtain $f'(t) = 2t$, $f''(t) = 2$, $g'(t) = 3t^2$, and $g''(t) = 6t$. Substituting in Theorem (15.13),

$$K = \frac{|(2t)(6t) - (3t^2)(2)|}{[(2t)^2 + (3t^2)^2]^{3/2}} = \frac{6t^2}{(4t^2 + 9t^4)^{3/2}}.$$

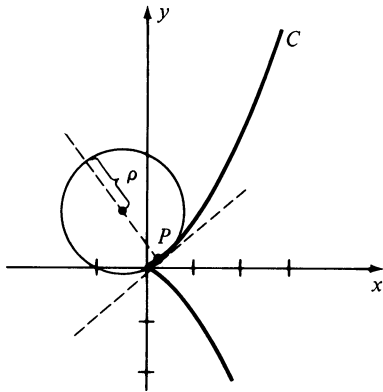


FIGURE 15.19

If $t = \frac{1}{2}$, then $K = (\frac{6}{4})/(\frac{25}{16})^{3/2} = \frac{96}{125} \approx 0.77$. The point corresponding to $t = \frac{1}{2}$ has coordinates $(\frac{1}{4}, \frac{1}{8})$ and the radius of curvature ρ at that point is $\frac{1}{K}$ or $\frac{125}{96} \approx 1.3$. The graph of C and the circle of curvature are sketched in Figure 15.19. ■

The definition of curvature $K = |d\theta/ds|$ introduced in (15.11) has no immediate analogue in three dimensions, because the unit tangent vector $\mathbf{T}(s)$ cannot be specified in terms of a single angle θ . Thus it is necessary to use a different approach for space curves. Of course, we then must show that if we specialize to vectors in a plane, the new definition of curvature agrees with (15.11).

To find a clue to a suitable definition, we first observe that in two dimensions,

$$\mathbf{T}(s) = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$$

where θ is the angle considered in Theorem (15.11). Regarding θ as a function of s and differentiating gives us

$$\mathbf{T}'(s) = \left(-\sin \theta \frac{d\theta}{ds}\right) \mathbf{i} + \left(\cos \theta \frac{d\theta}{ds}\right) \mathbf{j} = \frac{d\theta}{ds} (-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}).$$

$$\text{Hence} \quad |\mathbf{T}'(s)| = \left| \frac{d\theta}{ds} \right| |-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}| = \left| \frac{d\theta}{ds} \right| = K.$$

We shall use this fact to introduce the concept of curvature in three dimensions. Indeed, our plan is to describe the vector $\mathbf{T}(s)$ *without* referring to the angle θ , and then *define* K as $|\mathbf{T}'(s)|$. Let us proceed as follows.

Suppose a curve C is given by

$$x = f(s), \quad y = g(s), \quad z = h(s)$$

where s is arc length measured along C from a fixed point A to the point $P(s) = (f(s), g(s), h(s))$ and where $f''(s)$, $g''(s)$ and $h''(s)$ exist. As before, if $\mathbf{r}(s) = f(s)\mathbf{i} + g(s)\mathbf{j} + h(s)\mathbf{k}$ is the position vector of $P(s)$, then $\mathbf{r}'(s)$ is a unit tangent vector to C at $P(s)$ which we shall denote by $\mathbf{T}(s)$. Since $|\mathbf{T}(s)|$ is a constant, it follows that $\mathbf{T}'(s)$ is orthogonal to $\mathbf{T}(s)$ (see Example 4 of Section 15.2). If $\mathbf{T}'(s) \neq \mathbf{0}$, let

$$\mathbf{N}(s) = \frac{1}{|\mathbf{T}'(s)|} \mathbf{T}'(s).$$

The vector $\mathbf{N}(s)$ is a unit vector orthogonal to $\mathbf{T}(s)$ and is referred to as the **principal unit normal vector** to C at the point $P(s)$. These concepts are illustrated in Figure 15.20.

Having suitably arrived at the unit tangent vector $\mathbf{T}(s)$, we now follow our plan and define curvature in three dimensions as follows, where it is assumed that $\mathbf{T}'(s)$ exists.

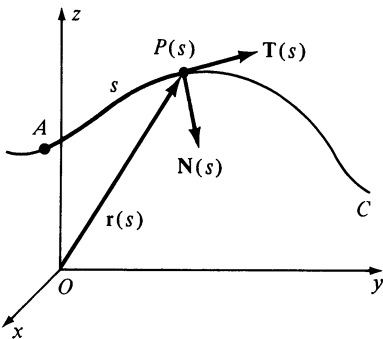


FIGURE 15.20

Definition (15.14)

Let a curve C be given by

$$x = f(s), \quad y = g(s), \quad z = h(s)$$

where s is the arc length parameter. Let $\mathbf{r}(s) = f(s)\mathbf{i} + g(s)\mathbf{j} + h(s)\mathbf{k}$ and let $\mathbf{T}(s) = \mathbf{r}'(s)$. The **curvature** K of C at the point $P(x, y, z)$ is

$$K = |\mathbf{T}'(s)|.$$

As we have already proved, the preceding definition reduces to (15.11) if C is a plane curve. Note that

$$\mathbf{N}(s) = \frac{1}{K} \mathbf{T}'(s), \quad \text{or} \quad \mathbf{T}'(s) = K\mathbf{N}(s).$$

In the next section we shall derive a formula that can be used to find the curvature of a space curve.

15.4 Exercises

In Exercises 1–12 find the curvature of the curve at the point P .

- 1 $y = 2 - x^3$; $P(1, 1)$
- 2 $y = x^4$; $P(1, 1)$
- 3 $y = e^{x^2}$; $P(0, 1)$
- 4 $y = \ln(x - 1)$; $P(2, 0)$
- 5 $y = \cos 2x$; $P(0, 1)$
- 6 $y = \sec x$; $P(\pi/3, 2)$
- 7 $x = t - 1, y = \sqrt{t}$; $P(3, 2)$
- 8 $x = t + 1, y = t^2 + 4t + 3$; $P(1, 3)$
- 9 $x = t - t^2, y = 1 - t^3$; $P(0, 1)$
- 10 $x = t - \sin t, y = 1 - \cos t$; $P(\pi/2 - 1, 1)$
- 11 $x = 2 \sin t, y = 3 \cos t$; $P(1, 3\sqrt{3}/2)$
- 12 $x = \cos^3 t, y = \sin^3 t$; $P(\sqrt{2}/4, \sqrt{2}/4)$

For the curves specified in Exercises 13–16 (a) find the radius of curvature for P ; (b) find the center of curvature for P ; and (c) sketch the graph and the circle of curvature for P .

- 13 $y = \sin x$; $P(\pi/2, 1)$
- 14 $y = \sec x$; $P(0, 1)$
- 15 $y = e^x$; $P(0, 1)$
- 16 $xy = 1$; $P(1, 1)$

For the curves described in Exercises 17–22 find the points at which the curvature is a maximum.

- 17 $y = e^{-x}$
- 18 $y = \cosh x$
- 19 $9x^2 + 4y^2 = 36$
- 20 $9x^2 - 4y^2 = 36$
- 21 $y = \ln x$
- 22 $y = \sin x$
- 23 Use the equation $y = mx + b$ to prove that the curvature at every point on a line is 0 (cf. Example 2).
- 24 Prove that the maximum curvature of a parabola occurs at the vertex.
- 25 Prove that for an ellipse, the maximum and minimum curvature occur at the ends of the major and minor axes, respectively.
- 26 Prove that for a hyperbola, the maximum curvature occurs at the ends of the transverse axis.

In Exercises 27–30 find the points on the indicated curve at which the curvature is 0.

- 27 $y = x^4 - 12x^2$
- 28 $y = \tan x$
- 29 $y = \sinh x$
- 30 $y = e^{-x^2}$